

# **Benchmarking Holistic vs Sentence-Level Reward-Model Preferences for Factual Summarization via Granular Credit Assignment (GCA)**

## **Master's Thesis**

in partial fulfillment of the requirements for  
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in Web and Data Science

submitted by

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## Abstract

Preference-based alignment of abstractive summarization models increasingly relies on automatically generated preference labels rather than human judgments. It remains unclear whether a single holistic judgment over a multi-sentence summary uses the available supervision as effectively as a judgment assembled from sentence-level evidence. This thesis studies that question as a controlled comparison of two preference-construction procedures: a holistic condition, where a complete candidate summary is scored against the source article, and a Granular Credit Assignment (GCA) condition, where each summary is decomposed into sentences, each is scored separately, and the sentence scores are aggregated into a summary-level preference. The candidate summaries, the factuality evaluator (AlignScore), the Bradley–Terry reward-model architecture, and the training procedure are held fixed across four research questions, so feedback granularity is isolated as the primary experimental variable.

**RQ1** asks whether GCA produces a more learnable preference signal, measured as reward-model pairwise validation accuracy. Across twenty independently seeded runs at  $n = 1,000$ , GCA led in every run, a mean advantage of 3.95 percentage points (bootstrap 95% CI  $[+3.19, +4.70]$ , run-level Wilcoxon  $p < 0.001$ , Cohen’s  $d = 2.22$ ).

**RQ3** asks whether this advantage is stable across dataset scale; it is not. Twenty-run campaigns at  $n = 5,000$  and  $n = 10,000$ , under the same corrected procedure, give mean gaps of  $-0.22$  and  $-0.11$  percentage points, both statistically indistinguishable from zero, with equivalence tests excluding any effect larger than  $\pm 0.61$  and  $\pm 0.36$  points: the advantage is absent, not merely undetected. Absolute accuracy improves for both conditions as data grows even as their difference vanishes.

**RQ2** asks how evaluator configuration and aggregation strategy affect this comparison. Both matter substantially: an early weakest-link penalty formula hurt GCA relative to a simple mean, and a sweep across four AlignScore modes finds GCA’s advantage appears only under modes that do not themselves decompose the input internally, suggesting external and evaluator-internal decomposition act as partial substitutes. Each mode was tested once, so this is associational, not confirmed.

**RQ4** asks whether the RQ1 advantage also holds against an independent, held-out ground truth: 500 articles disjoint from every training set, scored by the same judge but never used in training. It does, more sharply: at  $n = 1,000$ , GCA is significantly more precise than holistic on comparisons with a genuine quality gap (up to  $+9.25$  percentage points,  $p < 0.001$ ). It does not merely fail to persist, as RQ1 does; it reverses to a significant advantage for holistic at  $n = 10,000$  on three of four ground-truth constructions ( $p = 0.010$ – $0.036$ ). Two candidate explanations, near-zero-margin training pairs and a length/sentence-count shortcut, were tested and both rejected; the mechanism behind the reversal remains open.

These results characterize when granular feedback changes a preference signal’s structure, not that GCA is generally preferable: it is not, once data are abundant or the comparison is ambiguous. Higher accuracy shows a preference relation is more recoverable by this architecture, not that GCA labels better match human judgments or that trained models are more factually consistent, both future work.

# Acknowledgments

I would like to thank my first supervisor, Prof. Dr. Frank Hopfgartner of the Institute for Web Science and Technologies, for his guidance throughout this project and in particular for the feedback that led me to narrow the study to a single, cleanly controlled comparison rather than a broader but less conclusive pipeline. I am equally grateful to my second supervisor, Dr. Zeyd Boukhers of the Fraunhofer Institute for Applied Information Technology FIT, for his technical input on reward modelling and factuality evaluation.

The experiments in this thesis were carried out on the MOGON NHR cluster at Johannes Gutenberg University Mainz. I gratefully acknowledge the compute time made available there; the repeated-run and multi-scale campaigns reported in Chapter 6 would not have been feasible without access to that infrastructure.

Finally, I thank the University of Koblenz for providing the research environment in which this work was completed, and my family and friends for their support over the course of it.

## Declaration on the Use of AI Assistance

I acknowledge the use of large language models to support the writing of this thesis, specifically for improving the clarity, structure, and grammatical correctness of the text, and as a coding assistant during the implementation of the experimental pipeline. The research design, the experiments, the analysis of the results, and the conclusions drawn from them are my own. All generated text and code were reviewed, verified, and revised by me, and I take full responsibility for the content of this thesis.

# Contents

|                                                                                                            |           |
|------------------------------------------------------------------------------------------------------------|-----------|
| <b>Acknowledgments</b>                                                                                     | <b>v</b>  |
| <b>List of Abbreviations</b>                                                                               | <b>xi</b> |
| <b>1. Introduction</b>                                                                                     | <b>1</b>  |
| 1.1. Motivation . . . . .                                                                                  | 1         |
| 1.2. Problem Statement . . . . .                                                                           | 1         |
| 1.3. Thesis Objective . . . . .                                                                            | 2         |
| 1.4. Research Questions . . . . .                                                                          | 2         |
| 1.5. Contributions . . . . .                                                                               | 3         |
| 1.6. Scope . . . . .                                                                                       | 4         |
| 1.7. Thesis Structure . . . . .                                                                            | 5         |
| <b>2. Related Work</b>                                                                                     | <b>6</b>  |
| 2.1. Preference-Based Alignment: RLHF, RLAIF, and Offline Optimization . .                                 | 6         |
| 2.2. Reliability Risks in AI Supervision and LLM-as-a-Judge . . . . .                                      | 7         |
| 2.3. Feedback Granularity and Credit Assignment in Long-Form Generation                                    | 8         |
| 2.4. Factual Consistency in Summarization and Evaluation . . . . .                                         | 10        |
| 2.5. Synthesis: The Gap This Thesis Addresses . . . . .                                                    | 12        |
| 2.6. Positioning of This Thesis . . . . .                                                                  | 12        |
| <b>3. Theoretical Background</b>                                                                           | <b>15</b> |
| 3.1. Preference Optimization: The Bradley–Terry Model . . . . .                                            | 15        |
| 3.2. Direct Preference Optimization (DPO) . . . . .                                                        | 16        |
| 3.3. AlignScore: Factual Consistency Scoring . . . . .                                                     | 16        |
| 3.4. Statistical Validation: Bootstrap Confidence Intervals and the Wilcoxon<br>Signed-Rank Test . . . . . | 17        |
| <b>4. Methodology</b>                                                                                      | <b>19</b> |
| 4.1. Research Questions and Hypotheses . . . . .                                                           | 19        |
| 4.2. Study Design . . . . .                                                                                | 20        |
| 4.3. Granular Credit Assignment (GCA) . . . . .                                                            | 21        |
| 4.4. Scope of the Evaluation . . . . .                                                                     | 22        |
| 4.5. Reward-Model Evaluation Protocol . . . . .                                                            | 23        |
| 4.6. Exploratory and Confirmatory Experiments . . . . .                                                    | 24        |
| 4.7. Relationship Between the Dataset Sizes . . . . .                                                      | 25        |

|                                                                                                |           |
|------------------------------------------------------------------------------------------------|-----------|
| 4.8. Independent Ground-Truth Construction . . . . .                                           | 26        |
| 4.9. Evolution of the Study Design . . . . .                                                   | 28        |
| 4.10. Validity Considerations . . . . .                                                        | 29        |
| <b>5. Implementation</b>                                                                       | <b>33</b> |
| 5.1. Pipeline Overview . . . . .                                                               | 33        |
| 5.2. Data . . . . .                                                                            | 33        |
| 5.3. Candidate Generation . . . . .                                                            | 36        |
| 5.4. Preference Construction . . . . .                                                         | 37        |
| 5.5. Reward-Model Training . . . . .                                                           | 39        |
| 5.6. Compute Infrastructure . . . . .                                                          | 41        |
| 5.7. Reproducibility . . . . .                                                                 | 42        |
| <b>6. Results</b>                                                                              | <b>43</b> |
| 6.1. Early Prototype Results (Historical Context) . . . . .                                    | 44        |
| 6.2. Bradley–Terry Reward-Model Baseline . . . . .                                             | 44        |
| 6.3. GCA Formula Optimization . . . . .                                                        | 45        |
| 6.4. Does the Formula Change Alone Flip the Result? . . . . .                                  | 46        |
| 6.5. AlignScore Judge-Mode Sweep . . . . .                                                     | 47        |
| 6.6. Repeated Runs of the Selected Configuration ( $n = 1,000$ ) . . . . .                     | 49        |
| 6.7. Extended Campaign: Twenty Independent Runs . . . . .                                      | 51        |
| 6.8. Scale Robustness: 5,000 and 10,000 Samples . . . . .                                      | 54        |
| 6.9. Effect Sizes, Equivalence, and Statistical Power . . . . .                                | 56        |
| 6.10. Independent Ground-Truth Precision Evaluation . . . . .                                  | 58        |
| 6.10.1. Results at $n = 1,000$ . . . . .                                                       | 59        |
| 6.10.2. Scale Dependence: $n = 5,000$ and $n = 10,000$ . . . . .                               | 61        |
| 6.10.3. Locating the Crossover: $n = 2,000$ and $n = 3,000$ . . . . .                          | 63        |
| 6.10.4. A First Candidate Tested and Rejected: Near-Tie Training Pairs . . . . .               | 65        |
| 6.10.5. How Often Holistic and GCA Agree . . . . .                                             | 66        |
| 6.10.6. A Second Candidate Tested and Rejected: A Length/Sentence-<br>Count Shortcut . . . . . | 67        |
| 6.10.7. What the RQ4 Investigation Establishes . . . . .                                       | 68        |
| 6.11. Summary of Results . . . . .                                                             | 69        |
| <b>7. Discussion</b>                                                                           | <b>71</b> |
| 7.1. What the Experiments Establish, and What They Do Not . . . . .                            | 71        |
| 7.2. Why Granularity Interacts With Evaluator Design . . . . .                                 | 72        |
| 7.3. How Much Weight the Statistical Evidence Can Carry . . . . .                              | 74        |
| 7.4. Both Reward Models Are Weak in Absolute Terms . . . . .                                   | 76        |
| 7.5. Interpreting the Larger-Scale Results . . . . .                                           | 77        |
| 7.6. The Ground-Truth Reversal: A Stronger and Stranger Pattern . . . . .                      | 79        |
| 7.7. Implications for RLAIIF Preference Construction . . . . .                                 | 81        |
| 7.8. Judge Reliability Without Human Auditing . . . . .                                        | 81        |
| 7.9. Stochastic Variation Between Runs . . . . .                                               | 82        |

|                                                  |            |
|--------------------------------------------------|------------|
| 7.10. Confounds That Remain . . . . .            | 83         |
| 7.11. Generalizability . . . . .                 | 84         |
| <b>8. Conclusion</b>                             | <b>85</b>  |
| 8.1. Summary . . . . .                           | 85         |
| 8.2. Answers to the Research Questions . . . . . | 86         |
| 8.3. Status of the Working Hypotheses . . . . .  | 88         |
| 8.4. Limitations . . . . .                       | 89         |
| 8.5. Future Work . . . . .                       | 92         |
| <b>A. Data and Code Availability</b>             | <b>100</b> |
| <b>B. Reproduction Configuration</b>             | <b>101</b> |
| B.1. Summarization Prompt . . . . .              | 101        |
| B.2. Pipeline Parameters . . . . .               | 101        |
| B.3. Command-Line Invocations . . . . .          | 103        |
| <b>C. Example Preference Record</b>              | <b>105</b> |



## List of Figures

|                                                                               |    |
|-------------------------------------------------------------------------------|----|
| 4.1. Worked example: holistic scoring vs. GCA aggregation . . . . .           | 22 |
| 4.2. Nesting of the three dataset subsets . . . . .                           | 26 |
| 5.1. The final experimental pipeline . . . . .                                | 34 |
| 6.1. Agreement across the nine aggregation strategies . . . . .               | 46 |
| 6.2. Evaluator-mode sweep . . . . .                                           | 48 |
| 6.3. Per-run GCA–Holistic accuracy gap at $n = 1,000$ . . . . .               | 50 |
| 6.4. Per-run GCA–Holistic accuracy gap, twenty-run campaign . . . . .         | 54 |
| 6.5. Accuracy at the three dataset sizes . . . . .                            | 56 |
| 6.6. Ground-truth precision at $n = 1,000$ , four constructions . . . . .     | 60 |
| 6.7. Absolute ground-truth success rate by training-set size . . . . .        | 63 |
| 6.8. GCA-minus-holistic ground-truth precision gap by training-set size . . . | 64 |
| 7.1. Holistic score vs. mean of per-sentence scores . . . . .                 | 73 |

# List of Tables

|                                                                                 |     |
|---------------------------------------------------------------------------------|-----|
| 2.1. Where sentence-level information enters the pipeline in related work . .   | 13  |
| 4.1. Final experimental configuration . . . . .                                 | 20  |
| 4.2. Overlap between the three dataset subsets . . . . .                        | 26  |
| 4.3. Decoding-temperature asymmetry in the preference labels . . . . .          | 30  |
| 4.4. Surface-feature baselines against the judge’s decisions . . . . .          | 31  |
| 4.5. Threats to validity . . . . .                                              | 32  |
| 5.1. Input and output of each pipeline stage . . . . .                          | 35  |
| 5.2. CNN/DailyMail subsets used across the project . . . . .                    | 36  |
| 6.1. Bradley–Terry reward-model baseline at $n = 1,000$ . . . . .               | 45  |
| 6.2. Agreement with the holistic decision across nine aggregation strategies .  | 45  |
| 6.3. Effect of the aggregation-formula change alone . . . . .                   | 47  |
| 6.4. AlignScore evaluator-mode sweep . . . . .                                  | 47  |
| 6.5. Per-run results for the selected configuration . . . . .                   | 49  |
| 6.6. Averaged result across six runs at $n = 1,000$ . . . . .                   | 50  |
| 6.7. Progression of the pooled estimate as runs were added . . . . .            | 51  |
| 6.8. Per-run results, extended twenty-run campaign . . . . .                    | 53  |
| 6.9. Run-level statistics, extended twenty-run campaign . . . . .               | 53  |
| 6.10. Repeated-run statistics at all three dataset sizes . . . . .              | 55  |
| 6.11. Selected configuration at three dataset sizes, repeated-run means . . . . | 55  |
| 6.12. Effect sizes, equivalence tests, and detectable effects . . . . .         | 57  |
| 6.13. The four ground-truth constructions . . . . .                             | 59  |
| 6.14. Ground-truth precision at $n = 1,000$ , four constructions . . . . .      | 60  |
| 6.15. Ground-truth success rate by condition and training-set size . . . . .    | 61  |
| 6.16. Ground-truth precision gap across training-set size . . . . .             | 62  |
| 6.17. Ground-truth precision gap, all five training-set sizes . . . . .         | 64  |
| 6.18. Near-tie training-pair fraction, holistic vs. GCA . . . . .               | 66  |
| 6.19. Holistic/GCA label agreement by scale and configuration . . . . .         | 66  |
| 6.20. Reward-model score correlation with summary length, by training-set size  | 67  |
| 6.21. Summary of findings against the research questions . . . . .              | 70  |
| 7.1. Truncation ablation, RoBERTa-base arms . . . . .                           | 77  |
| B.1. Pipeline parameters for the final $n = 1,000$ comparison . . . . .         | 102 |

# List of Abbreviations

| Abbreviation | Full Term                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------|
| AI           | Artificial Intelligence                                                                         |
| CI           | Confidence Interval                                                                             |
| CV           | Cross-Validation                                                                                |
| DPO          | Direct Preference Optimization                                                                  |
| GCA          | Granular Credit Assignment                                                                      |
| GPU          | Graphics Processing Unit                                                                        |
| LLM          | Large Language Model                                                                            |
| MDE          | Minimum Detectable Effect                                                                       |
| MoDPO        | Multi-Objective Direct Preference Optimization                                                  |
| NHR          | Nationales Hochleistungsrechnen (National High-Performance Computing)                           |
| NLI          | Natural Language Inference                                                                      |
| RLAIF        | Reinforcement Learning from AI Feedback                                                         |
| RLHF         | Reinforcement Learning from Human Feedback                                                      |
| RM           | Reward Model                                                                                    |
| ROUGE        | Recall-Oriented Understudy for Gisting Evaluation                                               |
| SESOI        | Smallest Effect Size of Interest                                                                |
| TOST         | Two One-Sided Tests (equivalence testing procedure)                                             |
| _sp          | Suffix marking the AlignScore evaluation modes that split the input internally (nli_sp, bin_sp) |

# 1. Introduction

## 1.1. Motivation

Large language models (LLMs) are increasingly used for multi-sentence generation tasks such as news summarization. In this setting, the perceived quality of an output is often dominated not by its overall fluency but by a small number of high-impact factual errors: a wrong entity, a flipped relation, or a fabricated claim can outweigh otherwise well-written text. Throughout this thesis, *reliability* refers specifically to factual consistency with the source document, not to stylistic polish or fluency.

Preference-based fine-tuning, most prominently reinforcement learning from human feedback (RLHF), updates a model using comparative judgments between candidate outputs rather than token-level supervision [4, 31, 38]. Collecting high-quality human preferences at scale is expensive and slow, so reinforcement learning from AI feedback (RLAIF) replaces human annotators with an automatic judge [1, 24]. This trades annotation cost for a new source of label noise and bias.

A further question arises for long, multi-sentence outputs such as news summaries, one that is largely independent of whether the feedback comes from a human or an AI judge: how granular should the preference signal be? A single holistic judgment (“summary A is better than summary B”) collapses all of the evidence in a multi-sentence output into one binary decision. If a summary is mostly accurate but contains one fabricated sentence, a holistic judge may still prefer it over an alternative that is uniformly mediocre but contains no outright fabrication. The holistic label cannot express that distinction. Sentence-level judging can in principle localize exactly which claims are, and are not, supported by the source, and that localized evidence can then be aggregated into a summary-level decision. This thesis asks whether that extra granularity actually produces a better training signal, and under what conditions.

## 1.2. Problem Statement

When constructing AI-generated preference data for training a factuality-oriented reward model, it is unclear whether coarse, holistic preferences are sufficient, or whether they systematically underuse the supervision signal available in long, multi-sentence outputs. The problem addressed here, under controlled conditions, is whether sentence-level

factuality judgments, aggregated into a summary-level preference through a mechanism referred to as *Granular Credit Assignment* (GCA), yield a more learnable preference signal for reward-model training than purely holistic judgments of the same candidate summaries.

**Learnability, as used throughout this thesis**

The pairwise validation accuracy of a Bradley–Terry reward model trained on the resulting preference pairs (Chapter 3, Section 3.1): given a held-out preference pair, how often does the trained reward model rank the summary the judge preferred above the alternative?

### 1.3. Thesis Objective

The objective of this thesis is to determine, through a controlled empirical comparison, whether Granular Credit Assignment produces a more learnable factuality-preference signal than holistic scoring. The two conditions differ only in how factuality scores are obtained and converted into a pairwise label:

**Holistic:** article + complete candidate summary → factuality score → pairwise preference.

**GCA:** article + individual sentences → sentence factuality scores → aggregation → pairwise preference.

Both preference sets are then used to train the same Bradley–Terry reward-model architecture under the same procedure. Holding the source articles, the candidate summaries and their generation settings, the evaluator family, the reward-model architecture, and the training procedure fixed allows feedback granularity to be studied as the primary experimental variable, rather than as one difference among several.

The thesis does not propose a new reward-model architecture, a new factuality metric, or a new training objective. Its subject is the construction of the preference data itself.

### 1.4. Research Questions

Four research questions follow from this objective. They are stated here in summary form; Chapter 4, Section 4.1 gives them together with the working hypotheses and the scope conditions under which each can be answered, and Chapter 8, Section 8.2 answers them against the evidence collected.

- **RQ1.** Does sentence-level factuality feedback aggregated via GCA produce a more learnable pairwise preference signal than holistic summary-level feedback, as measured by reward-model pairwise validation accuracy?

- **RQ2.** How do evaluator configuration and sentence-score aggregation strategy affect the relative learnability of holistic and GCA-derived preferences?
- **RQ3.** How stable are the observed differences across repeated reward-model training runs and dataset sizes?
- **RQ4.** Does the learnability advantage measured in RQ1 also hold when the trained reward models are evaluated against an independent, held-out ground truth, rather than against their own training-style labels, and does that hold as the training-set size grows?

RQ1 is the confirmatory question and is addressed by a repeated-run comparison. RQ2 is addressed by an exploratory sweep, and therefore supports statements about association between configuration and outcome rather than about causal mechanism. RQ3 motivates both the repeated runs at a single dataset size and the multi-scale check at  $n = 5,000$  and  $n = 10,000$ . RQ4 was added after that multi-scale check, to test the learnability finding against data the trained models never saw and never influenced their selection (Chapter 4, Section 4.8).

## 1.5. Contributions

This work makes the following contributions:

1. A controlled framework for comparing holistic and sentence-level, GCA-based construction of factuality preferences for abstractive summarization, in which the candidate pool, evaluator, reward-model architecture, and training procedure are held fixed (Chapter 4).
2. An empirical study of how evaluator mode and aggregation strategy interact with feedback granularity. Neither the aggregation formula nor the evaluator configuration is neutral: a penalty-weighted aggregation performs worse than a simple mean, and in an exploratory sweep the evaluator mode is associated with whether an advantage for GCA is observed at all (Chapter 6, Sections 6.3 and 6.5).
3. A repeated-run, multi-scale evaluation establishing a statistically significant GCA advantage at  $n = 1,000$  (20 independent runs, mean +3.95 percentage points, run-level Wilcoxon  $p < 0.001$ ; found during final review to have been trained under AlignScore’s `nli_sp` mode rather than the `nli` mode intended, Chapter 5, Section 5.4), and confirming, through twenty-run campaigns at  $n = 5,000$  and  $n = 10,000$  matching that same resolution, that this advantage is genuinely absent rather than merely unreproduced at larger scale: both confidence intervals are centred on zero and neither overlaps the  $n = 1,000$  result (Sections 6.6 and 6.8).

4. An exploratory observation that the benefit of external sentence-level decomposition may be reduced when the underlying evaluator already performs comparable internal decomposition, offered as a hypothesis motivated by the mode comparison rather than as an established mechanism (Chapter 7, Section 7.2).
5. An independent, held-out ground-truth evaluation (RQ4) showing that the  $n = 1,000$  advantage is not an artefact of validating against the trained models' own training-style labels: it holds, more sharply, against 500 articles disjoint from every training set used in this thesis, and it reverses rather than merely vanishing as the training set grows, becoming a significant advantage for holistic scoring at  $n = 10,000$  (Chapter 6, Section 6.10). Two candidate explanations for the reversal were tested directly and both rejected (Sections 6.10.4 and 6.10.6).
6. A transparent account of instability, stochastic variation across runs, methodological limitations, and the reproducibility of the pipeline (Chapters 5, 7, and 8).

These contributions concern the construction and learnability of preference data. The thesis does not claim that GCA yields objectively more accurate factuality labels, nor that it improves the factual consistency of generated summaries; the scope of what the experiments support is stated precisely in Chapter 4, Section 4.4.

## 1.6. Scope

- **Task:** single-document abstractive news summarization, using the CNN/DailyMail dataset [16, 30].
- **Candidate generation:** a single instruction-tuned base model, Mistral-7B-Instruct-v0.3, sampled at two temperatures ( $T = 0.7$  and  $T = 1.0$ ) to produce an A/B candidate pair per source article.
- **Judge:** a single fixed, deterministic factuality judge, `yzha/AlignScore` [41], intended throughout in its `nli` evaluation mode, though the confirmatory campaigns were, due to an implementation oversight found during final review, actually run under its `nli_sp` mode (Chapter 5, Section 5.4). No generative LLM-as-judge and no external API dependency is used in the final pipeline.
- **Optimization target:** the learnability of the resulting preference data for Bradley-Terry reward-model training, measured by pairwise validation accuracy under 5-fold cross-validation. A downstream fine-tuned generation policy is not evaluated in the final comparison; an early Direct Preference Optimization (DPO) prototype was explored but dropped from the final scope (Chapter 4).

## 1.7. Thesis Structure

Chapter 2 situates this work within the literature on preference-based alignment, LLM-as-a-judge reliability, fine-grained feedback for long-form generation, and automatic factual-consistency evaluation. Chapter 3 introduces the technical mechanisms this thesis builds on: the Bradley–Terry preference model, Direct Preference Optimization, the AlignScore factuality metric, and the statistical tools used to validate the results. Chapter 4 presents the experimental design, including the GCA aggregation formula itself, which is this thesis’s own contribution rather than prior work, and an explicit account of how the study that was actually carried out diverges from the original proposal, and why. Chapter 5 describes the pipeline, codebase, and compute infrastructure. Chapter 6 reports the full experimental campaign, from the initial reward-model baseline through the formula and judge-mode optimization to the final multi-seed and multi-scale validation, and then to the independent ground-truth precision evaluation that answers RQ4. Chapter 7 interprets these results, including why the ground-truth result reverses rather than merely fading where the learnability result only fades, and Chapter 8 answers the research questions, states the limitations of this work, and outlines directions for future work.



## 2. Related Work

This chapter reviews four bodies of work that this thesis draws on: preference-based alignment and how preference data becomes a training signal (Section 2.1); the reliability of AI judges when they replace human annotators (Section 2.2); the granularity of feedback in long-form generation, which is the variable this thesis manipulates (Section 2.3); and automatic factual-consistency evaluation, which supplies the judge (Section 2.4). Section 2.5 draws these together into the specific gap this thesis addresses, and Section 2.6 states what separates the present work from the studies closest to it.

### 2.1. Preference-Based Alignment: RLHF, RLAI, and Offline Optimization

Preference-based alignment collects pairwise comparisons between candidate model outputs and converts those comparisons into a training signal. The approach originates in preference learning for reinforcement learning, where human annotators compare short trajectory segments and a reward model is fitted to reproduce their comparisons, allowing an agent to be trained on objectives that are easy to judge but hard to specify numerically [4]. The choice to elicit comparisons rather than absolute scores is deliberate: relative judgments between two concrete outputs are more consistent between annotators than absolute ratings on an abstract scale, and the Bradley–Terry model (Chapter 3, Section 3.1) provides a principled way to recover a scalar reward from them.

Stiennon et al. [38] adapted this recipe to summarization, and their pipeline is the direct ancestor of the one used in this thesis. They collect human comparisons between candidate summaries, train a reward model on those comparisons, and then fine-tune a policy against the learned reward with reinforcement learning. Two of their findings matter here. First, the reward model’s own held-out agreement with human preferences is reported as a diagnostic in its own right, which establishes the precedent for treating reward-model accuracy as a meaningful measurement rather than merely an intermediate artifact, as this thesis does. Second, they observe that optimizing too hard against a learned reward degrades true quality, an early instance of the reward-overoptimization problem that motivates careful attention to the quality of the preference signal itself. RLHF was subsequently scaled to general instruction-following [31], establishing the reward-model-plus-policy-optimization pattern as the default alignment recipe.

Because human comparison data is expensive, reinforcement learning from AI feedback (RLAIF) substitutes a model for the human annotator. Constitutional AI generates critiques and revisions from a written set of principles and derives preference labels from them, removing human annotation from the harmlessness training loop almost entirely [1]. Lee et al. [24] compare RLAIF and RLHF directly on summarization and dialogue tasks and report broadly comparable downstream performance, which is the empirical result that makes AI-generated preference data a credible substitute rather than a compromise. They also emphasize that the outcome depends substantially on how the judge is prompted and calibrated, a dependency that recurs throughout Section 2.2 and, in a different form, in the evaluator-configuration effects observed here (Chapter 6, Section 6.5).

On the optimization side, RLHF pipelines have typically used on-policy algorithms such as Proximal Policy Optimization [35], which require sampling from the policy during training and maintaining a separate reward model. Direct Preference Optimization (DPO) removes both requirements by reparameterizing the RLHF objective so that the optimal policy can be recovered in closed form directly from preference pairs [33]. DPO was the objective originally proposed for this thesis, chosen because it isolates the effect of the preference data from the dynamics of reinforcement learning. As Chapter 4 describes, DPO was implemented and produced early results before being dropped from the final comparison in favor of evaluating Bradley–Terry reward models directly. That change narrows the comparison further than DPO alone would: it removes both the optimization dynamics that motivated adopting DPO and the decoding variance introduced by generating and scoring a fine-tuned policy’s outputs, leaving a more targeted question about whether the preference data itself is more learnable.

## 2.2. Reliability Risks in AI Supervision and LLM-as-a-Judge

Substituting a language model for a human annotator introduces a distinct class of error. Benchmark studies of LLM-as-a-judge document sensitivity to prompt wording, position bias favoring whichever candidate appears first, verbosity bias favoring longer answers regardless of quality, inconsistent scoring across repeated calls, and self-enhancement effects in which a judge prefers outputs from models similar to itself [43]. Liu et al. [27] document related calibration problems in LLM-based evaluation of generated text, and a recent survey synthesizes the accumulated evidence on these failure modes and the mitigations proposed for them, including rubric enforcement, pairwise rather than absolute evaluation, and position swapping [13].

These issues are more consequential in RLAIF than in evaluation. An unreliable evaluation metric produces a noisy estimate of a model’s quality; an unreliable judge used to construct training data produces systematically mislabeled supervision, and the model is then optimized to fit that mislabeling. Casper et al. [3] catalogue failure modes of this kind across the RLHF pipeline, including reward misspecification and the divergence

between training loss and intended behavior, and argue that problems in the preference data propagate into the trained policy in ways that are difficult to detect after the fact.

One response to this literature is to treat the judge as a fallible instrument and wrap it in a reliability protocol: confidence gating, order randomization, evidence-grounded rationale requirements, and prompt-robustness checks. The design adopted here removes the exposure at its source instead, by using a fixed, deterministic factuality model rather than a generative judge (Section 2.4). A deterministic model cannot exhibit prompt sensitivity, position bias, or run-to-run inconsistency, because it is never prompted and returns no natural-language output. This eliminates the category of risk the protocol above is designed to manage, at two costs: it does not make the evaluator correct, and it removes the rationale text that a human audit would examine. Chapter 4, Section 4.4 states what this implies for the claims the thesis can support.

### **2.3. Feedback Granularity and Credit Assignment in Long-Form Generation**

The variable this thesis manipulates is how localized the supervision signal is. A single preference over a multi-sentence output provides one bit of information about a text that may contain a dozen independently checkable claims, and it does not indicate which claim caused the preference. This is a credit assignment problem, and it is present even in the original preference-learning formulation: Christiano et al. [4] have annotators compare short trajectory segments rather than whole episodes precisely to make the resulting supervision denser and easier to attribute.

The closest prior work to this thesis is Wu et al. [40], who study fine-grained human feedback for long-form question answering. Their approach differs from holistic RLHF along two dimensions simultaneously. First, feedback is collected at the density of individual sentences or sub-sentence spans rather than whole responses, so an annotator marks the specific span that is irrelevant, untruthful, or incomplete. Second, they train several distinct reward models, one per error category, and combine their outputs into a dense reward signal during reinforcement learning, so the policy receives a reward at each segment rather than a single scalar at the end of the sequence. They report that this improves both the resulting policy and the reward models themselves relative to holistic feedback, and that the multiple reward models allow the relative weight of different objectives to be adjusted after training.

Three differences separate their setting from this thesis. Their fine-grained labels come from trained human annotators, whereas this thesis asks whether the same localization strategy is still useful when the localized judgments come from an automatic judge, which is cheaper by orders of magnitude but noisier per label. Their fine-grained signal is consumed as a dense per-segment reward during reinforcement learning, whereas GCA aggregates local scores back into a single summary-level preference pair so that the

resulting dataset remains compatible with standard pairwise preference optimization, requiring no change to the training objective. And they vary feedback density and objective decomposition together, whereas this thesis holds the objective fixed at factual consistency alone and varies only granularity, which isolates the granularity question at the cost of addressing a narrower one.

Three more recent lines of work place sentence- or step-level signals at other points in the alignment pipeline, and each should be distinguished carefully from the present study, since none leaves the question examined here open in general.

Gu et al. [14] introduce Mask-DPO, which incorporates sentence-level factuality signals directly into the DPO objective. Rather than treating a whole response as preferred or dispreferred, Mask-DPO uses sentence-level factuality information to learn only from the factually correct sentences within a preferred response, and to avoid penalizing factually correct content that appears inside a dispreferred one. The motivation is that response-level factuality supervision is noisy, because preferred and dispreferred responses each mix accurate and inaccurate sentences. The intervention point is the policy-optimization objective: the preference pair remains a response-level object, and the granular signal is applied as a mask during training. The present thesis intervenes earlier and leaves the training objective untouched. It asks how automatic factuality judgments at different granularities should be converted into the pairwise label in the first place, and measures the learnability of the resulting labels under a fixed Bradley–Terry architecture.

Qiu et al. [32] modify the reward model itself, proposing a sentence-level reward model that assigns scores at sentence boundaries and aggregates them into a response-level score, while still learning from response-level human preferences. Their aim is to address reward sparsity by producing an intermediate granularity between response-level and token-level reward modeling. The distinction from this thesis is architectural: they change what the reward model computes, and the preference labels they learn from are conventional response-level human judgments. This thesis does not modify the reward-model architecture at all. The Bradley–Terry model, its backbone, and its training procedure are held fixed across both conditions precisely so that differences in outcome can be attributed to the preference labels rather than to the model consuming them.

Closer still to GCA’s decompose-then-aggregate structure is process reward modeling for multi-step reasoning [25]. Rather than training a single model to judge a complete solution as correct or incorrect, Lightman et al. [25] have human annotators label the correctness of each individual reasoning step in a chain-of-thought solution, train a reward model to reproduce those per-step judgments, and show that this process-level supervision outperforms a response-level outcome reward for selecting correct solutions among many sampled candidates. The structural resemblance to GCA is real: both decompose a long output into smaller units, score each unit, and use the per-unit scores rather than a single holistic judgment. Two differences remain. Process reward models keep the per-step scores as the object consumed downstream, typically for reranking or search over sampled solutions, whereas GCA aggregates its per-sentence

scores back into one summary-level score so that the output is a conventional pairwise preference, compatible with unmodified Bradley–Terry training. And the per-step labels in Lightman et al. [25] come from trained human annotators judging mathematical correctness, a domain with a comparatively well-defined notion of a correct step, whereas GCA’s per-sentence labels come from an automatic factuality judge applied to open-ended news summaries, where what counts as a faithful sentence is judged by a model rather than verified against a solution. The result that step-level supervision outperforms outcome-level supervision in their setting is consistent with the motivation for GCA, but it is evidence from a different domain, a different label source, and a different point of consumption in the pipeline, not a direct precedent for the automatic, aggregated case this thesis studies.

Reward models are also increasingly evaluated in their own right rather than only as components of a larger pipeline. RewardBench assembles prompt-chosen-rejected trios across several domains and evaluates reward models by how often they rank the preferred response above the dispreferred one [23]. That pairwise-ranking accuracy is the same quantity used as the dependent variable in this thesis (Chapter 4, Section 4.4), which places the measurement on established ground; the difference is that RewardBench varies the reward model against fixed data, whereas this thesis varies the data construction against a fixed reward model.

## 2.4. Factual Consistency in Summarization and Evaluation

News summarization remains a standard benchmark domain, with CNN/DailyMail widely used for both training and evaluation [16, 30]. Abstractive systems have long produced fluent output that nonetheless contradicts or embellishes the source [36], and this persistent gap between fluency and faithfulness is what makes factual consistency a worthwhile optimization target rather than a solved preprocessing concern.

Automatic factuality evaluation has developed along two main lines. Entailment-based methods treat the source as a premise and the summary as a hypothesis: FactCC trains a classifier on synthetically corrupted summaries to detect factual conflicts [20]. Question-answering-based methods instead generate questions from the summary and check whether the source yields the same answers, an approach developed in FEQA [9] and refined in QAFactEval, which systematically compares components of the QA-based pipeline [11].

**Decomposition and aggregation in SummaC.** The work most directly relevant to GCA’s design is SummaC [21], which observes that natural language inference models trained on sentence-level data transfer poorly to document-level consistency checking, partly because the inputs are far longer than anything seen in training. Their solution is to decompose the problem: they compute an entailment matrix between each source

sentence and each summary sentence, then aggregate that matrix into a single score. Critically for this thesis, they show that *how* the matrix is aggregated changes performance substantially. Their zero-shot variant takes, for each summary sentence, the maximum entailment score across all source sentences, and then averages these maxima across summary sentences. Their learned variant instead trains a convolutional layer over the distribution of entailment scores per summary sentence, retaining information that the maximum discards, and performs better.

That aggregation strategy is a design decision with measurable consequences is therefore established in the evaluation literature before this thesis takes it up. The contribution here is to ask the same question one step upstream, at the point where scores are converted into training labels rather than into an evaluation metric, and Chapter 6, Section 6.3 finds the same lesson holds: the choice between a mean and a penalty-weighted aggregation changes the outcome more than the decision to decompose at all.

**AlignScore.** The judge used throughout this thesis, AlignScore, was not part of the literature surveyed in the original proposal. Zha et al. [41] recast a broad collection of natural language processing tasks, including natural language inference, question answering, paraphrase detection, and fact verification, into a single “information alignment” format, and train one model on the union. The resulting function takes a context and a claim and returns a scalar in  $[0, 1]$  estimating whether the context supports the claim. Because it is trained across many task formats rather than on entailment data alone, it generalizes across factuality benchmarks better than the single-task methods above.

AlignScore also inherits the decomposition question from SummaC, and resolves it internally: in its split evaluation modes, the context is divided into chunks and the claim into sentences, and the final score aggregates the resulting matrix by taking a maximum over chunks and then a mean over claim sentences, which is structurally the same rule as SummaC’s zero-shot variant. This internal behavior turns out to matter a great deal for the present work. Chapter 6, Section 6.5 finds that whether GCA outperforms holistic scoring depends on which AlignScore mode is used, and Chapter 7, Section 7.2 argues that this is because the split modes already perform, inside the judge, much of the decomposition that GCA performs outside it.

**Fine-grained evaluation.** More recent work evaluates summaries at sub-summary granularity rather than assigning a single score. FineSurE assesses faithfulness at the sentence level and content coverage against a list of extracted key facts, producing a diagnostic profile rather than one number [37]. Like SummaC, and unlike this thesis, it is positioned as an evaluation framework: it tells a practitioner where a summary went wrong, but does not convert that localized judgment into supervision for training.

FactScore decomposes a generation into individual atomic facts and checks each against a knowledge source, then reports the fraction supported as a single precision score [29]. Its decompose-score-aggregate structure is the closest evaluation-side analogue to GCA’s own aggregation step (Chapter 4, Section 4.3): both split a generation into small checkable units, score each independently, and combine the results into one number. The differences are the same ones that separate this thesis from SummaC and FineSurE. FactScore decomposes into atomic facts extracted from the generation itself and verifies them against an external knowledge source, whereas GCA decomposes into sentences and scores each against the source article with a single fixed judge; and FactScore’s aggregate score is reported as a diagnostic metric, not converted into a pairwise preference for training a reward model. The aggregation choice it makes, a simple fraction of atomic facts supported, is nonetheless a further data point for the same lesson this thesis draws in Section 6.3: how per-unit scores are combined is a design decision with measurable consequences, examined here independently by two unrelated methods on two different granularities of decomposition.

## 2.5. Synthesis: The Gap This Thesis Addresses

Table 2.1 summarizes the closest lines of work along the dimension that distinguishes the present study: the point in the pipeline at which sentence-level information is introduced, and what each approach holds fixed while doing so.

Table 2.1 organizes this literature by *where* sentence-level information is injected, which is the dimension along which the present study is distinguished. Sentence-level factuality signals are clearly not unexplored: they have been used to mask the policy-optimization objective [14], to restructure the reward model [32], to supply dense rewards during reinforcement learning [40], to train a reward model that outputs per-step judgments directly [25], and to build evaluation metrics [21, 29, 37]. What distinguishes this thesis is the insertion point. Each of the training-oriented works above takes the preference labels as given, in most cases as response-level human judgments, and modifies what happens downstream of them. The step examined here is the construction of those labels: how the output of an automatic factuality evaluator, queried at different granularities, is converted into a pairwise preference, and how learnable the resulting labels are when everything downstream is held constant. This is attractive because automatic labels are cheap, and uncertain because they are noisier per label than human ones, so whether the additional granularity recovers more signal than noise is not obvious in advance.

## 2.6. Positioning of This Thesis

The contribution of this thesis is not sentence-level reward modeling itself, nor sentence-level factuality alignment in general, both of which are active areas with established

Table 2.1.: Where sentence-level information enters the pipeline in related work.

| Work                 | Sentence-level signal enters at             | Held fixed                                       |
|----------------------|---------------------------------------------|--------------------------------------------------|
| Stiennon et al. [38] | not used                                    | –                                                |
| Lee et al. [24]      | not used                                    | –                                                |
| Wu et al. [40]       | reward signal during RL (human labels)      | response-level objective                         |
| Gu et al. [14]       | the DPO objective, as a sentence mask       | preference pair construction                     |
| Qiu et al. [32]      | the reward-model architecture               | preference labels (human, response-level)        |
| Lightman et al. [25] | the reward model’s own output (per-step)    | training objective; labels (human, per-step)     |
| Laban et al. [21]    | evaluation metric only                      | –                                                |
| Song et al. [37]     | evaluation metric only                      | –                                                |
| Min et al. [29]      | evaluation metric only                      | –                                                |
| Zha et al. [41]      | internal to the evaluator                   | –                                                |
| This thesis          | <b>construction of the preference label</b> | reward-model architecture and training procedure |

results. It is a controlled comparison of *preference-construction procedures*: the downstream Bradley–Terry architecture and training procedure are held fixed while the granularity at which automatic factuality evidence is converted into pairwise preference labels is varied. No new factuality scorer is introduced; AlignScore is used off the shelf, and the measurement instrument is the learnability of the resulting labels.

Six adjacent lines of work are distinguished as follows:

- **RLAIF and LLM-as-a-judge research** [1, 13, 24, 43] studies judge reliability at the level of a whole-response verdict, rather than the effect of querying the judge at a finer granularity.
- **Fine-grained human feedback** [40] establishes that localized supervision helps when labels are reliable; the present study asks the cheaper and less certain version, in which each local label is produced automatically.
- **Sentence-level factuality alignment** [14] applies granular factuality information inside the policy-optimization objective while leaving preference construction unchanged; here the objective is unchanged and preference construction varies.
- **Sentence-level reward modeling** [32] changes the reward model’s architecture while learning from conventional response-level preferences; here the architecture is fixed and the preferences vary.



- **Process reward modeling** [25] trains a reward model to output human-labeled, per-step correctness judgments directly, consumed downstream for reranking or search; here the reward model outputs a single scalar per summary, unchanged from the holistic condition, and what varies is only how that scalar’s training label was constructed upstream.
- **Fine-grained evaluation frameworks** [21, 29, 37] decompose summaries in order to diagnose them rather than to supervise with them.

The comparison is conducted on preference data generated by one evaluator on one corpus, so the findings describe behaviour under specific conditions rather than a general property of granular supervision (Chapter 7, Section 7.11).

## 3. Theoretical Background

### 3.1. Preference Optimization: The Bradley–Terry Model

The Bradley–Terry model [2] is a standard probabilistic model of pairwise comparison. It assumes that every item  $i$  has an underlying, unobserved scalar quality  $r_i$ , and that the probability of item  $i$  being preferred over item  $j$  in a pairwise comparison is

$$P(i \succ j) = \sigma(r_i - r_j), \quad \sigma(x) = \frac{1}{1 + e^{-x}}. \quad (3.1)$$

Given a dataset of observed pairwise preferences  $(w, l)$ , where  $w$  (the “winner”) was judged preferable to  $l$  (the “loser”), the model parameters are fit by maximizing the likelihood of the observed comparisons under Equation 3.1. Here the parameters belong to a neural network  $R_\theta$  that maps an (article, summary) pair to a scalar reward, and maximizing the likelihood is equivalent to minimizing the negative log-likelihood

$$\mathcal{L}(\theta) = -\mathbb{E}_{(w,l) \sim \mathcal{D}} \left[ \log \sigma(R_\theta(w) - R_\theta(l)) \right], \quad (3.2)$$

where  $\mathcal{D}$  is the preference dataset, holistic or GCA in this thesis. Equation 3.2 is exactly the loss implemented in `src/reward_model/train.py` (`bradley_tery_loss`), and it belongs to the same family of reward-modeling loss used inside RLHF pipelines more broadly [31, 38]. The difference here is that the reward model  $R_\theta$  is not an intermediate component on the way to fine-tuning a generation policy; it is the final object of comparison. The research question this thesis asks, whether GCA produces a more learnable preference signal than holistic scoring, is operationalized directly: does a Bradley–Terry reward model trained on GCA-derived preferences achieve higher held-out pairwise accuracy than one trained on holistic-derived preferences, under an otherwise identical training recipe?

This corresponds to the inverse reinforcement learning perspective: given a set of preference demonstrations, the task is to recover the latent reward function that best explains them. Comparing two reward models trained on two differently constructed preference datasets is then a controlled test of which construction procedure yields the more consistently recoverable latent signal, without requiring a downstream generation policy to be fine-tuned and re-evaluated (Section 3.2).

The architecture used throughout this thesis (`BradleyTerryRewardModel` in `src/reward_model/train.py`) encodes the concatenation of a truncated source article and a

candidate summary with a pretrained transformer encoder, mean-pools the encoder’s final hidden states over non-padding tokens, and maps the pooled representation to a scalar reward through a single linear layer. Reward-model accuracy is defined as the fraction of held-out pairs for which the model correctly ranks the chosen summary above the rejected one, i.e.  $R_\theta(w) > R_\theta(l)$  (`_evaluate` in `src/reward_model/train.py`); this is the metric reported throughout Chapter 6.

## 3.2. Direct Preference Optimization (DPO)

This section is included as background on the study’s earlier design rather than as part of its final methodology; DPO is not used in the main experiments (Chapter 4, Section 4.9).

Direct Preference Optimization (DPO) replaces RLHF’s reward-model-plus-reinforcement-learning pipeline with a single closed-form supervised objective on a policy  $\pi_\theta$ , defined relative to a fixed reference policy  $\pi_{\text{ref}}$  [33]:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x,w,l) \sim \mathcal{D}} \left[ \log \sigma \left( \beta \log \frac{\pi_\theta(w | x)}{\pi_{\text{ref}}(w | x)} - \beta \log \frac{\pi_\theta(l | x)}{\pi_{\text{ref}}(l | x)} \right) \right], \quad (3.3)$$

where  $x$  is the source article,  $w$  and  $l$  are the chosen and rejected summaries, and  $\beta$  bounds divergence from the reference policy. The relevant point for this thesis is structural: Equation 3.3 consumes exactly the same pairwise preference pairs used to train the reward models in Section 3.1, but converts them directly into a policy update, so the quality of the preference data and the behavior of the optimizer both influence the resulting summaries. Separating those two influences is what motivated measuring the preference data on its own (Section 4.9). Early DPO results obtained during the project are reported for context in Chapter 6, Section 6.1.

## 3.3. AlignScore: Factual Consistency Scoring

AlignScore is a unified alignment function trained across a mixture of natural language processing tasks recast into a single “information alignment” format [41]. Given a context  $c$ , the source article in this thesis, and a claim  $a$ , a candidate summary or a single sentence from one, AlignScore produces a scalar score in  $[0, 1]$  estimating the degree to which  $a$  is supported by  $c$ . This thesis uses AlignScore unmodified as the sole factuality judge for both preference-construction regimes (`src/judging/reward_model_judge.py`, class `RewardModelJudge`).

AlignScore exposes several evaluation modes, of which four are used in this thesis (the `alignscore-evaluation-mode` argument of `build_reward_preferences.py`):

- `nli_sp`: the context is first split into sentences before scoring, and the claim is scored against this sentence-split representation via AlignScore’s NLI (three-way entailment/neutral/contradiction) output head. This was the default mode inherited from the earliest AlignScore-based experiments in the project.
- `nli`: the same NLI output head, without the sentence-splitting preprocessing step.
- `bin_sp` / `bin`: the same two preprocessing variants, but using AlignScore’s binary (aligned / not aligned) output head instead of the three-way NLI head.

Chapter 6, Section 6.5 reports an empirical sweep across all four modes. The choice of mode is associated with substantial changes in the outcome, alongside the GCA aggregation formula (Chapter 4, Section 4.3). The `nli` mode produced the largest GCA advantage in that sweep and was selected as the intended final configuration for the validated  $n = 1,000$  campaign and the scale-robustness checks (Section 6.6), though Chapter 5, Section 5.4 reports that the confirmatory twenty-run campaigns were, due to an implementation oversight, actually trained under `nli_sp` instead. This should be read as a judge-*configuration* finding specific to this pipeline and this task, not as a general property of sentence-level aggregation independent of the underlying judge; Chapter 7 discusses what can and cannot be concluded from it.

### 3.4. Statistical Validation: Bootstrap Confidence Intervals and the Wilcoxon Signed-Rank Test

Two statistical tools are used throughout Chapter 6 to assess whether an observed GCA-versus-holistic accuracy gap is distinguishable from noise.

**Bootstrap confidence intervals.** The bootstrap [10] estimates the sampling distribution of a statistic by resampling the observed data. Given a set of paired fold-level accuracy differences  $\{d_1, \dots, d_m\}$ , where  $d_k = \text{acc}_{\text{GCA},k} - \text{acc}_{\text{Holistic},k}$  for cross-validation fold  $k$ , pooled across independent training seeds, a bootstrap 95% confidence interval for the mean difference  $\bar{d}$  is constructed by resampling  $\{d_1, \dots, d_m\}$  with replacement  $B$  times, computing the mean of each resample, and taking the 2.5th and 97.5th percentiles of the resulting distribution of means. This thesis uses  $B = 10,000$  resamples with a fixed NumPy random seed of 42.

**Wilcoxon signed-rank test.** As a complementary non-parametric test suited to a small number of paired, non-normally-distributed differences, the two-sided Wilcoxon signed-rank test [39] is reported on the same paired differences. Unlike a paired  $t$ -test it does not assume normality, which is appropriate given the small number of observations and the fact that fold accuracies are bounded proportions.

**An important qualification: the unit of replication.** Both procedures assume that the observations they are applied to are independent, and this determines which level

the tests may legitimately be applied at. Fold-level differences are *not* independent: the five folds within a run partition a single dataset and therefore share most of their training data, so applying either procedure to 30 such observations treats correlated measurements as independent replications, which understates uncertainty and produces intervals that are too narrow and  $p$ -values that are too small. Run-level differences, by contrast, come from separately seeded training runs and satisfy the assumption.

This thesis therefore reports two kinds of figure, and the distinction should be kept in view when reading Chapter 6. Fold-level intervals and  $p$ -values, which appear only for the early six-run campaign (Section 6.6), are anti-conservative and are presented as exploratory summaries of spread, never as hypothesis tests. All confirmatory statistics (the twenty-run campaigns at each dataset size, their bootstrap intervals, Wilcoxon tests, effect sizes, and equivalence tests) are computed over run-level differences, where the independence assumption holds. Chapter 7, Section 7.3 traces how the analysis moved from the former to the latter.

**Effect size.** A  $p$ -value indicates whether a difference is distinguishable from zero, not how large it is, so both are reported. For paired differences  $\{d_1, \dots, d_n\}$  the standardised effect size is Cohen's  $d = \bar{d}/s_d$ , the mean difference expressed in units of its own standard deviation [7]. Alongside it, the matched-pairs rank-biserial correlation is reported as the effect size native to the Wilcoxon test: it is the difference between the sums of positive and negative signed ranks, divided by their total, and therefore ranges from  $-1$  (every run favours one condition) to  $+1$  (every run favours the other), capturing consistency of direction independently of magnitude.

**Equivalence testing.** A non-significant result does not by itself establish that an effect is absent; it may equally reflect insufficient data. Establishing absence requires reversing the burden of proof, which is what the two one-sided tests (TOST) procedure does [22]. Given an equivalence bound  $\pm\delta$  representing the smallest effect that would be practically meaningful, TOST tests the two null hypotheses that the true effect is at least  $+\delta$  and at most  $-\delta$ . Rejecting both places the true effect inside  $(-\delta, +\delta)$  with the stated confidence, which is positive evidence of a negligible effect rather than mere failure to detect one. This procedure is applied in Chapter 6, Section 6.9 to the  $n = 5,000$  and  $n = 10,000$  campaigns, where the substantive claim is precisely that no advantage remains.

Pooling across runs rather than reporting a single run is nonetheless deliberate, and responds to an early observation (Section 6.4): a promising single-run result, a  $+1.4$  percentage-point advantage from one hyperparameter configuration, did not reproduce when the same configuration was run again. Single runs at this dataset scale are therefore not reliable evidence on their own, whatever statistics are computed from their folds.

## 4. Methodology

### 4.1. Research Questions and Hypotheses

- **RQ1.** Does sentence-level factuality feedback aggregated via GCA produce a more learnable pairwise preference signal than holistic summary-level feedback, as measured by reward-model pairwise validation accuracy?
- **RQ2.** How do evaluator configuration and sentence-score aggregation strategy affect the relative learnability of holistic and GCA-derived preferences?
- **RQ3.** How stable are the observed differences across repeated reward-model training runs and dataset sizes?
- **RQ4.** Does the learnability advantage measured in RQ1 also hold when the trained reward models are scored against an independent, held-out ground truth constructed from the same factuality judge, rather than against the models' own training-style labels, and does that hold as the training-set size grows?

RQ1–RQ3 are addressed by the experiments in Chapter 6 and answered in Chapter 8. RQ2 is addressed by an exploratory sweep rather than a controlled ablation, so it supports statements about association between configuration and outcome, not about causal mechanism (Section 4.6). RQ4 was added after the  $n = 1,000/5,000/10,000$  campaign, in direct response to a limitation that campaign left open: pairwise validation accuracy is measured against labels supplied by the same evaluator that constructed the preferences, so a reward model could in principle learn to recover those labels' idiosyncrasies rather than genuine factuality signal (Section 4.4). Section 4.8 describes the independent evaluation constructed to address this, and Section 6.10 reports its results.

Three working hypotheses guided the design. H1: sentence-level aggregated preferences are more learnable than holistic preferences, because they localize the factuality signal within a long output. H2: the evaluator's configuration affects whether any such difference is observable. H3: effects observed at one dataset scale need not persist at another. Their status is assessed in Chapter 8.

## 4.2. Study Design

The comparison is between two procedures for converting factuality scores into pairwise preference labels. For every source article, two candidate summaries are generated once and reused by both conditions:

**Holistic.** The complete candidate summary is scored against the source article, producing one score per candidate. The two scores are compared to yield a pairwise preference.

**GCA.** The candidate summary is segmented into sentences; each sentence is scored separately against the source article; the sentence scores are aggregated into one summary-level score. The two aggregated scores are compared to yield a pairwise preference.

The following are held identical across conditions: the source articles; the two candidate summaries and the decoding settings that produced them; the evaluator family and checkpoint; the reward-model architecture; and the reward-model training and evaluation procedure. The manipulated variable is the granularity at which the evaluator is queried, together with the aggregation step that GCA requires and holistic scoring does not. Table 4.1 lists the final configuration.

Table 4.1.: Final experimental configuration.

| Factor               | Value                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------|
| Task                 | Single-document abstractive summarization (news)                                                |
| Dataset              | CNN/DailyMail; runs at $n = 1,000, 5,000$ , and $10,000$ (subset seed 200)                      |
| Candidate generation | Mistral-7B-Instruct-v0.3, two-temperature sampling ( $T = 0.7 / T = 1.0$ )                      |
| Evaluator            | yzha/AlignScore, mode <code>nli</code> <sup>1</sup>                                             |
| Manipulated variable | Feedback granularity: holistic vs. sentence-level + GCA                                         |
| GCA aggregation      | Simple mean ( $\alpha = 0.0$ ), $n = 5,000 / n = 10,000$ only <sup>1</sup>                      |
| Margin               | 0 (all pairs used)                                                                              |
| Reward model         | Bradley-Terry, FacebookAI/roberta-base backbone, mean-pool + linear scalar head                 |
| RM training          | epochs = 5, lr = $2 \times 10^{-5}$ , batch size = 8, max length = 512, 5-fold cross-validation |
| Dependent variable   | Reward-model pairwise validation accuracy                                                       |

<sup>1</sup>This is the configuration selected in Chapter 6, Section 6.5 and intended as the locked design. The confirmatory campaigns that produced the results in Chapter 6 were, due to an implementation oversight found during final review, actually trained under evaluation mode `nli_sp` throughout, and, at  $n = 1,000$  specifically, under  $\alpha = 0.5$  rather than  $\alpha = 0.0$ . This discrepancy is reported in full in

### 4.3. Granular Credit Assignment (GCA)

Granular Credit Assignment (GCA) is the aggregation layer that converts per-sentence AlignScore scores into a single summary-level score, which is then compared against the corresponding score for the other candidate summary to construct a preference label. A candidate summary is first split into sentences with a regex-based sentence splitter (`src/judging/sentence_level.py`, `split_into_sentences`). Each sentence is then scored independently against the full source article using AlignScore (`RewardModelJudge.score_sentences`), yielding a list of per-sentence scores  $s = (s_1, \dots, s_n)$ . These per-sentence scores are aggregated into a single summary-level score via (`src/judging/gca.py`, `aggregate_sentence_scores`):

$$\text{gca}(s) = \bar{s} \cdot \left( \frac{\min(s)}{\bar{s}} \right)^\alpha, \quad \bar{s} = \frac{1}{n} \sum_{k=1}^n s_k, \quad (4.1)$$

where  $\alpha \geq 0$  is a penalty exponent. When  $\alpha = 0$ , Equation 4.1 reduces to the simple mean,  $\text{gca}(s) = \bar{s}$ ; larger  $\alpha$  increasingly penalizes summaries containing at least one low-scoring, “weakest-link” sentence, even when the average sentence score is otherwise high. This aggregation rule, GCA itself, is the contribution of this thesis rather than an established method it builds on, which is why it is defined here alongside the rest of the study design rather than in Chapter 3.

Two distinct configurations of Equation 4.1 were used at different points in the project. The original, proposal-era configuration used  $\alpha = 0.5$ , a moderate weakest-link penalty intended to make GCA sensitive to exactly the failure mode motivating this thesis: a summary that is mostly accurate but contains one fabricated or unsupported sentence. Empirically, this formula achieved only 82.2% agreement with the corresponding holistic decision on the same 1,000 candidate pairs, and when used to construct reward-model training data it caused the GCA reward model to *underperform* the holistic reward model (56.0% vs. 57.2% mean pairwise accuracy; Chapter 6, Section 6.2). A subsequent optimization campaign (Chapter 6, Section 6.3) tested nine alternative aggregation strategies and found that the simple mean,  $\alpha = 0.0$ , achieved 97.6% agreement with holistic decisions, the highest of all strategies tested, and, combined with the AlignScore `nli` mode (Chapter 3, Section 3.3), was intended as the configuration behind the repeated-run GCA advantage (Chapter 6, Section 6.6).  $\alpha = 0.0$  was accordingly locked as the final configuration; whether the `nli` mode was too is qualified in Chapter 5, Section 5.4 and Chapter 7, Section 7.2.

Figure 4.1 illustrates the difference between the two settings of  $\alpha$  on a real candidate pair drawn from the  $n = 200$  preference set. Summary A contains four well-supported sentences and one that the evaluator scores at 0.011; summary B is more uniformly mediocre. Under the simple mean the single weak sentence is diluted by the other four and A is preferred, matching the holistic decision. Under the penalty formula

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Chapter 5, Section 5.4, and its consequences are discussed in Chapter 7, Section 7.2.



the minimum term dominates and the decision reverses. This is the behaviour the penalty was designed to produce, and also the behaviour that made it noisier in practice (Chapter 6, Section 6.3).

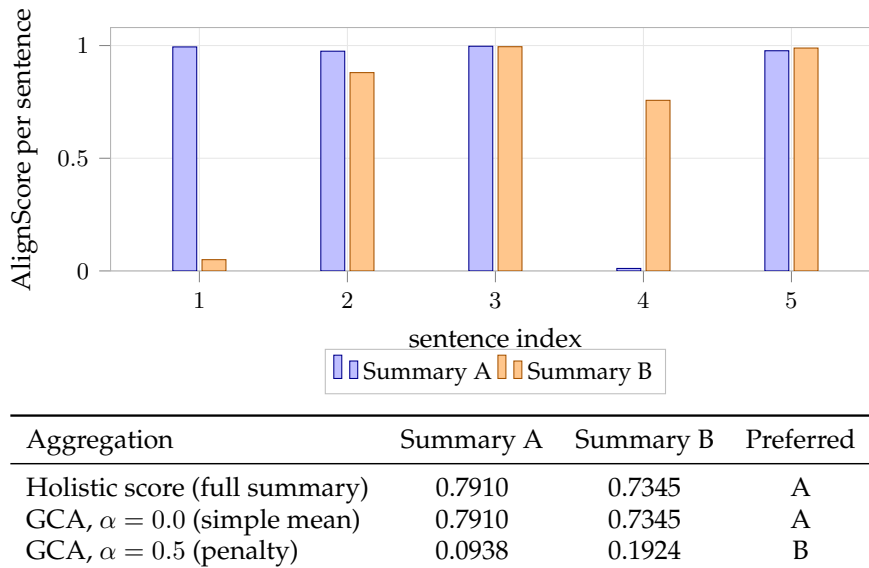


Figure 4.1.: Worked example on pair `cnn_00233` from the  $n = 200$  preference set (evaluator mode `nli_sp`). Summary A has one sentence scored 0.011 among four strong ones. The simple mean preserves the holistic ordering, while the  $\alpha = 0.5$  penalty inverts it. Values are computed from the stored per-sentence scores in `data/preferences/gca_reward_preferences_200.jsonl`.

**No cross-summary sentence alignment.** One design point is worth stating explicitly, since sentence-level methods in this area sometimes align the sentences of one candidate to those of another before comparing them. GCA as defined here performs no such cross-summary alignment. Each candidate summary’s sentences are scored independently against the source article, and the two aggregated scores are compared only afterwards. The method therefore contains no alignment step that could introduce matching errors, and correspondingly offers no opportunity to study alignment quality as a variable.

## 4.4. Scope of the Evaluation

The dependent variable throughout this thesis is the pairwise validation accuracy of a Bradley–Terry reward model trained on a given preference set. Within this thesis, *learnability* denotes exactly this quantity: the extent to which the preference relation

encoded in a dataset can be recovered by the chosen reward-model architecture under a fixed training procedure.

Because this term is easily over-read, it is worth stating precisely what a higher accuracy does and does not establish. A higher value indicates that the generated preference relation is more consistently recoverable by the selected reward-model architecture. It does not by itself establish any of the following:

- that the preference labels agree better with human factuality judgments;
- that the labels are objectively more accurate as factuality annotations;
- that a summarization model trained on those preferences produces more factually consistent summaries;
- that GCA is generally preferable to holistic scoring.

The first two are inaccessible here because no independent human annotation was collected, so the labels produced by AlignScore [41] function as the reference standard rather than as verified ground truth. Substituting an automatic judge for human annotators is the defining move of RLAIIF [1, 24], and it carries this limitation with it: the resulting labels are cheap and consistent, but unaudited. The third is inaccessible because the final study does not include a downstream policy-optimization stage (Section 4.9). The fourth is addressed empirically and answered in the negative: Chapter 6 reports configurations in which GCA does not lead. Claims made in later chapters are confined to learnability in the sense defined here.

## 4.5. Reward-Model Evaluation Protocol

Reward-model accuracy is evaluated by 5-fold cross-validation [19] rather than a single fixed held-out split. At the dataset sizes used here, a fixed split would leave either too little data for training or too little for a stable accuracy estimate, whereas cross-validation uses the full dataset for both and yields five accuracy readings per run.

Cross-validation alone does not account for variance introduced by the reward model’s initialization and by which pairs fall into which fold. This became apparent during the study: a first run of the final configuration produced a GCA advantage of +6.0 percentage points, while a rerun of the identical configuration *at the same seed value* produced +1.3 percentage points (Chapter 6, Section 6.5). That two same-seed runs diverge follows from an implementation detail documented in Chapter 5, Section 5.5: the `-seed` argument controls the cross-validation fold assignment, while PyTorch’s generator is left unseeded, so weight initialization, batch ordering and dropout all vary between executions. A seed value therefore does not pin down a run completely, and repeated runs at one seed value are not redundant.

The  $n = 1,000$  result was accordingly first pooled over *six training runs* spanning *five distinct seed values* (42, 42, 7, 100, 314, 2026; seed 42 appears twice, as the original run and its rerun), giving 30 fold-level accuracy readings per condition.

Two properties of this pooling constrain the statistical claims that can be made from it, and are revisited in Chapter 7, Section 7.3. Six runs across five seed values is a modest basis for a variance estimate. More consequentially, the 30 fold-level differences are not 30 independent observations: the five folds within a run partition a single dataset, so their training sets overlap heavily and their errors are correlated. Procedures assuming independent observations therefore overstate the effective sample size. Fold-level uncertainty estimates are reported in Chapter 6 as exploratory and anti-conservative, not as confirmatory tests.

Both constraints motivated the design actually used for the reported headline result. The *run* is treated as the unit of independent replication throughout, and the campaign was extended to *twenty independently seeded runs* (seeds 1–20, fixed before any was executed) under a corrected training procedure in which PyTorch and CUDA are seeded per fold, so that a run is fully determined by its seed (Chapter 5, Section 5.5). All confirmatory statistics reported in this thesis (the bootstrap interval, the Wilcoxon test, the effect sizes, and the equivalence tests of Section 6.9) are computed over these run-level differences, for which the independence assumption does hold. Fold-level figures are retained only as descriptive summaries of spread. Twenty runs is also the smallest campaign size that permits a two-sided Wilcoxon result below the conventional threshold with a realistic margin: with  $n$  paired same-signed observations the smallest attainable two-sided  $p$ -value is  $2/2^n$ , which at  $n = 6$  is 0.031 and leaves no room for a single discordant run, while at  $n = 20$  it is  $1.9 \times 10^{-6}$  (Chapter 6, Section 6.7).

The  $n = 5,000$  and  $n = 10,000$  configurations were initially each executed once rather than repeated, for reasons of computational budget, and at that stage carried correspondingly less weight than the pooled  $n = 1,000$  result. Both were subsequently extended to twenty-run campaigns under the same corrected, deterministic training procedure used for the  $n = 1,000$  extension, matching its resolution exactly (Chapter 6, Section 6.8), so all three dataset sizes are now supported by equally powered, repeated-run evidence rather than only  $n = 1,000$ .

## 4.6. Exploratory and Confirmatory Experiments

The experiments in Chapter 6 do not all have the same evidential status, and the distinction is material to how the results should be read.

The aggregation-formula comparison (Section 6.3), the evaluator-mode sweep (Section 6.5), and the hyperparameter search (Section 6.4) were all carried out on the  $n = 1,000$  setting, and the final configuration was selected after observing their outcomes. These experiments are exploratory. The configuration they produced was not specified

in advance, and reporting it as though it had been pre-registered would overstate the evidence: with enough configurations examined, some separation between conditions is expected by chance alone.

The repeated runs at  $n = 1,000$  (Section 6.6) test whether the selected configuration produces a stable effect under repetition, but they reuse the dataset on which that configuration was chosen and therefore cannot establish that it generalizes beyond that setting.

The  $n = 5,000$  and  $n = 10,000$  experiments provide a weaker check than would be ideal, for a reason that must be stated explicitly. All three subsets are drawn from the same CNN/DailyMail test split by the same deterministic procedure at the same seed, differing only in the number of samples requested, and they are consequently nested rather than disjoint (Section 4.7). The larger-scale runs therefore re-use almost all of the data on which the configuration was selected, and are best understood as measurements of how the effect behaves as the dataset grows, not as out-of-sample replication on independent data. No experiment in this thesis evaluates the selected configuration on data disjoint from the set used to choose it.

## 4.7. Relationship Between the Dataset Sizes

The three dataset sizes used in the final experiments are not independent samples, and this constrains how the larger-scale runs can be interpreted.

All three subsets are produced by the same deterministic procedure (`src/data/subset.py`, `select_subset`) from the same CNN/DailyMail test split, under the same length filters and the same subset seed (200). Only the number of requested samples differs. Because the selection draws from one eligible pool of 10,896 articles with a fixed random seed, the resulting subsets overlap heavily. Reproducing the selection with the recorded parameters gives the intersections in Table 4.2, shown schematically in Figure 4.2.<sup>2</sup>

The  $n = 5,000$  subset is contained entirely within the  $n = 10,000$  subset, and the  $n = 1,000$  subset is almost entirely contained within both. The  $n = 10,000$  set additionally covers 91.8% of the whole eligible pool, so at that size little of the test split remains unused in any case.

Two consequences follow, and both are carried through the rest of the thesis. The larger-scale experiments do not constitute out-of-sample replication: they re-use almost all

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<sup>2</sup>Computed by re-running `select_subset` with the parameters recorded in `configs/subset_1000.yaml` and `slurm/generate_candidates_scale.sh` (`seed = 200`, `test split`, `article limit 8,000` characters, `summary limit 2,000` characters) against the local copy of the test split in `data/cnn_dailymail_test_full`, and comparing the resulting SHA-256-derived sample IDs. The candidate and preference files for these runs are not retained in the repository, so the comparison reproduces the selection rather than reading the stored artifacts; the selection is deterministic given these parameters.

Table 4.2.: Overlap between the subsets used in the final experiments. All three use subset seed 200 and differ only in sample count.

| Pair                         | Shared sample IDs | Share of the smaller set |
|------------------------------|-------------------|--------------------------|
| $n = 1,000$ and $n = 5,000$  | 980               | 98.0%                    |
| $n = 1,000$ and $n = 10,000$ | 999               | 99.9%                    |
| $n = 5,000$ and $n = 10,000$ | 5,000             | 100%                     |

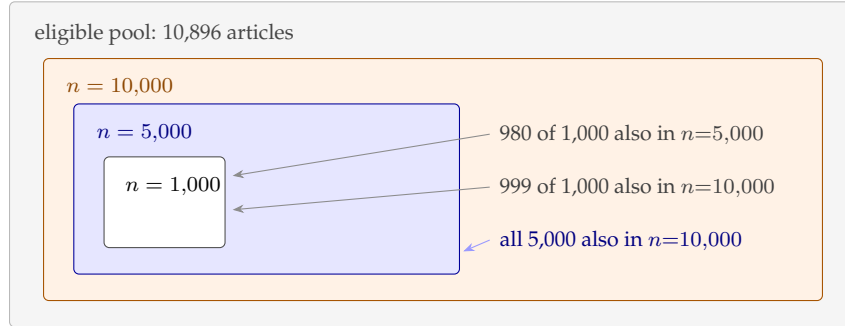


Figure 4.2.: The three subsets are nested rather than independent. Box sizes are schematic; the counts are the measured intersections reported in Table 4.2.

of the data on which the configuration was selected, and therefore measure how the effect behaves as further data is added to the same pool rather than whether it transfers to new data. And because the samples are nested, the three results are not statistically independent of one another, so the differences between them should not be read as three separate estimates of the same quantity. Drawing the larger subsets with different seeds would have made them independent, at the cost of the nesting property the generation script was written to provide; this is recorded as a limitation in Chapter 8.

## 4.8. Independent Ground-Truth Construction

The evaluation described above has two properties that limit what it can establish, both noted in Section 4.6 and Section 8.4: the labels a trained reward model is scored against are supplied by the same evaluator that constructed its training preferences, and the  $n = 5,000/n = 10,000$  checks reuse almost all of the data the configuration was selected on rather than evaluating it on disjoint data. RQ4 was formulated to address both at once, by scoring the already-trained reward models against a held-out set built specifically to be disjoint from every training pool used in this thesis, under an evaluation criterion that does not depend on how any individual training pair happened to be labelled.

**Held-out article selection.** 500 articles were drawn from the same CNN/DailyMail test split, using `select_disjoint_subset (src/data/subset.py)`: the function reconstructs the exact `rng.sample()` call that produced the  $n = 10,000$ , seed-200 draw used throughout this thesis, removes those indices from the eligible pool, and draws 500 samples from what remains under a new seed (999). Because the  $n = 5,000$  subset is fully contained in the  $n = 10,000$  draw and the  $n = 1,000$  subset is 99.9% contained in it (Table 4.2), excluding the  $n = 10,000$  draw is sufficient to make the held-out set disjoint from all three training pools used in the main campaign; two further training pools introduced later in this work, at  $n = 2,000$  and  $n = 3,000$  (Section 6.10.3), were verified by direct index comparison to be exact subsets of the  $n = 10,000$  draw and are therefore disjoint from the held-out set for the same reason.

**Candidate generation and scoring.** Two candidate summaries were generated per held-out article with the same Mistral-7B-Instruct-v0.3 pipeline and the same two decoding temperatures used throughout this thesis (Section 4.2), giving 1,000 candidate summaries in total. Every candidate was scored with AlignScore under the locked configuration used for the main campaign: sentence-level scoring, GCA aggregation with  $\alpha = 0.0$  (Section 6.3). This GCA-aggregated score is used only to construct the evaluation criterion below; it is never used as a training signal for the held-out set, and no reward model is trained on these 500 articles.

**Evaluation criterion.** Two ground-truth constructions were used. The first pairs each article’s own two candidates directly, mirroring the structure of the training preferences; the two candidates are generated by the same model at similar temperatures, so their quality gap is typically small (mean GCA score gap approximately 0.04). The second pools all 1,000 scored candidates across all 500 articles, sorts them by GCA score, and compares the top  $X\%$  against the bottom  $X\%$  for  $X \in \{25, 10, 5\}$ , which produces progressively larger, more unambiguous quality gaps between the two sides at the cost of fewer summaries per comparison (Section 6.10). At  $X = 5$ , every high-scoring summary is compared against every low-scoring summary (2,500 pairs from 50 summaries per side) rather than one random pairing, to remove pairing-order noise from the smallest, highest-signal condition. A trained reward model’s *success rate* on a given construction is the fraction of comparisons in which it scores the higher-GCA-score summary above the lower-GCA-score one; this is the dependent variable throughout Section 6.10, and it is computed identically for the holistic and GCA reward models, using the same fixed ranking for both.

That last point requires a sharper qualification than an earlier version of this section gave it. The ranking itself, the ground-truth criterion both models are scored against, is constructed using GCA’s own aggregation rule (sentence-level AlignScore, mean-pooled at  $\alpha = 0.0$ ), the same rule the GCA reward model is trained to predict. The holistic reward model is trained to predict a different rule (whole-summary AlignScore) and is

then scored against a criterion built from the rule it never saw. The *data* are genuinely held out for both conditions, no reward model trains on these 500 articles, but the *criterion* is in-distribution for GCA and out-of-distribution for holistic in a way that plain data held-out-ness does not fix. This does not mean the RQ4 result is void: a higher success rate for GCA under a GCA-defined ranking is still evidence that GCA’s aggregated score is a more decisive signal by its own standard, and the two same-model comparisons (GCA vs. its own criterion, holistic vs. the same criterion) are still informative relative to each other. But it is not, contrary to what this section previously claimed, a symmetric test that treats the two conditions identically, and the possibility that part of GCA’s RQ4 advantage reflects this asymmetry rather than a general precision advantage cannot be ruled out with the data collected here. Scoring the held-out pool under a holistic ranking as well, and reporting GCA’s and holistic’s success against both criteria, would settle this and is recorded as future work (Chapter 8, Section 8.5); it was not attempted here because it requires rescoring 1,000 held-out summaries with AlignScore under a second configuration, which is new computation this thesis does not undertake at this stage.

**Checkpoint retention.** The training procedure used for RQ1–RQ3 (Section 4.5) discards each fold’s trained weights after recording its accuracy, since only the accuracy is needed there. RQ4 requires scoring the same trained model against a second, independent criterion, so the training procedure was changed for this evaluation to save the full reward-model checkpoint (encoder weights and reward head) for every run, rather than only its cross-validation accuracy.

## 4.9. Evolution of the Study Design

The study reported here is narrower in scope than the one outlined in the original research plan, and the reasons are methodological.

The initial design used the generated preferences directly for DPO-based policy optimization, evaluating factual consistency on the fine-tuned model’s own generations. Early downstream experiments made it difficult to attribute observed differences to any single stage: a change in generated-summary factuality could originate in the preference-construction procedure, in the quality of the reward signal, or in the policy-optimization step, and the design provided no way to separate these. Increasing the number of stages also increased the number of variance sources standing between the manipulated variable and the measured outcome.

The final study therefore isolates an earlier and more controlled diagnostic question: whether the preference data produced by each construction procedure is recoverable by a fixed reward-model architecture. Removing the policy-optimization stage removes both reinforcement-learning dynamics and decoding variance from the measurement

path, leaving a comparison in which granularity is the principal difference between conditions.

The claims of this thesis are correspondingly narrower. The work evaluates the learnability of the preference signal rather than demonstrating improvements in generated-summary factuality (Section 4.4). Two further consequences follow from the same change. Margin-based filtering of near-tie pairs was removed, since AlignScore produces continuous scores for which exact ties are effectively impossible, and the threshold discarded roughly one fifth of the data without a demonstrated benefit. The reliability protocol designed for a generative LLM judge, comprising confidence gating, order randomization, and rationale requirements, no longer applies once the evaluator is a deterministic model that is never prompted and returns no rationale; a planned human audit of judge rationales was not carried out and is recorded as a limitation in Chapter 8.

## 4.10. Validity Considerations

**Ethical and practical considerations.** All experiments use CNN/DailyMail [16, 30], a publicly available corpus of news articles and human-written summaries; no personally sensitive data is collected. Because both the evaluator’s judgments and the generated summaries can be incorrect, results are reported as measurements of preference-data learnability rather than as statements about factual correctness, and model outputs are not presented as verified fact. Compute use was bounded by working with fixed-size subsets.

**Candidate generation and decoding temperature.** The two candidates in each pair are produced by the same model at different decoding temperatures ( $T = 0.7$  and  $T = 1.0$ ). This introduces a potential confound: preference structure may correlate with systematic stylistic differences induced by temperature rather than with factuality alone. The preference data stored in the repository allow the asymmetry to be quantified directly, and Table 4.3 reports it.<sup>3</sup>

The asymmetry is substantial under both conditions and consistently larger under holistic scoring, and it is not an artifact of the margin threshold. Two implications follow. Features correlated with decoding temperature, such as summary length or lexical conservativeness, may carry a non-trivial share of the label information, so a reward

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<sup>3</sup>Computed from `data/preferences/holistic_reward_preferences_200.jsonl` and `data/preferences/gca_reward_preferences_200.jsonl`. The `margin-0` rows recompute the decision from the stored scores at the threshold used in the final configuration; the `margin-0.05` rows reproduce the stored decisions, whose aggregate counts also appear in `reports/reward_model_judging_summary.csv`. Preference files for the final  $n = 1,000$  configuration are not retained in the repository, so these proportions could not be recomputed under the final evaluator mode.



Table 4.3.: Proportion of pairs in which the lower-temperature candidate ( $T = 0.7$ ) is preferred, on the  $n = 200$  preference set (evaluator mode `nli_sp`,  $\alpha = 0.5$ ).

| Condition | Threshold     | $T=0.7$ preferred | $T=1.0$ preferred | Share |
|-----------|---------------|-------------------|-------------------|-------|
| Holistic  | margin = 0    | 137               | 63                | 68.5% |
| GCA       | margin = 0    | 126               | 74                | 63.0% |
| Holistic  | margin = 0.05 | 109               | 45                | 70.8% |
| GCA       | margin = 0.05 | 97                | 60                | 61.8% |

model could recover part of the preference relation without representing factuality as such. And because the asymmetry differs between conditions by roughly five percentage points, the two preference sets are not equivalent in this respect either, which means the comparison between them is not perfectly clean with respect to temperature.

These figures come from the  $n = 200$  set scored under the earlier evaluator configuration and need not hold exactly at the final one. They are reported as evidence that the confound is real and measurable, not as a quantification of its effect on the reported results. Controlling candidate generation more explicitly, by sampling both candidates at one temperature, would remove the confound at source and is left to future work (Chapter 8).

**How much label information do surface features actually carry?** The concern above is only as serious as the amount of label information that temperature-correlated surface features actually carry, and that is measurable directly on the same stored data. Table 4.4 reports the accuracy of the cheapest possible classifiers, ones that ignore the article entirely and decide from a single surface property of the two candidate summaries, against the same decisions the evaluator made.<sup>4</sup>

None of these predictors performs meaningfully above chance. The mean length of the chosen summary differs from that of the rejected summary by 2 characters under holistic scoring and by  $-6$  characters under GCA, on summaries averaging roughly 850 characters, and sentence counts differ by 0.01 and  $-0.28$  respectively. Length and sentence count, the two most obvious surface proxies for decoding temperature, therefore do not carry the preference signal, and a reward model cannot be recovering its accuracy from them.

This narrows the confound without eliminating it, and the distinction matters. The 68.5% and 63.0% asymmetries in Table 4.3 are not themselves a shortcut available to the reward model: the model is trained on (article, chosen) and (article, rejected) text pairs and never observes which candidate came from which temperature, or which position it

<sup>4</sup>Computed by `analysis/surface_feature_baselines.py` from the same two preference files as Table 4.3, restricted to the pairs carrying an A/B decision (154 holistic, 157 GCA).

Table 4.4.: Accuracy of surface-feature classifiers against the evaluator’s own decisions,  $n = 200$  preference set (mode `nli_sp`,  $\alpha = 0.5$ ). Each row states the accuracy of the better of the two opposing rules (prefer longer/shorter, prefer more/fewer sentences). Mean lengths are in characters.

| Predictor                  | Holistic | GCA   |
|----------------------------|----------|-------|
| Prefer the longer summary  | 0.494    | 0.465 |
| Prefer the shorter summary | 0.506    | 0.535 |
| Prefer more sentences      | 0.519    | 0.475 |
| Prefer fewer sentences     | 0.481    | 0.525 |
| Mean length, chosen        | 865.6    | 840.5 |
| Mean length, rejected      | 863.5    | 846.2 |

occupied (Chapter 5, Section 5.5). For the asymmetry to inflate reward-model accuracy, some *textual* property correlated with temperature would have to be learnable from the summary itself. Table 4.4 rules out the two most likely candidates for such a property. Subtler lexical or stylistic correlates are not ruled out, and testing them would require the feature-level analysis recorded as future work, but the simple version of the objection does not hold.

**Threats to validity.** Table 4.5 summarizes the threats considered and how each is handled.

Table 4.5.: Threats to validity and how each is addressed.

| Threat                                     | Status                                                                                                                                                                                                |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single evaluator (AlignScore)              | Not mitigated; all results are conditional on this evaluator’s judgments, which are not independent ground truth.                                                                                     |
| Evaluator and aggregation sensitivity      | Measured directly via the formula comparison and evaluator-mode sweep (Chapter 6).                                                                                                                    |
| Fold- and run-level noise                  | Addressed by twenty independently seeded runs at each of the three dataset sizes, under fully deterministic per-fold seeding (Section 4.5).                                                           |
| Non-independence of pooled folds           | Mitigated by testing at the run level, where the independence assumption holds; fold-level figures are reported only as descriptive summaries of spread (Section 4.5).                                |
| Configuration selected on observed results | Acknowledged, not remedied; the larger-scale subsets are nested supersets of the selection data (Section 4.7), so no run evaluates the configuration on independent data.                             |
| Scale sensitivity                          | Examined explicitly rather than assumed away (Section 6.8).                                                                                                                                           |
| Decoding-temperature confound              | Quantified where data permit; not controlled for in the design. Length and sentence count are excluded as carriers of the signal (Table 4.4); subtler lexical correlates remain untested (see above). |
| Source-document truncation                 | Documented in Chapter 5, Section 5.5; discussed as a limitation.                                                                                                                                      |
| Weak absolute accuracy                     | Both conditions remain near chance (Chapter 7, Section 7.4).                                                                                                                                          |

## 5. Implementation

### 5.1. Pipeline Overview

Figure 5.1 shows the final pipeline. It differs from the pipeline described in the original proposal in exactly the ways described in Chapter 4: there is no DPO fine-tuning stage, no MoDPO [44] branch, and no cross-summary sentence-alignment step. The pipeline has four stages. First, a fixed number of CNN/DailyMail articles are selected deterministically (Section 5.2). Second, two candidate summaries are generated per article at different sampling temperatures (Section 5.3). Third, the same candidate pool is scored twice, once holistically and once at the sentence level with GCA aggregation, to produce two separate preference datasets (Section 5.4). Fourth, a Bradley–Terry reward model is trained independently on each preference dataset and the two are compared (Section 5.5).

Table 5.1 states the input and output of every stage in Figure 5.1 explicitly. Stages 4A/5A form the holistic branch, stages 4B/5B/6B/7B form the GCA branch, and stage 9 is the only point at which the two branches are compared against each other.

### 5.2. Data

All experiments use CNN/DailyMail [16], version 3.0.0 as distributed through the Hugging Face `datasets` library. Subset selection (`src/data/subset.py`, function `select_subset`) filters the chosen split to articles under 8,000 characters and reference summaries under 2,000 characters, then draws a fixed-size sample from the eligible pool using Python’s `random.Random` seeded deterministically. Each sample is assigned an ID of the form `cnn_{index}_{hash}`, where `hash` is the first 12 hexadecimal characters of the SHA-256 digest of the article text (`make_sample_id`). This ties every sample’s identifier to its content rather than to its position in the dataset, so the same article always receives the same ID regardless of which subset size or seed selected it.

Table 5.2 lists every subset used across the project. The  $n = 1,000$ ,  $n = 5,000$ , and  $n = 10,000$  subsets all use subset seed 200 and the same filtering parameters, differing only in the number of samples requested. They are therefore *nested rather than disjoint*: the smaller sets are almost entirely contained in the larger ones. Chapter 4, Section 4.7

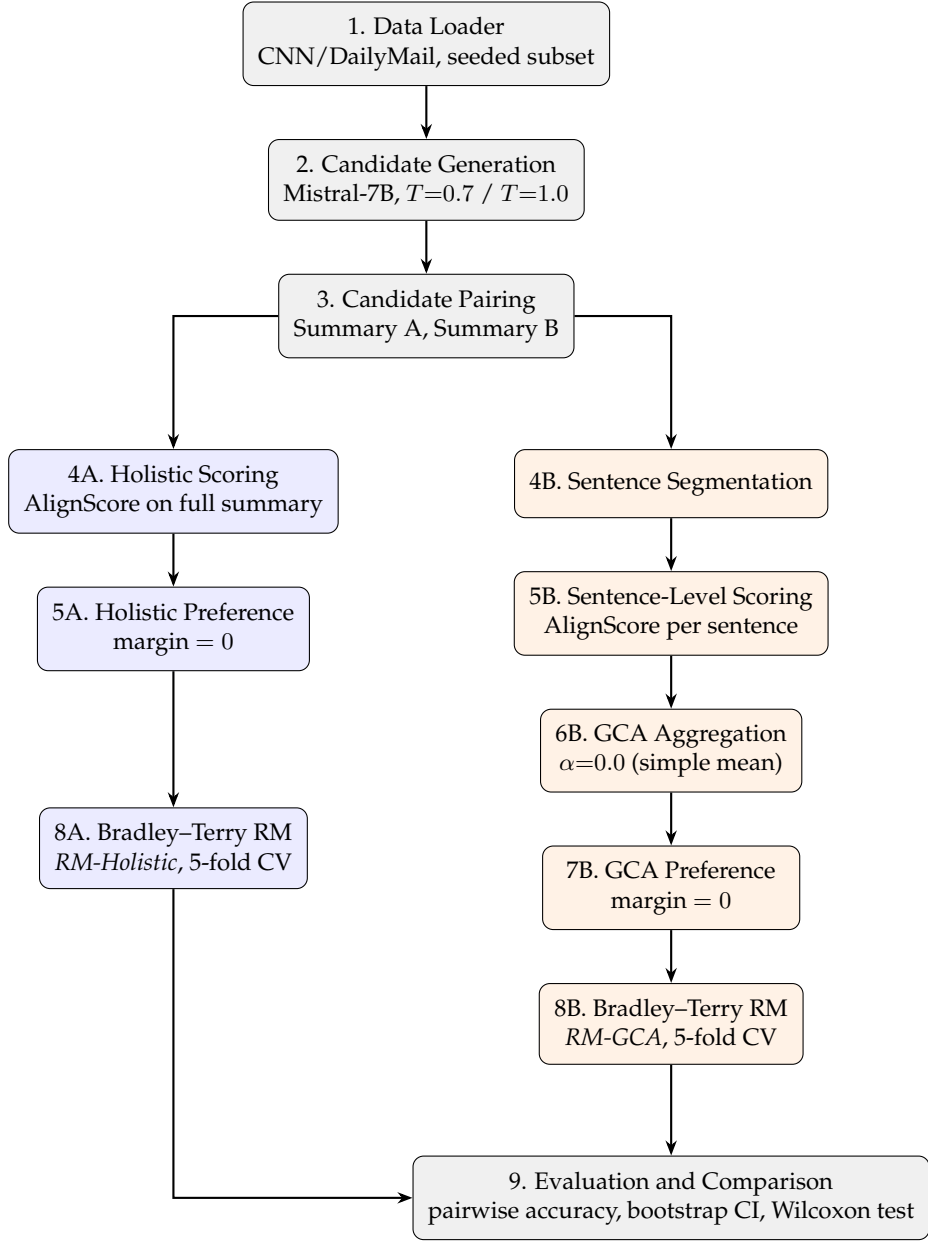


Figure 5.1.: The final pipeline. Shared setup (grey) produces one candidate pool from which two independent preference-construction branches run: holistic (blue) and GCA (orange). Each branch trains its own Bradley-Terry reward model under an identical recipe, and the two are compared in the final evaluation step. Compare to the pipeline described in the original proposal (Chapter 4), which additionally included a DPO fine-tuning stage and a cross-summary sentence-alignment step, neither of which appears here.

Table 5.1.: Input and output of each stage of the final pipeline shown in Figure 5.1.

| Stage                      | Input                                 | Output                                               |
|----------------------------|---------------------------------------|------------------------------------------------------|
| 1. Data Loader             | CNN/DailyMail dataset                 | $N$ seeded article samples with content-derived IDs  |
| 2. Candidate Generation    | Article text                          | 2 candidate summaries per article ( $T=0.7, T=1.0$ ) |
| 3. Candidate Pairing       | 2 candidate summaries                 | Summary A / Summary B pair                           |
| 4A. Holistic Scoring       | Article + full summary                | 1 AlignScore factuality score per summary            |
| 4B. Sentence Segmentation  | Full summary                          | List of sentences                                    |
| 5A. Holistic Preference    | Score(A), Score(B)                    | chosen/rejected label (or no_preference)             |
| 5B. Sentence-Level Scoring | Article + each sentence               | 1 AlignScore score per sentence                      |
| 6B. GCA Aggregation        | All sentence scores for one summary   | 1 aggregated score per summary ( $\alpha=0.0$ )      |
| 7B. GCA Preference         | Aggregated score(A), score(B)         | chosen/rejected label (or no_preference)             |
| 8A. Train RM-Holistic      | Holistic preference pairs             | Trained Bradley-Terry RM, 5-fold CV accuracy         |
| 8B. Train RM-GCA           | GCA preference pairs                  | Trained Bradley-Terry RM, 5-fold CV accuracy         |
| 9. Evaluation & Comparison | Both conditions' accuracies, 20 seeds | Mean gap, bootstrap 95% CI, Wilcoxon $p$ -value      |

quantifies the overlap and states its methodological consequence. Earlier subsets at seeds 42 and 100 were drawn with different seeds and are not part of the final comparison.

Table 5.2.: CNN/DailyMail subsets used across the project.

| $n$    | Subset seed | Purpose                             | Notes                                                                                                   |
|--------|-------------|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| 200    | 42          | Early DPO-based prototype           | Superseded; see Chapter 6, Section 6.1                                                                  |
| 495    | 100         | First Bradley–Terry RM training set | Drawn with a different seed from the $n = 200$ set                                                      |
| 1,000  | 200         | Final repeated-run campaign         | 6 RM training runs; seeds 42, 42, 7, 100, 314, 2026; later extended to 20 (Section 6.7)                 |
| 5,000  | 200         | Scale-robustness check              | Single training run; later extended to 20 (Section 6.8)                                                 |
| 10,000 | 200         | Scale-robustness check              | Single training run; later extended to 20 (Section 6.8)                                                 |
| 2,000  | 200         | RQ4 crossover check                 | 5 training runs; exact subset of the $n = 10,000$ draw (Section 6.10.3)                                 |
| 3,000  | 200         | RQ4 crossover check                 | 5 training runs; exact subset of the $n = 10,000$ draw (Section 6.10.3)                                 |
| 500    | 999         | RQ4 held-out ground truth           | Disjoint from the $n = 10,000$ draw by construction; never used to train any reward model (Section 4.8) |

### 5.3. Candidate Generation

Candidate summaries are generated with Mistral-7B-Instruct-v0.3 [18], loaded in `bfloat16` without quantization (`src/generation/candidates.py`). On the A100-SXM4-40GB GPUs used for this project, the model occupies roughly 14 GB in `bfloat16`, comfortably within budget, so 4-bit quantization, though supported in the code via `BitsAndBytesConfig`, is disabled in the configuration actually used (`load_in_4bit: false` in `configs/generation.yaml`). The tokenizer uses

left-padding, required for batched generation with a decoder-only model, and articles are truncated to 1,024 input tokens.

Each article is summarized twice, using the fixed prompt template "Summarize the following news article in a concise paragraph. . . .", with two different sampling configurations: a low-temperature setting ( $T = 0.7$ ,  $\text{top-}p = 0.9$ ) and a high-temperature setting ( $T = 1.0$ ,  $\text{top-}p = 0.95$ ), generating up to 256 new tokens each. These two outputs form the candidate pair ("summary A" and "summary B") that both the holistic and GCA preference construction stages score. Generation is resumable: the script checks which sample IDs already appear in the output file and only generates candidates for the remainder, which matters in practice on a shared cluster where a job may need to be resubmitted after a queue timeout.

## 5.4. Preference Construction

Both preference-construction regimes are implemented in `src/judging/build_reward_preferences.py` and share a single judge object, `RewardModelJudge` (`src/judging/reward_model_judge.py`), configured to use the `AlignScore` backend.

In the holistic regime (`judge_holistic_pair`), each full candidate summary is scored once against the article with `judge.score(article, summary)`, and the two resulting scores are compared with a margin guard (`make_decision`): summary A is preferred if its score exceeds summary B's by more than the margin, summary B is preferred in the opposite case, and the pair is marked `no_preference` otherwise. With the final margin of 0 (Chapter 4, Section 4.9), every pair with a nonzero score difference produces a decision.

In the GCA regime (`judge_gca_pair`), each candidate summary is first split into sentences with a regex-based splitter, then every sentence is scored independently against the article with `judge.score_sentences(article, sentences)`. The resulting list of per-sentence scores is passed to `aggregate_sentence_scores(scores, alpha)` (Chapter 4, Equation 4.1) to produce a single GCA score per summary, and the same margin-guarded comparison determines the preference. Every output record retains the full sentence-level detail, including each sentence's individual `AlignScore` score and a count of "low-faithfulness" sentences (score below 0.5), which supports the qualitative disagreement analysis in Chapter 6.

The locked final configuration is invoked as:

```
python src/judging/build_reward_preferences.py
  -mode both -alpha 0.0 -margin 0
  -alignscore-evaluation-mode nli
  -alignscore-backbone FacebookAI/roberta-base
```



**A discrepancy between this intended configuration and what the confirmatory campaigns actually ran, found during final review.** This is reported here in full because it materially affects how several results in this thesis should be read, and because finding it and reporting it precisely is more useful to the reader than leaving it undiscovered.

The command above is what *should* produce the preference data behind every confirmatory campaign, and it is what a small number of exploratory runs earlier in the project, including the judge-mode sweep itself (Chapter 6, Section 6.5) and the six-run campaign that followed it (Section 6.6), were in fact built from. It is not, however, what the twenty-run extended campaigns at  $n = 1,000$ ,  $n = 5,000$ , and  $n = 10,000$  were trained on, nor what the  $n = 2,000/n = 3,000$  crossover checks (filtered from the  $n = 10,000$  file) or the RQ4 checkpoints (trained on the same files as RQ1–RQ3) were trained on. Checking the stored preference files directly against the training scripts that consumed them shows the following.

- `slurm/build_preferences_scale.sh`, used to build the  $n = 5,000$  and  $n = 10,000$  preference files, never passes `-alignscore-evaluation-mode` on its command line. The flag’s default in `src/judging/build_reward_preferences.py` is `nli_sp`, not `nli`, so both files were built under `nli_sp`. The aggregation exponent is unaffected: both files correctly carry  $\alpha = 0.0$ , since `-alpha 0.0` is passed explicitly.
- The  $n = 1,000$  file at `data/preferences_1000/` is dated to the original, pre-optimization run of `slurm/build_preferences_1000.sh` (Section 6.2), which explicitly passes `-alpha 0.5` and, like the scale script, never sets `-alignscore-evaluation-mode`, so it too defaults to `nli_sp`. A separate file built correctly under `nli`,  $\alpha = 0.0$  exists at `data/preferences_1000_alpha0_mode_nli/`. This correct file is what the six-run campaign (Section 6.6) actually used: its four “seed validation” runs were each built fresh by `slurm/mode_nli_seed_confirm.sh`, which does pass `-alignscore-evaluation-mode nli` explicitly, and the run-metadata JSON files for those four runs (`outputs/mode_sweep_alpha0/mode_nli_seed\{7, 100, 314, 2026\}/`) confirm both the correct configuration and an exact match to the six-run figures reported in Table 6.5. The twenty-run extended campaign that supersedes it (Section 6.7) does not share this property: its own stored run-metadata, for every one of its twenty seeds, records `data/preferences_1000/` as the input, i.e. the stale  $\alpha = 0.5/nli\_sp$  file, not `data/preferences_1000_alpha0_mode_nli/`. The RQ4 checkpoints at  $n = 1,000$  were trained from the same stale file. In short: the six-run campaign at  $n = 1,000$  is correctly configured; the twenty-run campaign that supersedes it, and every RQ4 result built on it, is not.
- The RQ4 held-out ground truth (Chapter 4, Section 4.8) is not affected in the same way: `slurm/build_ground_truth.sh` does pass `-alignscore-evaluation-mode`

`nli` explicitly, and this is the only stage in the pipeline for which the intended and executed configurations agree.

The practical consequence is that every reward model whose accuracy is reported as a confirmatory result in this thesis, the RQ1/RQ3 twenty-run campaigns at all three scales, the  $n = 2,000/n = 3,000$  crossover checks, and the RQ4 checkpoints, was trained on preference data built under `nli_sp`, not the `nli` configuration that Section 6.5 selected and that this thesis describes elsewhere as locked. At  $n = 1,000$  specifically, the aggregation exponent is additionally the original  $\alpha = 0.5$  penalty formula rather than the locked  $\alpha = 0.0$  simple mean. The RQ4 ground-truth criterion itself was scored under the correctly-locked `nli`,  $\alpha = 0.0$  configuration, so the training data and the ground-truth criterion in RQ4 were built under different judge configurations from one another, a further consequence addressed in Chapter 6, Section 6.10 and recorded as a limitation in Chapter 8, Section 8.4.

Two things are unaffected by this. First, every reported number is a genuine measurement of the data that was actually used: the run-metadata files trace every seed’s result back to the exact preference file consumed, and the accuracies, gaps, and statistics reported throughout Chapter 6 are correctly computed from those runs. What is incorrect is the description, repeated at several points in this thesis prior to this discovery, of that data as the `nli`,  $\alpha = 0.0$  configuration. Second, the comparison between holistic and GCA within each campaign remains fair: both conditions in a given campaign were built from the same candidate pool under the same (if mislabelled) judge configuration, so the discrepancy does not advantage one condition over the other within a run. What it does undermine is the specific mechanistic account in Chapter 7, Section 7.2, that the `nli` mode’s absence of internal decomposition is why GCA shows an advantage: the confirmatory campaigns were not run under that mode, and that account is revisited in light of this finding at the same location.

This was found during a final review pass after the experimental phase of the project had concluded. Re-running the affected campaigns under the correctly-configured data would settle whether the reported effects persist under the intended configuration, and is recorded as the first item under future work (Chapter 8, Section 8.5); it was not attempted here, both because of the time remaining before submission and because it constitutes new experimental work beyond what had already been agreed with the supervisors for this stage of the project.

## 5.5. Reward-Model Training

Reward-model training is implemented in `src/reward_model/train.py` (`BradleyTerryRewardModel`, `bradley_terry_loss`) and orchestrated by `src/reward_model/run_training.py`, which trains the holistic and GCA conditions from a single invocation. Two training modes are used in this thesis,

selected by whether `run_training.py` is invoked with `-kfold`. Every RQ1–RQ3 result (Chapter 6, Sections 6.2 to 6.9) uses  $k$ -fold cross-validation (`_kfold_cv` in `run_training.py`): the preference pairs are shuffled using a seeded `random.Random` instance, split into  $k = 5$  contiguous folds, and for each fold a freshly initialized reward model is trained on the other four folds and evaluated on the held-out one; only each fold’s accuracy is kept, and the trained weights are discarded. The RQ4 ground-truth evaluation (Chapter 6, Section 6.10) instead omits `-kfold`, which routes through `train_reward_model` in `train.py`: a single model is trained on the full preference set for a given seed, and its checkpoint (encoder weights and reward head) is written to disk rather than discarded, since RQ4 needs the trained model itself, not only a cross-validation accuracy (Section 4.8). Both modes share the same architecture, loss, optimizer, and hyperparameters; they differ only in whether the data is split into folds and in whether the trained weights are kept. One implementation detail is worth stating precisely, since it affects how “training seed” should be interpreted. The `-seed` argument is passed to Python’s `random` module, which governs the fold-assignment shuffle in the  $k$ -fold path (and the train/seed initialization in the non- $k$ -fold path).

In the version of `_kfold_cv` used for the original six-run campaign (Chapter 6, Section 6.6), PyTorch’s own generator was not seeded anywhere in the cross-validation path, so every other source of stochasticity was uncontrolled: the randomly initialized reward head, the shuffling of training batches by the `DataLoader`, and the dropout masks applied during training. Changing `-seed` therefore changed only the fold assignment, while these other sources varied between process executions regardless of the seed value, which is why two executions launched with the same seed value produced different results (Section 6.6) and why the original campaign could not be extended past six runs without first addressing this.

This was fixed before the twenty-run campaign that produces every headline result in this thesis: `_kfold_cv` now seeds PyTorch and CUDA explicitly at the start of each fold, supplies an explicit generator to the training `DataLoader`, and seeds the `DataLoader` worker processes. Under this corrected procedure, a run is fully determined by its seed value on fixed hardware, and this is the training procedure that produced every twenty-run campaign reported in Chapter 6 (Sections 6.7 and 6.8 onward) and the reproduction instructions in Appendix B. The original, unseeded behaviour is retained here as the explanation for why the six-run campaign’s individual runs differ even at a repeated seed value, and is discussed further as a source of run-to-run variation in Chapter 7, Section 7.9.

The final training configuration, used for every reported reward-model result, is: FacebookAI/roberta-base [28] backbone, 5 epochs, batch size 8, learning rate  $2 \times 10^{-5}$  with a linear warmup over 10% of total steps, maximum sequence length 512 tokens (article truncated to 2,000 characters before tokenization, concatenated with the summary), and 5-fold cross-validation. This is confirmed by `slurm/train_rm_1000.sh`, the Slurm script used for the headline  $n = 1,000$  result. One inconsistency in the codebase is worth flagging explicitly, since it could otherwise look like an error

in the reported results: the in-code default for `-backbone` in both `train.py` and `run_training.py` is `microsoft/deberta-v3-base`, a leftover from an earlier version of the training script. Every Slurm script used to produce a result reported in this thesis overrides this default and passes `-backbone FacebookAI/roberta-base` explicitly; the `deberta-v3-base` default was never actually used to produce a reported number. The one exception is the very first Bradley–Terry result on the  $n = 495$  set (Chapter 6, Section 6.2), which predates the standardization on `roberta-base` and is reported as such.

**Source-document truncation.** The reward model does not see the full source article. Two limits apply in sequence. The article string is first truncated to its leading 2,000 characters (`-max-article-chars`, applied in `PreferencePairDataset`), and the truncated article is then concatenated with the candidate summary and tokenized under a 512-token cap (`-max-length`), so the article prefix is shortened further to make room for the summary. Subset selection admits articles up to 8,000 characters (Section 5.2), so for the longer articles in the pool the reward model observes well under half of the source text.

This matters for interpreting the reward-model results. The evaluator that produced the preference labels scores summaries against the article under its own chunking scheme, whereas the reward model must reconstruct those labels from a prefix only. Where the evidence supporting or contradicting a claim lies beyond the retained prefix, that evidence is simply unavailable to the reward model, and the corresponding preference is not recoverable from its input in principle rather than merely in practice. How large this effect is was not measured here, and the near-chance accuracies reported in Chapter 6 should not be attributed to truncation without evidence; it is one plausible contributing factor among several discussed in Chapter 7, Section 7.4. Varying `-max-article-chars` while holding everything else fixed would isolate it, and is recorded as future work.

## 5.6. Compute Infrastructure

All experiments run on MOGON NHR (`mogon-nhr-01.zdv.uni-mainz.de`), the University of Mainz’s high-performance computing cluster, using Slurm partition `a100dl` under account `nhr-haloed`, on NVIDIA A100-SXM4-40GB GPUs. All Slurm scripts producing results reported in Chapter 6 exclude one cluster node found to have a faulty CUDA runtime, via `#SBATCH -exclude`; this affects scheduling only and not the computation performed.

## 5.7. Reproducibility

The pipeline is config-driven throughout: dataset, generation, and judging parameters live in `configs/*.yaml` and can be overridden per run from the command line, and every script accepts explicit flags for the parameters that matter for a given experiment (margin,  $\alpha$ , AlignScore evaluation mode, reward-model hyperparameters, and so on) rather than hardcoding them.

Seeding is now complete for the procedure used to produce every headline result. Subset selection and cross-validation fold assignment are explicitly seeded through Python’s `random` module, and, since the fix described in Section 5.5, PyTorch and CUDA are seeded per fold as well, with an explicit generator for the training `DataLoader` and seeded worker processes. Under this corrected procedure a run is fully determined by its seed value on fixed hardware. This was not always the case: the original six-run campaign at  $n = 1,000$  (Chapter 6, Section 6.6) predates the fix, and two executions with identical arguments could differ under that earlier version of the code, which is why that campaign’s per-run figures do not exactly reproduce from the seed value alone. Every pipeline script writes a JSON run-metadata file (`src/utis/logging.py, save_run_metadata`) recording its configuration, output artifact paths, and a timestamped run ID, so that any reported number can in principle be traced back to the exact invocation that produced it.

The configuration required to reproduce the  $n = 1,000$  result is given in full in Table 4.1 (Chapter 4), together with the twenty training seeds (1–20) listed in Chapter 6, Table 6.8, and is implemented by `slurm/train_rm_1000.sh` and `slurm/seed_campaign.sh`. The six seeds in Table 6.5 (42, 42, 7, 100, 314, 2026) reproduce only the earlier, superseded six-run campaign under the pre-fix code and are retained for that historical comparison, not as the current reproduction path. The codebase, including all scripts referenced in this chapter, is available at the project’s GitHub repository: <https://github.com/hasnat23/master-thesis-rlaif-gca>.

## 6. Results

### Reading this chapter: where the validated results are

This chapter is organised by chronology, not by importance, because the exploratory sections are themselves evidence for a methodological argument made later in the chapter (Section 6.6): that single runs at this dataset scale are not reliable, which is why every confirmatory result in this thesis is a twenty-run campaign rather than one training run. A reader who wants the validated headline numbers first, and the exploratory process that led to them only as needed, can go directly to Section 6.7 (the RQ1 headline result), Section 6.8 (RQ3, scale robustness), and Section 6.10 (RQ4, the independent ground-truth evaluation and its reversal). Sections 6.1 to 6.6 below cover the exploratory process that selected the locked configuration and motivated the repeated-run protocol; they are referenced from those later sections and from Chapter 7 rather than needing to be read first.

This chapter reports the experiments in the order they were carried out. Two intermediate findings did not survive repetition: a hyperparameter configuration that initially appeared promising (Section 6.4) and a two-run estimate that proved inconclusive (Section 6.5). Both are reported, since they motivated the repeated-run protocol described in Chapter 4, Section 4.5, and omitting them would misrepresent how much of the final estimate rests on distinguishing an effect from run-to-run variance. Sections 6.3 to 6.5 are exploratory in the sense of Chapter 4, Section 4.6; Section 6.8 examines how the selected configuration behaves as the dataset grows. Section 6.10 then re-evaluates the same trained reward models against an independent, held-out ground truth (Chapter 4, Section 4.8, RQ4), rather than against the training-style labels used everywhere above.

That final section follows a single line of enquiry across five steps, and it may help to state the sequence in advance. It first establishes that GCA is substantially more precise than holistic at  $n = 1,000$  (Section 6.10.1); then finds that this advantage does not merely fade but reverses as the training set grows (Section 6.10.2); then narrows down where in between the two conditions change places (Section 6.10.3); and finally tests two candidate mechanisms for the reversal against data already collected, rejecting both (Sections 6.10.4 and 6.10.6). Section 6.10.7 states what the sequence collectively establishes and what it leaves open.

## 6.1. Early Prototype Results (Historical Context)

An earlier version of the pipeline produced results under a DPO-based design before the study narrowed to reward-model comparison (Chapter 4, Section 4.9). They are reported here as context for that change, not as findings of the present study.

A FLAN-T5 [5] baseline on a 20-sample pilot gave ROUGE-1 0.3278, ROUGE-2 0.1201, ROUGE-L 0.2341 [26], and BERTScore F1 0.8612 [42]. On a 200-sample subset (subset seed 42), the original AlignScore-based preference construction, using the proposal-era margin of 0.05 and  $\alpha = 0.5$ , produced 154 usable holistic pairs out of 200 (77.0%, mean scores 0.7324 for the low-temperature candidate and 0.6530 for the high-temperature one) and 157 usable GCA pairs (78.5%, mean GCA scores 0.4066 and 0.3337). The two methods agreed on 121 of 200 decisions (60.5%).

Mistral-7B-Instruct-v0.3 was then fine-tuned with DPO on both preference sets (Chapter 3, Section 3.2), using LoRA adapters [17] on the attention and feed-forward projections rather than full fine-tuning: training loss reached 0.6902 for the holistic condition and 0.6913 for the GCA condition, each in roughly 134.5 seconds on an A100. On 200 held-out articles, baseline AlignScore was 0.7965, DPO-Holistic reached 0.8017, and DPO-GCA reached 0.7996. Only DPO-GCA’s ROUGE-L differed from baseline at the conventional threshold (Wilcoxon  $p = 0.015$ ); the AlignScore differences between conditions were small and not distinguishable from noise. These downstream results illustrate the attribution problem described in Chapter 4, Section 4.9: differences of this magnitude cannot be traced to the preference-construction procedure rather than to the optimization stage, which is why the final study measures the preference data directly.

## 6.2. Bradley–Terry Reward-Model Baseline

The first Bradley–Terry reward-model result used the  $n = 495$  set (subset seed 100) with a `microsoft/deberta-v3-base` backbone [15], predating the standardization on `FacebookAI/roberta-base` (Chapter 5, Section 5.5). Under 5-fold cross-validation, the holistic reward model reached 58.1% mean pairwise accuracy and the GCA reward model reached 54.6%, both above the 50% chance level.

The first result on the  $n = 1,000$  set, with the `roberta-base` backbone and the original configuration ( $\alpha = 0.5$ , margin 0.05), is summarized in Table 6.1. This is the baseline the rest of this chapter’s optimization campaign works from: holistic outperforms GCA by 1.2 percentage points.

Table 6.1.: Bradley–Terry reward-model baseline at  $n = 1,000$  ( $\alpha = 0.5$ , margin 0.05, roberta-base).

| Condition   | Mean pairwise accuracy | Gap (GCA – Holistic) |
|-------------|------------------------|----------------------|
| Holistic RM | 57.20%                 | –1.20 pp             |
| GCA RM      | 56.00%                 |                      |

### 6.3. GCA Formula Optimization

To understand why GCA underperformed, disagreement between the holistic and GCA decisions at  $\alpha = 0.5$  was analyzed directly on the  $n = 1,000$  pairs (`analysis/gca_vs_holistic_analysis.py`). The two methods disagreed on 17.8% of pairs. GCA scores showed higher discrimination between candidates than holistic scores (mean absolute score difference 0.2392 versus 0.1527), but only 82.2% of GCA decisions agreed with the corresponding holistic decision, suggesting that the extra discrimination was at least partly noise rather than signal.

Nine aggregation strategies were then compared by how well each agreed with the holistic decision on the same 1,000 pairs (`analysis/test_gca_formulas.py`, `analysis/test_advanced_aggregation.py`). Table 6.2 reports the results and Figure 6.1 plots them in rank order. The simple mean ( $\alpha = 0.0$ ) had the highest agreement of any strategy tested, 15.4 percentage points above the original penalty formula.

Table 6.2.: Agreement with the holistic decision across nine GCA aggregation strategies,  $n = 1,000$  pairs.

| Strategy                            | Agreement with holistic | vs. $\alpha = 0.5$ baseline |
|-------------------------------------|-------------------------|-----------------------------|
| Simple mean ( $\alpha = 0.0$ )      | 97.6%                   | +15.4 pp                    |
| Confidence-weighted                 | 96.0%                   | +14.0 pp                    |
| Ensemble                            | 93.3%                   | +11.1 pp                    |
| Length-weighted                     | 91.5%                   | +9.3 pp                     |
| Weighted by position                | 88.2%                   | +6.0 pp                     |
| Robust mean (10% trim)              | 85.4%                   | +3.2 pp                     |
| Harmonic mean                       | 82.9%                   | +0.7 pp                     |
| Original penalty ( $\alpha = 0.5$ ) | 82.2%                   | baseline                    |
| Minimum                             | 78.7%                   | –3.5 pp                     |

On the strength of this result, the default  $\alpha$  was changed from 0.5 to 0.0 throughout the codebase (`src/judging/gca.py`, `src/judging/build_reward_preferences.py`, and every Slurm script).



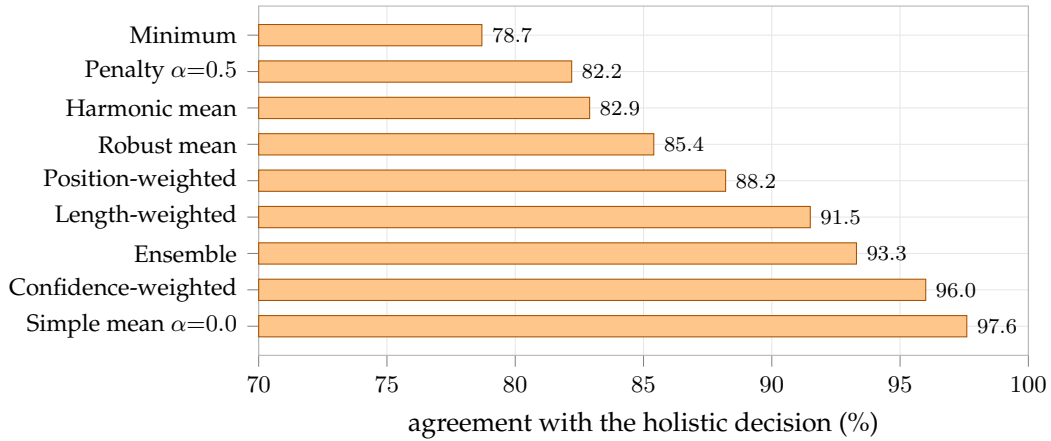


Figure 6.1.: Agreement with the holistic decision across the nine aggregation strategies of Table 6.2 ( $n = 1,000$  pairs). Agreement measures consistency between two procedures driven by the same evaluator, not external factual correctness.

Two qualifications apply to this selection. First, the criterion used in Table 6.2 is agreement with the holistic decision, which is a measure of internal consistency between two procedures driven by the same evaluator, not a measure of external factual correctness. A strategy could agree closely with holistic scoring while reproducing the same errors, and nothing in this comparison would distinguish those cases. The criterion is used here as a diagnostic for whether the penalty term was amplifying disagreement, not as evidence that the simple mean produces better labels. Second, this comparison and the evaluator-mode sweep that follows are exploratory in the sense defined in Chapter 4, Section 4.6: the final configuration was chosen after observing  $n = 1,000$  outcomes, so results on that setting cannot also serve as independent confirmation of the choice. The  $n = 5,000$  and  $n = 10,000$  experiments do not fully remedy this, because their subsets are nested supersets of the  $n = 1,000$  set rather than independent samples (Chapter 4, Section 4.7).

## 6.4. Does the Formula Change Alone Flip the Result?

The formula change was then validated the same way the baseline was measured: by training reward models on the resulting preferences and checking pairwise accuracy, holding the judge mode fixed at the original `nli_sp` setting. Table 6.3 shows that  $\alpha = 0.0$  did not, on its own, close the gap. Holistic accuracy improved more than GCA accuracy did, and the gap between them widened slightly rather than closing.

A nine-configuration hyperparameter search was run next, sweeping learning rate over  $\{1 \times 10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}\}$  and epoch count over  $\{3, 5, 7\}$ , still at  $\alpha = 0.0$ . One configuration, learning rate  $1 \times 10^{-5}$  with 7 epochs, produced a GCA advantage of

Table 6.3.: Effect of the formula change alone ( $n = 1,000$ , judge mode `nli_sp` unchanged).

| Metric                 | $\alpha = 0.5$ (baseline) | $\alpha = 0.0$ | Change         |
|------------------------|---------------------------|----------------|----------------|
| Holistic mean accuracy | 0.572                     | 0.586          | +0.014         |
| GCA mean accuracy      | 0.560                     | 0.561          | +0.001         |
| Gap (Holistic – GCA)   | 0.012                     | 0.025          | +0.013 (wider) |

+0.014 (GCA 0.586 versus holistic 0.572). A same-seed confirmation rerun of that exact configuration, however, produced the opposite result: holistic 0.577 versus GCA 0.560, a gap of  $-0.017$ . The apparent advantage did not reproduce. This result is reported in full because of what it implies: at this dataset size, a single training run, even with 5-fold cross-validation, can produce a result that looks like a real effect but is actually within the range of ordinary fold-to-fold and initialization noise. This directly motivated moving away from hyperparameter tuning toward the judge-mode sweep below, and toward the multi-seed validation protocol used for the headline result.

## 6.5. AlignScore Judge-Mode Sweep

With  $\alpha = 0.0$  fixed, four AlignScore evaluation modes (Chapter 3, Section 3.3) were compared under the same  $n = 1,000$  training setup. Table 6.4 shows that the choice of mode, not only the aggregation formula, is associated with whether GCA leads at all. The `nli` mode produced the largest GCA advantage of any configuration tested so far; `nli_sp`, the mode used for every result up to this point, is actually one of the two modes that favor holistic.

Table 6.4.: AlignScore judge-mode sweep ( $n = 1,000$ ,  $\alpha = 0.0$ ).

| Mode                | Holistic mean acc. | GCA mean acc. | Gap (GCA – Holistic) |
|---------------------|--------------------|---------------|----------------------|
| <code>nli</code>    | 0.523              | 0.583         | +0.060               |
| <code>bin</code>    | 0.510              | 0.564         | +0.054               |
| <code>bin_sp</code> | 0.554              | 0.562         | +0.008               |
| <code>nli_sp</code> | 0.578              | 0.557         | $-0.021$             |

The `nli_sp` row above (0.578 holistic, 0.557 GCA) is a separate single run from the  $\alpha = 0.0/\text{nli\_sp}$  row in Table 6.3 (0.586 holistic, 0.561 GCA), not a repetition of it: the two experiments were run independently, at different points in the exploratory process, before either PyTorch or CUDA was seeded. The 0.8 and 0.4 percentage-point differences between them, on what is nominally the same configuration, illustrate exactly the run-to-run instability documented directly in Section 6.6, where the same

seed value (42) produced a 4.7 percentage-point swing between the original sweep and its confirmation rerun. Neither single-run table should be read as a precise measurement of the underlying configuration; both are superseded by the twenty-run campaign (Section 6.7), which is why this chapter reports them as exploratory context rather than as confirmatory results.

Figure 6.2 plots the same values grouped by whether the evaluator performs its own internal splitting.

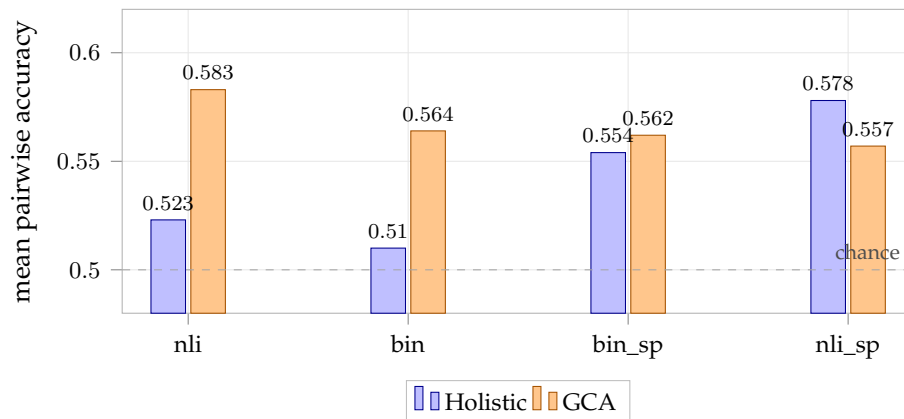


Figure 6.2.: Evaluator-mode sweep ( $n = 1,000$ ,  $\alpha = 0.0$ ). The two left-hand modes do not split the input internally; the two right-hand modes do. Each bar is a single run, so differences of this size fall within the run-to-run variation documented in Section 6.6, and the comparison is exploratory.

The ordering in Table 6.4 follows a systematic pattern rather than varying arbitrarily. The two modes in which GCA leads by a clear margin, `nli` and `bin`, are precisely the two that operate on the context and claim without splitting them. The two modes in which the advantage is negligible or negative, `bin_sp` and `nli_sp`, are the split variants, in which the evaluator itself segments the input and aggregates over the segments (Chapter 3, Section 3.3). Holistic accuracy also rises from 0.510–0.523 in the unsplit modes to 0.554–0.578 in the split modes, which is consistent with the split modes supplying a stronger holistic signal rather than with GCA becoming worse.

The pattern suggests that external sentence-level decomposition may contribute most where the evaluator does not already decompose internally, and little where it does; on this reading GCA and the evaluator’s own splitting act partly as substitutes. This sweep is exploratory and each mode was run once, so the comparison across modes should be read as suggestive rather than established: Section 6.6 shows that single-run differences of this magnitude fall within the range of run-to-run variation, and only the `nli` configuration was subsequently validated across repeated runs in that six-run campaign (though, as Chapter 5, Section 5.4 reports, the twenty-run campaign that later supersedes it did not, in fact, remain on `nli`). Chapter 7, Section 7.2 develops the

interpretation and states what would be required to test it.

Because a single run had already proved insufficient in Section 6.4, the `nli` result was rerun at the same seed value before any conclusion was drawn. The rerun gave holistic 0.543 and GCA 0.556, a gap of +0.013: smaller than the original +0.060, but in the same direction. Over the fold-level differences from both runs, the 95% interval was  $[+0.003, +0.064]$  and the corresponding Wilcoxon  $p$ -value was 0.084. Under the independence caveat discussed above, neither figure was treated as decisive, and further runs were carried out rather than concluding from two.

## 6.6. Repeated Runs of the Selected Configuration ( $n = 1,000$ )

The `nli` mode,  $\alpha = 0.0$ , margin 0 configuration was fixed and run across four further seed values, in addition to the original sweep and its confirmation rerun, for six runs and 30 cross-validation folds in total. Table 6.5 lists every run. Note that the six runs span five distinct seed values, not six: seed 42 appears twice, as the original sweep and its rerun. As discussed in Chapter 4, Section 4.5, those two runs are not duplicates, because the seed argument controls only fold assignment and not the reward model’s weight initialization.

Table 6.5.: Per-run results for the selected `nli`-mode configuration ( $n = 1,000$ ).

| Run               | Seed | Holistic | GCA   | Gap (GCA – Holistic) |
|-------------------|------|----------|-------|----------------------|
| Original sweep    | 42   | 0.523    | 0.583 | +0.060               |
| Confirmation      | 42   | 0.543    | 0.556 | +0.013               |
| Seed validation 1 | 7    | 0.556    | 0.546 | −0.010               |
| Seed validation 2 | 100  | 0.510    | 0.561 | +0.051               |
| Seed validation 3 | 314  | 0.520    | 0.552 | +0.032               |
| Seed validation 4 | 2026 | 0.525    | 0.564 | +0.039               |

GCA leads in five of the six runs; seed 7 is the only run in which holistic is ahead. Figure 6.3 plots the per-run gaps against the pooled mean, which makes the spread easier to judge than the table alone: the individual runs range from  $-0.010$  to  $+0.060$ , a spread wide enough that any single run, taken on its own, would have been weak evidence in either direction. Averaging over all 30 fold-level readings gives the estimate this campaign supported, shown in Table 6.6: a mean gap of +3.08 percentage points. This was superseded by the twenty-run campaign of Section 6.7, which is the headline result of this thesis; the six-run figures are retained here as the record of what the original campaign established and of why it had to be extended.

At the fold level, GCA leads in 22 of the 30 folds (73%), with no ties. The interval and  $p$ -value in Table 6.6 are computed as described in Chapter 3, Section 3.4, using 10,000 bootstrap resamples with a fixed NumPy seed of 42.

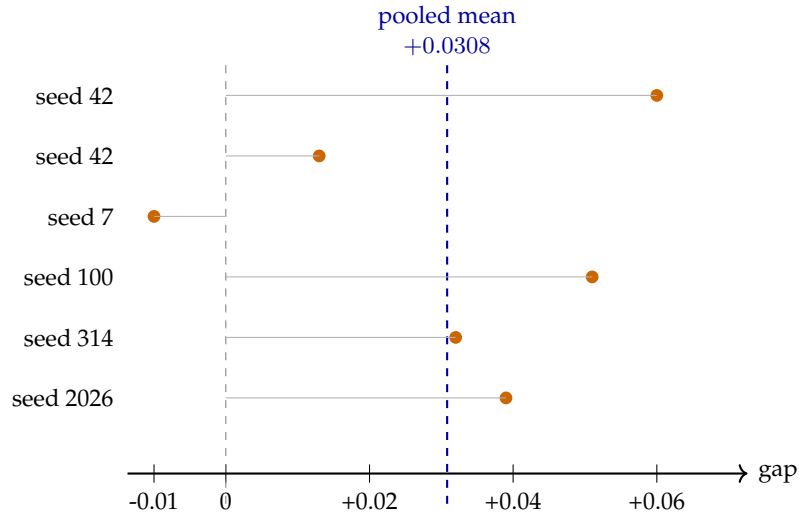


Figure 6.3.: Per-run GCA–Holistic accuracy gap at  $n = 1,000$ , against the pooled mean. Five of six runs favor GCA, but the spread across runs is wide relative to the pooled effect, which is why no single run was treated as conclusive.

Table 6.6.: Averaged result across 6 runs / 30 cross-validation folds at  $n = 1,000$ . The interval and  $p$ -value are computed over folds, which are not independent observations, and are therefore exploratory and anti-conservative (Chapter 4, Section 4.5).

| Holistic mean | GCA mean | Gap            | 95% fold-level CI | Fold-level Wilcoxon $p$ |
|---------------|----------|----------------|-------------------|-------------------------|
| 0.5295        | 0.5603   | <b>+0.0308</b> | [+0.013, +0.047]  | 0.0034                  |

These uncertainty estimates require an explicit caveat and are not presented as confirmatory. The 30 folds are not 30 independent observations: the five folds within a run partition one dataset, so their training sets overlap heavily and their errors are correlated. A procedure that assumes independence therefore treats the evidence as stronger than it is, and the quoted  $p$ -value is anti-conservative. What the fold-level analysis supports is that the difference is consistent in direction across runs and stable in magnitude, not that it meets a particular significance threshold. Chapter 7, Section 7.3 examines what a more conservative treatment would and would not permit.

A second property should be read alongside the gap. Both conditions sit close to chance in absolute terms: 0.5295 and 0.5603 are 2.95 and 6.03 percentage points above the 0.5 chance level. The comparison is therefore between two weak preference signals, and the finding concerns which is more consistently recoverable, not whether either yields a strong reward model (Section 7.4).

Table 6.7 tracks how the estimate changed as runs were added. The interval narrows at each stage while the point estimate stays near three percentage points. Stability of the effect size as precision improves is consistent with a real underlying difference rather than with an artifact that grows as data accumulates, though the same independence caveat applies to every row of the table.

Table 6.7.: Progression of the pooled statistical estimate as runs were added. The  $p$ -values are subject to the independence caveat discussed in Chapter 7, Section 7.3.

| Stage    | Runs | Folds | Pooled gap | 95% CI           | Wilcoxon $p$ |
|----------|------|-------|------------|------------------|--------------|
| Two-run  | 2    | 10    | +0.034     | [+0.003, +0.064] | 0.084        |
| Five-run | 5    | 25    | +0.029     | [+0.009, +0.048] | 0.0123       |
| Six-run  | 6    | 30    | +0.031     | [+0.013, +0.047] | 0.0034       |

At six runs, the fold-pooled  $p$ -value in Table 6.7 had already fallen to 0.0034, yet the run-level test in Table 6.6’s surrounding discussion could not fall below 0.0312 no matter how consistent the six runs were, because that is the smallest two-sided value the Wilcoxon signed-rank test can return with six paired observations. The six-run campaign was therefore not merely inconclusive by chance; it was structurally unable to establish run-level significance at the conventional threshold, independent of the true size of the effect. Resolving this required more runs, which in turn required the reward-model training procedure to be fully deterministic under a given seed, a property it did not previously have, as described next.

## 6.7. Extended Campaign: Twenty Independent Runs

Before further runs could be added, a limitation in the cross-validation implementation had to be fixed. The seed argument used for the six-run campaign controlled only the

fold-assignment shuffle; PyTorch’s own random-number generator was never seeded, so the reward head’s weight initialization, the training `DataLoader`’s batch order, and dropout were all left uncontrolled (Chapter 5, Section 5.5). Two executions of the same seed could therefore still differ, which is why the original sweep and its confirmation rerun in Table 6.5, both at seed 42, produced different results. This was corrected by seeding PyTorch and CUDA per fold, supplying an explicit generator to the training `DataLoader`, and seeding the `DataLoader` workers, so that a run is now fully determined by its seed on fixed hardware.

Under this corrected procedure, the campaign was extended to twenty runs at seeds 1 through 20, fixed in advance of running any of them so that no seed could be selected after seeing its result. Table 6.8 lists every run.

#### **The preference data for this campaign, found during final review**

The intention, consistent with the six-run campaign it extends, was to train this campaign on the `nli-mode`,  $\alpha = 0.0$  preference file at `data/preferences_1000_alpha0_mode_nli/`. The stored run-metadata for all twenty seeds shows the training script instead read `data/preferences_1000/`, the earlier, pre-optimization file built under `nli_sp` and  $\alpha = 0.5$  (Chapter 5, Section 5.4). Every number in this section and Table 6.8 is an accurate report of the runs as executed; the comparison between holistic and GCA remains fair, since both conditions were built from the same (mislabelled) file, but the campaign does not test the configuration Section 6.5 selected. Chapter 7, Section 7.2 discusses what this means for the mode-based interpretation offered there.

GCA led in all twenty runs. Figure 6.4 plots every per-run gap against the pooled mean. Table 6.9 reports the run-level statistics, which is the appropriate unit of independent replication (Section 6.6 above; Chapter 7, Section 7.3).

#### **Headline learnability result (RQ1)**

At  $n = 1,000$ , under the `nli_sp-mode`,  $\alpha = 0.5$ , margin-0 configuration this campaign was actually trained on (not the locked `nli`/ $\alpha = 0.0$  configuration intended, see the caveat above), GCA preference construction produces a more learnable reward-model signal than holistic construction: mean gap +3.95 percentage points across twenty independently seeded runs, GCA ahead in all twenty, run-level Wilcoxon  $p < 0.001$ , Cohen’s  $d = +2.22$ .

The mean gap under the extended campaign, +3.95 percentage points, is close to the six-run estimate of +3.08 percentage points, so the additional runs refined the precision of the estimate rather than revealing a different effect. Section 7.3 in the next chapter discusses what this result does and does not establish.

Table 6.8.: Per-run results for the extended campaign, seeds 1–20,  $n = 1,000$ , as executed under `nli_sp` mode,  $\alpha = 0.5$ , margin 0 (see caveat above; `nli`/ $\alpha = 0.0$  was intended).

| Seed | Holistic | GCA    | Gap (GCA – Holistic, pp) |
|------|----------|--------|--------------------------|
| 1    | 0.5280   | 0.5920 | +6.40                    |
| 2    | 0.5400   | 0.5450 | +0.50                    |
| 3    | 0.5200   | 0.5580 | +3.80                    |
| 4    | 0.5360   | 0.5700 | +3.40                    |
| 5    | 0.5470   | 0.5770 | +3.00                    |
| 6    | 0.5380   | 0.5610 | +2.30                    |
| 7    | 0.5050   | 0.5700 | +6.50                    |
| 8    | 0.5340   | 0.5430 | +0.90                    |
| 9    | 0.5300   | 0.5830 | +5.30                    |
| 10   | 0.5270   | 0.5550 | +2.80                    |
| 11   | 0.5090   | 0.5830 | +7.40                    |
| 12   | 0.5260   | 0.5730 | +4.70                    |
| 13   | 0.5320   | 0.5680 | +3.60                    |
| 14   | 0.5490   | 0.5690 | +2.00                    |
| 15   | 0.5220   | 0.5710 | +4.90                    |
| 16   | 0.5280   | 0.5690 | +4.10                    |
| 17   | 0.5290   | 0.5740 | +4.50                    |
| 18   | 0.5400   | 0.5740 | +3.40                    |
| 19   | 0.5220   | 0.5730 | +5.10                    |
| 20   | 0.5310   | 0.5750 | +4.40                    |

Table 6.9.: Run-level statistics across the extended twenty-run campaign. The bootstrap interval resamples runs (not folds) 10,000 times with a fixed seed of 0. The two-sided Wilcoxon  $p$ -value is `scipy`’s default computation; an independent exact permutation test (enumerating all  $2^{20}$  sign assignments) gives  $p \approx 1.9 \times 10^{-6}$ , the theoretical floor for twenty same-signed observations. The two figures differ because one pair of runs (seeds 4 and 18) produced an identical gap of +3.4 percentage points, and `scipy`’s default method falls back from the exact permutation calculation to a normal approximation whenever a tie is present; both figures are far below the conventional threshold, so the substantive conclusion does not depend on which is reported.

| Runs | GCA ahead | Mean gap        | SD      | 95% bootstrap CI  | Wilcoxon $p$ (two-sided) |
|------|-----------|-----------------|---------|-------------------|--------------------------|
| 20   | 20/20     | <b>+3.95 pp</b> | 1.78 pp | [+3.19, +4.70] pp | $8.8 \times 10^{-5}$     |



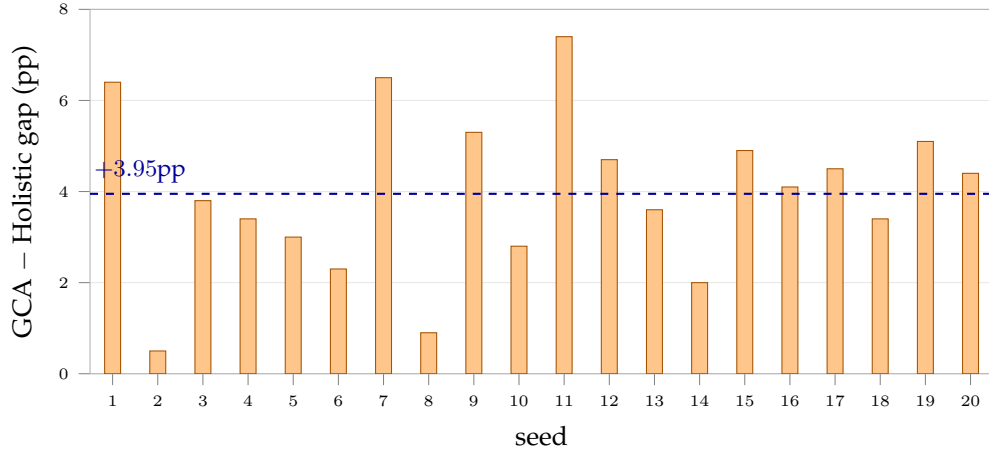


Figure 6.4.: Per-run GCA–Holistic accuracy gap across all twenty runs of the extended campaign, against the pooled mean. Every run favors GCA; the spread (1.78 percentage points, standard deviation) is narrower than in the six-run campaign, consistent with the earlier spread having reflected sampling noise from too few runs rather than genuine instability of the effect.

## 6.8. Scale Robustness: 5,000 and 10,000 Samples

The selected configuration was first run once at  $n = 5,000$  and once at  $n = 10,000$ . These subsets are nested supersets of the  $n = 1,000$  set rather than independent samples (Chapter 4, Section 4.7), so they measure how the effect behaves as data is added to the same pool rather than providing out-of-sample replication. The single-run advantage was small and inconsistent in direction:  $-0.0042$  at  $n = 5,000$ ,  $+0.0035$  at  $n = 10,000$ , against  $+0.0308$  (later  $+0.0395$  under the extended campaign) at  $n = 1,000$ . As with the original  $n = 1,000$  campaign, a single run cannot distinguish a genuine dataset-size effect from ordinary run-to-run variation, so each scale was subsequently extended, first to six and then to twenty independently seeded runs, under the same corrected, deterministic training procedure used for the  $n = 1,000$  extension (Section 6.7). Table 6.10 reports the resulting run-level statistics at all three scales.

The  $n = 5,000$  and  $n = 10,000$  preference files carry the same discrepancy documented for  $n = 1,000$  in the caveat above, though only in the evaluation mode: `slurm/build_preferences_scale.sh` never sets `-alignscore-evaluation-mode`, so both files were built under `nli_sp` rather than the intended `nli` (Chapter 5, Section 5.4). The aggregation exponent at these two scales is unaffected, since `-alpha 0.0` is passed explicitly; unlike  $n = 1,000$ , there is no correctly-configured six-run precursor at these scales, as both were extended directly from a single run. Every  $n = 5,000/n = 10,000$  result in this thesis, including the  $n = 2,000/n = 3,000$  crossover checks filtered from the  $n = 10,000$  file, is therefore reported under `nli_sp`, not `nli`.

Table 6.10.: Repeated-run statistics at all three dataset sizes. All three rows are now twenty-run campaigns using the same corrected training procedure; the  $n = 1,000$  row is repeated from Table 6.9.

| Dataset size | Runs | GCA ahead | Mean gap | 95% bootstrap CI | Wilcoxon $p$         |
|--------------|------|-----------|----------|------------------|----------------------|
| 1,000        | 20   | 20/20     | +3.95pp  | [+3.19, +4.70]pp | $8.8 \times 10^{-5}$ |
| 5,000        | 20   | 7/20      | -0.22pp  | [-0.66, +0.21]pp | 0.3410               |
| 10,000       | 20   | 9/20      | -0.11pp  | [-0.39, +0.18]pp | 0.5882               |

At  $n = 5,000$  and  $n = 10,000$  the mean gap is close to zero in both cases, GCA leads in a minority of runs at each scale (7 of 20 and 9 of 20) rather than consistently, and the two-sided Wilcoxon tests give  $p = 0.3410$  and  $p = 0.5882$ , results indistinguishable from chance. With twenty runs at each scale, exactly matching the resolution of the  $n = 1,000$  campaign, this is not an underpowered result in either direction: with twenty paired observations the test could have detected an effect as small as the  $n = 1,000$  one had it been present, and it did not.

The comparison against  $n = 1,000$  is the test proposed when only single runs were available (Chapter 7, Section 7.5): if the scale movement were statistical noise, the confidence intervals at  $n = 5,000$  and  $n = 10,000$  should overlap the  $n = 1,000$  estimate; if it reflects a real, scale-dependent change, they should not. They do not: the  $n = 1,000$  interval, [+3.19, +4.70]pp, does not overlap [-0.66, +0.21]pp at  $n = 5,000$  or [-0.39, +0.18]pp at  $n = 10,000$ , and the gap between the nearest edges is more than two and a half percentage points in both cases. This resolves the question the single runs left open: the GCA advantage observed at  $n = 1,000$  is not present at  $n = 5,000$  or  $n = 10,000$ , and this is now supported by repeated, properly powered evidence at each scale, all three at the same twenty-run resolution, rather than by runs of unequal weight.

Table 6.11 gives the underlying mean accuracies behind these gaps, and Figure 6.5 plots them.

Table 6.11.: Selected configuration at three dataset sizes. All three are now averaged over twenty runs each.

| Dataset size | Holistic mean | GCA mean | Gap (GCA – Holistic) |
|--------------|---------------|----------|----------------------|
| 1,000        | 0.5297        | 0.5691   | +0.0395              |
| 5,000        | 0.5842        | 0.5820   | -0.0022              |
| 10,000       | 0.5859        | 0.5848   | -0.0011              |

Absolute accuracy continues to rise with dataset size for both conditions, from roughly 0.53–0.57 at  $n = 1,000$  to roughly 0.582–0.586 at the larger scales, while the gap between conditions collapses from a significant 3.95 percentage points to a non-significant fraction of a point. Reporting only  $n = 1,000$  would have substantially overstated how generally

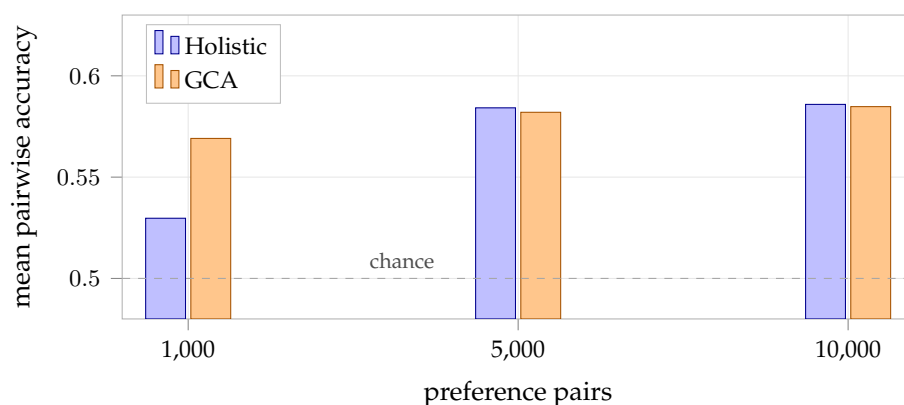


Figure 6.5.: Accuracy at the three dataset sizes, all now averaged over twenty repeated runs each. Absolute accuracy rises for both conditions as data is added, while the two conditions converge to statistically indistinguishable means at the larger scales.

the GCA advantage holds; reporting only a single run at each of the larger scales, as the design originally did, would have left the alternative open, namely that the larger-scale movement was itself just noise. Both possibilities are now closed by the repeated-run evidence.

One property of the training setup is worth naming explicitly here, since it bears on the first half of that claim. Epoch count, batch size, and learning rate schedule are held fixed at every scale (Table 4.1), but the number of optimizer steps is not: `src/reward_model/train.py` sets `total_steps = len(train_loader) * epochs`, so at a fixed batch size the number of gradient updates scales with the number of preference pairs. A model trained at  $n = 10,000$  therefore receives roughly ten times as many optimizer steps as one trained at  $n = 1,000$ , not only ten times as much distinct data. The two are coupled by this design and nothing here separates them, so part of the rise in absolute accuracy with  $n$  may reflect more optimization rather than more information alone. This does not bear on the GCA-versus-holistic comparison itself, since both conditions receive the same step count at a given  $n$ , but it qualifies what “more data helps both conditions” is evidence of (Chapter 7, Section 7.5).

## 6.9. Effect Sizes, Equivalence, and Statistical Power

The tests reported so far establish whether each gap is distinguishable from zero. Two further questions bear directly on how the results should be read, and neither is answered by a  $p$ -value. How large is the  $n = 1,000$  effect in standardised terms? And at  $n = 5,000$  and  $n = 10,000$ , does the failure to reject the null reflect a genuinely negligible effect, or merely insufficient data to detect one? The second question matters because the claim

made in Section 6.8, that the advantage is *absent* at the larger scales rather than merely unconfirmed, is an assertion about absence, and a non-significant result does not by itself support one. Table 6.12 reports the relevant quantities, computed over the run-level differences by `analysis/effect_sizes_and_equivalence.py`.

Table 6.12.: Effect sizes, equivalence tests, and minimum detectable effects across the three twenty-run campaigns. Cohen’s  $d$  and the rank-biserial correlation  $r_{rb}$  are computed over run-level gaps. The TOST column gives the larger of the two one-sided  $p$ -values at an equivalence bound of  $\pm 1.0$  percentage points; a value below 0.05 establishes equivalence at that bound. The tightest bound is the smallest  $\delta$  at which equivalence still holds ( $p < 0.05$ ). MDE is the smallest true effect detectable at 80% power given each campaign’s observed standard deviation.

| $n$    | Mean gap | SD     | Cohen’s $d$ | $r_{rb}$ | TOST $p$ ( $\pm 1pp$ ) | Tightest bound | MDE    |
|--------|----------|--------|-------------|----------|------------------------|----------------|--------|
| 1,000  | +3.95pp  | 1.78pp | +2.22       | +1.00    | –                      | –              | 1.12pp |
| 5,000  | –0.22pp  | 1.03pp | –0.21       | –0.24    | 0.0015                 | $\pm 0.61pp$   | 0.64pp |
| 10,000 | –0.11pp  | 0.67pp | –0.16       | –0.14    | $< 0.0001$             | $\pm 0.36pp$   | 0.42pp |

**The  $n = 1,000$  effect is large in standardised terms.** Cohen’s  $d = +2.22$  places the mean gap more than two standard deviations of the run-to-run spread above zero, which is a large effect by any conventional reading of the scale [7]. The rank-biserial correlation reaches its maximum value of +1.00, reflecting that all twenty runs favour GCA without exception. These two figures should be read together with the absolute accuracies, however, and they measure different things: the effect is large *relative to the variability of the measurement*, while remaining small in absolute terms, since it moves a reward model from roughly 0.53 to roughly 0.57 pairwise accuracy against a chance level of 0.5 (Chapter 7, Section 7.4). A large  $d$  here indicates a highly reproducible difference, not a practically decisive one.

**The larger-scale nulls are positive evidence of absence.** At both  $n = 5,000$  and  $n = 10,000$ , the two one-sided tests reject the hypothesis that the true effect is as large as  $\pm 1$  percentage point in either direction ( $p = 0.0015$  and  $p < 0.0001$  respectively). The effect at these scales is therefore not merely undetected; it is positively bounded. Tightening the bound until equivalence just fails gives the strongest statement the data support: at  $n = 5,000$  any true effect larger than  $\pm 0.61$  percentage points can be ruled out, and at  $n = 10,000$  any effect larger than  $\pm 0.36$  percentage points. Both bounds are far below the +3.95-point advantage measured at  $n = 1,000$ , and below the  $\pm 1$ -point threshold at which an effect would begin to be practically interesting at all. This is what licenses the language of confirmed absence used in Section 6.8, which a non-significant Wilcoxon result alone would not.

**The design had ample power to detect the effect it did not find.** Given each campaign’s observed run-to-run standard deviation, a paired test over twenty runs detects an effect of 1.12 percentage points at  $n = 1,000$ , 0.64 points at  $n = 5,000$ , and 0.42 points at  $n = 10,000$ , each at 80% power and  $\alpha = 0.05$ . Had an effect of the size observed at  $n = 1,000$  been present at either larger scale, it would have exceeded the detection threshold there by a factor of more than six. The larger-scale campaigns are therefore not underpowered with respect to the effect they were testing for, which is the claim Section 6.8 makes and this analysis substantiates. Note that the smaller standard deviations at the larger scales, 1.03 and 0.67 points against 1.78 at  $n = 1,000$ , also mean those campaigns are intrinsically more sensitive, not less, so the failure to find an effect there cannot be attributed to noisier measurement.

## 6.10. Independent Ground-Truth Precision Evaluation

The experiments above establish that GCA preferences are more learnable at  $n = 1,000$ : a reward model recovers them more consistently under cross-validation. They do not establish that the recovered signal is more accurate, because the validation labels are supplied by the same evaluator that built the training preferences (Section 4.4). This section reports RQ4, evaluated against the independent, held-out ground truth described in Section 4.8: 500 articles disjoint from every training pool used in this thesis, scored by the same locked AlignScore-GCA configuration but never used to train any reward model. The reward models evaluated here are trained on the same preference data, seeds, and hyperparameters as RQ1–RQ3, but they are not the same trained weights: RQ1–RQ3 train five fold models per run under  $k$ -fold cross-validation and discard all five after recording their accuracy, so no single reusable model exists from that procedure (Chapter 5, Section 5.5). RQ4 therefore required one additional, non- $k$ -fold training pass per seed and scale, using the same preference files, that trains a single model on the full preference set and retains its checkpoint (Chapter 4, Section 4.8). Every such checkpoint was trained once for this evaluation; every result that follows below reuses those same checkpoints without retraining them again, including the two hypothesis tests in Sections 6.10.4 and 6.10.6.

One consequence of this difference is worth flagging before the two sets of results are read together. The RQ1–RQ3 accuracies report each fold model trained on 80% of the preference set and evaluated on the held-out 20%, whereas the RQ4 models trained below see 100% of the same preference set during training. The comparison between RQ1–RQ3 learnability and RQ4 ground-truth precision (Chapter 7, Sections 7.5 and 7.6) is therefore a comparison across two things that differ in more than just the evaluation target: the training data fraction differs as well, and RQ4’s numbers alone cannot separate what is due to scoring against an independent ground truth from what is due to training on the full set rather than four fifths of it. Nothing in this thesis isolates that second factor, and it is recorded as a limitation in Chapter 8, Section 8.4.

Table 6.13 states the four ground-truth constructions concretely, since the difference between them drives most of the pattern reported below. All four draw on the same pool of 1,000 scored held-out summaries and differ only in which pairs of that pool are compared, and therefore in how large a quality gap separates the two sides.

Table 6.13.: The four ground-truth constructions, all built from the same pool of 1,000 scored held-out candidate summaries (500 articles, two candidates each). “Top/bottom  $X\%$ ” sorts the whole pool by AlignScore-GCA score and compares the highest-scoring  $X\%$  against the lowest-scoring  $X\%$ , a global split across articles rather than a per-article comparison. Narrowing  $X$  gives fewer summaries per side but a larger and less ambiguous quality gap between them.

| Construction              | Summaries/side | Score gap      | Comparison made                                                                                                       |
|---------------------------|----------------|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Same-article pairs        | 500 pairs      | $\approx 0.04$ | Each article’s own two candidates, mirroring the training pairs                                                       |
| Top/bottom 25%            | 250            | $\geq 0.36$    | Best 250 of the pool against worst 250                                                                                |
| Top/bottom 10%            | 100            | $\geq 0.66$    | Best 100 against worst 100                                                                                            |
| Top/bottom 5% (all-pairs) | 50             | $\geq 0.79$    | Best 50 against worst 50, every high summary against every low one (2,500 comparisons) rather than one random pairing |

### 6.10.1. Results at $n = 1,000$

Table 6.14 reports the success rate (the fraction of comparisons in which the reward model scores the higher-quality summary above the lower-quality one, Section 4.8) for both reward models, averaged over the same twenty seeds used in Section 6.7, across all four ground-truth constructions.

The four constructions differ only in how large a quality gap separates the two summaries being compared (Section 4.8): same-article pairs are typically close in quality (mean GCA score gap  $\approx 0.04$ ), while the percentile-split constructions compare progressively more clear-cut cases as  $X$  shrinks. The pattern across the four rows is consistent with that ordering, and Figure 6.6 shows it directly: the two conditions sit on top of one another on the closest comparison and separate progressively as the quality gap widens. On the closest comparison, same-article pairs, the two reward models are statistically

Table 6.14.: Ground-truth precision at  $n = 1,000$ : success rate against four independent ground-truth constructions, averaged over the same twenty seeds used throughout Section 6.7.

| Construction              | Holistic | GCA    | Gap (GCA – Hol.) | 95% CI               | Wilcoxon $p$ |
|---------------------------|----------|--------|------------------|----------------------|--------------|
| Same-article pairs        | 55.16%   | 55.30% | +0.14pp          | $[-0.76, +1.06]$ pp  | 0.760        |
| Top/bottom 25%            | 71.94%   | 75.18% | +3.24pp          | $[-0.40, +7.36]$ pp  | 0.191        |
| Top/bottom 10%            | 76.75%   | 86.00% | +9.25pp          | $[+5.70, +13.15]$ pp | $< 0.001$    |
| Top/bottom 5% (all-pairs) | 76.01%   | 83.39% | +7.38pp          | $[+3.22, +11.69]$ pp | 0.004        |

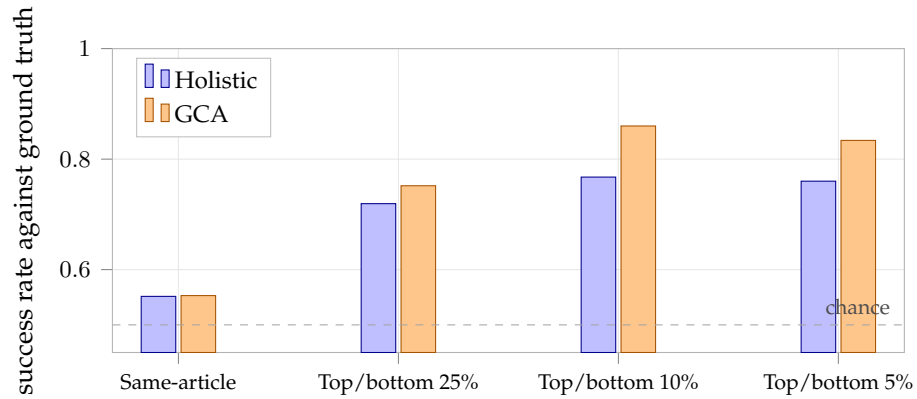


Figure 6.6.: Ground-truth precision at  $n = 1,000$  (Table 6.14). GCA is indistinguishable from holistic on the closest comparison and ahead on all three constructions with a genuine quality gap, significantly so on two of the three.

indistinguishable ( $p = 0.760$ ); a TOST equivalence test rules out any true difference larger than  $\pm 0.96$  percentage points at the 5% level. On the three constructions with a genuine, sizeable quality gap, GCA is ahead, and the advantage is significant on two of the three (top/bottom 10%,  $d = +1.06$ , a large effect by conventional standards [7]; and top/bottom 5% all-pairs,  $d = +0.74$ ). The top/bottom 25% gap is directionally the same but does not reach significance at this seed count ( $p = 0.191$ ). Neither reward model approaches 100% on any construction, plausibly because both are limited by the same 2,000-character article truncation (Chapter 5, Section 5.5), which bounds how much evidence either model can check a summary against regardless of how large the quality gap between the two candidates is.

The ranking used to compute success rate throughout this section is built from GCA’s own aggregation rule, not a rule neutral to both conditions (Chapter 4, Section 4.8). Every GCA-versus-holistic gap reported here, including the +9.25 and +7.38 point advantages above, should be read with that in mind: they are established against a GCA-defined criterion, and the same evaluation under a holistic-defined criterion has not been run (Chapter 8, Section 8.5).

### 6.10.2. Scale Dependence: $n = 5,000$ and $n = 10,000$

The same four ground-truth constructions were then evaluated against reward models trained at  $n = 5,000$  and  $n = 10,000$ , again over twenty seeds each, using the same checkpoints trained for Section 6.8. Only the training-set size changes; the held-out set, the four constructions, and the evaluation procedure are identical to Section 6.10.1.

Table 6.15 reports the underlying success rates for both conditions at all three sizes, and Table 6.16 reports the GCA-minus-holistic gaps derived from them. Both are given because the gaps alone are actively misleading about what happens as the training set grows, for reasons the absolute rates make immediately visible.

Table 6.15.: Ground-truth success rate (%) for each condition, by construction and training-set size. All values are twenty-seed means. Both conditions improve substantially as the training set grows; the gap between them is reported separately in Table 6.16.

| Construction              | $n = 1,000$ |       | $n = 5,000$ |       | $n = 10,000$ |       |
|---------------------------|-------------|-------|-------------|-------|--------------|-------|
|                           | Hol.        | GCA   | Hol.        | GCA   | Hol.         | GCA   |
| Same-article pairs        | 55.16       | 55.30 | 55.04       | 54.61 | 58.10        | 56.65 |
| Top/bottom 25%            | 71.94       | 75.18 | 75.06       | 74.64 | 81.16        | 79.04 |
| Top/bottom 10%            | 76.75       | 86.00 | 83.25       | 83.35 | 91.95        | 90.00 |
| Top/bottom 5% (all-pairs) | 76.01       | 83.39 | 82.00       | 82.73 | 90.84        | 88.72 |



Table 6.16.: GCA-minus-holistic ground-truth precision gap, by construction and training-set size, derived from Table 6.15. All three columns are twenty-seed means.  $p$ -values are for the  $n = 10,000$  column.

| Construction              | $n = 1,000$ | $n = 5,000$ | $n = 10,000$ | Wilcoxon $p$ ( $n=10,000$ ) |
|---------------------------|-------------|-------------|--------------|-----------------------------|
| Same-article pairs        | +0.14pp     | −0.43pp     | −1.45pp      | 0.010                       |
| Top/bottom 25%            | +3.24pp     | −0.42pp     | −2.12pp      | 0.036                       |
| Top/bottom 10%            | +9.25pp     | +0.10pp     | −1.95pp      | 0.058                       |
| Top/bottom 5% (all-pairs) | +7.38pp     | +0.73pp     | −2.12pp      | 0.025                       |

The advantage found at  $n = 1,000$  does not merely fail to persist, as the learnability advantage did (Section 6.8); it reverses. At  $n = 5,000$  every gap is within noise of zero, matching the learnability result at that scale.

#### The ground-truth reversal (RQ4)

At  $n = 10,000$ , holistic is significantly ahead of GCA on three of the four ground-truth constructions ( $p = 0.010$  to  $0.036$ ), with the fourth (top/bottom 10%) narrowly short of the conventional threshold ( $p = 0.058$ ) while pointing the same way. This is the first result in this thesis, on any metric, in which holistic significantly outperforms GCA rather than the two being indistinguishable.

**The reversal is holistic improving faster, not GCA degrading.** Table 6.15 makes a point that Table 6.16 cannot, and that matters for how the reversal should be understood. GCA does not get worse as the training set grows. On the top/bottom 10% construction its success rate rises from 86.00% at  $n = 1,000$  to 90.00% at  $n = 10,000$ , and the same upward movement holds on every construction except the same-article one. What produces the negative gap at  $n = 10,000$  is that holistic improves *faster* over the same interval, from 76.75% to 91.95% on that construction, a gain of 15.2 percentage points against GCA’s 4.0. Both conditions benefit from additional training data; they simply benefit at different rates, and holistic’s steeper trajectory carries it past GCA somewhere between  $n = 3,000$  and  $n = 5,000$  (Section 6.10.3). Figure 6.7 plots this directly.

This reframing does not soften the finding for practical purposes: if the question is which construction procedure to use at  $n = 10,000$ , holistic is the better choice, and significantly so. But it does constrain what any explanation of the reversal has to account for. A mechanism that predicts GCA becoming actively worse with more data is inconsistent with Table 6.15. What requires explaining is why GCA’s returns to additional training data are so much flatter than holistic’s, which is the form the two candidate mechanisms tested in Sections 6.10.4 and 6.10.6 take.

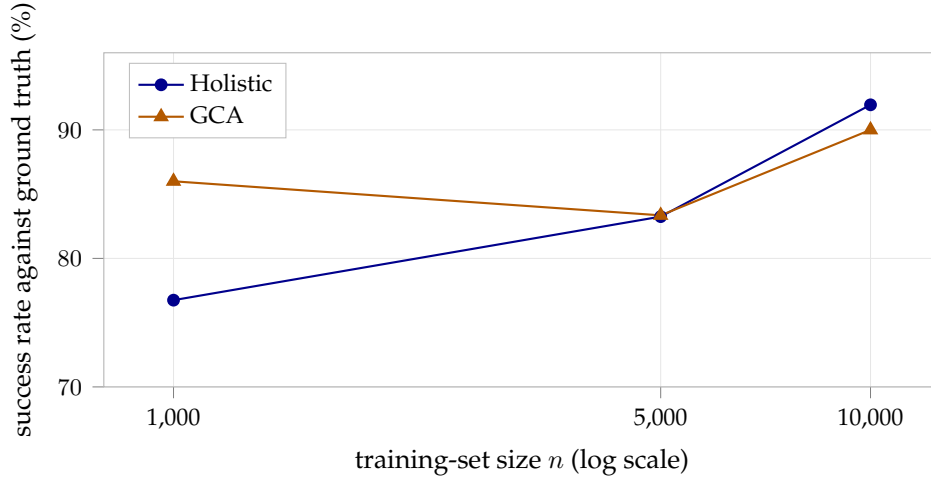


Figure 6.7.: Absolute ground-truth success rate on the top/bottom 10% construction, by training-set size (Table 6.15). Both conditions improve as training data grow. GCA starts far ahead and gains 4.0 percentage points overall; holistic starts lower and gains 15.2, overtaking GCA between  $n = 5,000$  and  $n = 10,000$ . The reversal in Table 6.16 is this difference in slope, not a decline in GCA’s own performance.

### 6.10.3. Locating the Crossover: $n = 2,000$ and $n = 3,000$

Two further training-set sizes were run to locate where the crossover from a GCA advantage to a holistic advantage occurs.  $n = 2,000$  and  $n = 3,000$  were verified by direct index comparison to be exact subsets of the  $n = 10,000$  draw used throughout this thesis (seed 200), so their preference pairs were constructed by filtering the existing  $n = 10,000$  preference files by article identifier rather than by regenerating candidates or re-running AlignScore. Given the number of scales already covered and the time remaining before submission, these two intermediate points were each run at five seeds rather than twenty; they are reported as a supplementary, lower-precision check on where the crossover falls, not as an equally-powered addition to Table 6.16. Table 6.17 places all five sizes side by side, and Figure 6.8 plots the same values as a curve per construction.

For the three constructions with a genuine effect at  $n = 1,000$  (top/bottom 25%/10%/5%), the gap declines steadily from  $n = 1,000$  through  $n = 2,000$  and  $n = 3,000$ , sits within noise of zero at  $n = 5,000$ , and is significantly negative at  $n = 10,000$ . The same-article row, which never showed a reliable effect at any scale, stays close to zero throughout and does not display a comparable trend, consistent with there being nothing directional to decline in that condition.

The transition should be read as a region rather than a point, and the data do not support

Table 6.17.: GCA-minus-holistic ground-truth precision gap at all five training-set sizes tested. The  $n = 2,000$  and  $n = 3,000$  columns are five-seed means; the remaining three are twenty-seed means, repeated from Table 6.16.

| Construction              | $n=1,000$ | $n=2,000$ | $n=3,000$ | $n=5,000$ | $n=10,000$ |
|---------------------------|-----------|-----------|-----------|-----------|------------|
| Same-article pairs        | +0.14     | -1.72     | +0.48     | -0.43     | -1.45      |
| Top/bottom 25%            | +3.24     | -0.16     | -0.88     | -0.42     | -2.12      |
| Top/bottom 10%            | +9.25     | +3.40     | +3.80     | +0.10     | -1.95      |
| Top/bottom 5% (all-pairs) | +7.38     | +1.83     | +1.42     | +0.73     | -2.12      |

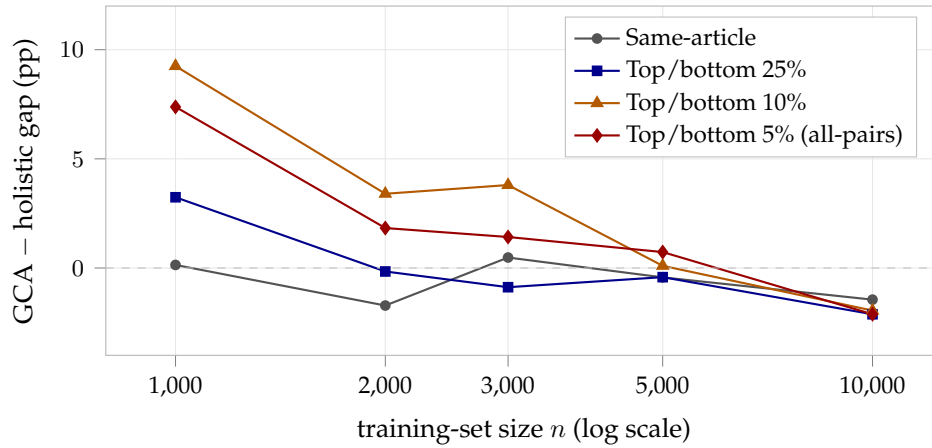


Figure 6.8.: GCA-minus-holistic ground-truth precision gap (Table 6.17) across all five training-set sizes tested. The  $n = 2,000$  and  $n = 3,000$  points are five-seed means; the rest are twenty-seed means. The three constructions with a genuine effect at  $n = 1,000$  decline together and are all negative by  $n = 10,000$ , passing through a null region around  $n = 3,000$ – $5,000$  in which none is distinguishable from zero.

locating it more precisely than that. Statistical significance marks it out only at the two ends: the GCA advantage is significant at  $n = 1,000$  and the holistic advantage is significant at  $n = 10,000$ , while no gap at  $n = 2,000$ ,  $n = 3,000$ , or  $n = 5,000$  is distinguishable from zero in either direction. The point estimates cross zero at different places depending on the construction, which is itself a reason not to over-read them: top/bottom 25% is already marginally negative by  $n = 2,000$ , whereas top/bottom 10% and top/bottom 5% remain marginally positive at  $n = 5,000$  and turn over only between there and  $n = 10,000$ . With five seeds at the two intermediate points, differences of this size are well inside the run-to-run variation documented in Section 6.7. What the intermediate scales establish is therefore the *shape* of the decline and the fact that it passes through a null region around  $n = 3,000$ – $5,000$ , not a single crossover value.

#### 6.10.4. A First Candidate Tested and Rejected: Near-Tie Training Pairs

With the reversal established and its location narrowed to a region, the question becomes what produces it. Two candidate mechanisms were tested directly against data already on disk. Both were chosen because they make a specific, checkable prediction about *where* the effect should be largest, so that the data can contradict them rather than merely fail to confirm them, and because neither required any new reward-model training.

The first concerns the training data rather than the evaluation criterion. GCA’s sentence-level aggregation might produce a larger share of near-zero-margin, effectively noise-level preference pairs than holistic scoring does, since every training preference file in this thesis is built with no margin filter, following prior feedback to use all candidate pairs rather than the smaller, margin-filtered set used in an earlier proposal-era configuration (`slurm/build_preferences_1000.sh`). If GCA’s training pairs contain disproportionately more near-tie comparisons, a reward model trained on more of them might fit that noise rather than genuine signal. That would predict a GCA disadvantage that grows with the training set, which is the pattern Table 6.16 shows.

This was checked directly against the stored per-pair score gaps in the existing preference files, which required no additional model training or AlignScore scoring. Table 6.18 reports the fraction of training pairs with a score gap below 0.05, for both conditions at all three main training-set sizes, recomputed directly from the files each campaign actually trained on (Chapter 5, Section 5.4) rather than from the intended `nli/α = 0.0` files, which is why the  $n = 1,000$  column differs from an earlier version of this table computed before that discrepancy was found.

The mechanism does not hold up against this data, though the shape of the rejection differs from a first look at the raw preference files (which used the intended rather than the executed configuration). At  $n = 5,000$  and  $n = 10,000$ , the near-tie fraction is essentially identical between the two conditions, not larger for GCA, so there is no growing asymmetry at the scales where GCA actually loses. At  $n = 1,000$ , the scale at which GCA *leads*, GCA has *fewer* near-tie pairs than holistic (21.3% versus 24.6%), not

Table 6.18.: Fraction of training preference pairs with a score gap below 0.05, by condition and training-set size, computed from the preference files each campaign actually trained on.

| $n$    | Holistic (frac. < 0.05) | GCA (frac. < 0.05) | Holistic (count) | GCA (count) |
|--------|-------------------------|--------------------|------------------|-------------|
| 1,000  | 24.6%                   | 21.3%              | 246              | 213         |
| 5,000  | 24.6%                   | 24.4%              | 1,232            | 1,218       |
| 10,000 | 24.2%                   | 24.2%              | 2,416            | 2,425       |

more. The hypothesis requires GCA’s near-tie share to be disproportionately larger, and to grow relative to holistic’s as  $n$  grows, so that fitting that noise increasingly hurts GCA at scale; instead the largest gap between conditions occurs at the one scale where GCA wins, and it points the wrong way, while the two scales where GCA loses show next to no gap at all. Near-tie training-pair prevalence is therefore rejected as an explanation for the reversal.

### 6.10.5. How Often Holistic and GCA Agree

A related question, raised independently of the two hypotheses above, is how often the two conditions’ labels agree in the first place. Holistic and GCA are scored from the same candidate pool, so on any pair where both conditions reach the same A/B decision, the two reward models are trained on literally the same (article, chosen, rejected) triple; the labels can only differ on pairs where the two decisions disagree. Table 6.19 reports this directly, computed from the same preference files as Table 6.18, at every scale for which the actually-trained files are available, and additionally for the intended  $nli/\alpha = 0.0$  configuration at  $n = 1,000$  (data/preferences\_1000\_alpha0\_mode\_nli/), where it can be measured without new AlignScore scoring.

Table 6.19.: Fraction of preference pairs on which holistic and GCA reach the same A/B decision, by training-set size and evaluator configuration.

| $n$    | Configuration                       | Pairs compared | Agreement |
|--------|-------------------------------------|----------------|-----------|
| 1,000  | nli, $\alpha = 0.0$ (intended)      | 1,000          | 63.9%     |
| 1,000  | nli_sp, $\alpha = 0.5$ (as trained) | 1,000          | 82.2%     |
| 5,000  | nli_sp, $\alpha = 0.0$ (as trained) | 4,998          | 97.5%     |
| 10,000 | nli_sp, $\alpha = 0.0$ (as trained) | 9,991          | 97.9%     |

Two patterns stand out. First, agreement rises sharply with training-set size under the configuration actually used for the confirmatory campaigns: from 82.2% at  $n = 1,000$  to 97.5–97.9% at  $n = 5,000/n = 10,000$ . This bears directly on the RQ1 result. Because agreeing pairs give both conditions identical training signal, the disagreeing pairs,

roughly 18% of the training set at  $n = 1,000$  but only 2–3% at  $n = 5,000$  and  $n = 10,000$ , are the only pairs that can possibly produce a difference in what the two reward models learn, absent second-order effects from training jointly on a slightly different overall set. A shrinking pool of disagreeing pairs is consistent with, and offers a direct mechanistic account of, a shrinking gap between conditions: there is simply less room at large  $n$  for the two label sets to diverge, independent of any claim about which construction procedure is more informative per label. Second, evaluator mode affects agreement by more than dataset size does at  $n = 1,000$ : the intended `nli` configuration agrees only 63.9% of the time, well below `nli_sp`’s 82.2%, consistent with Section 7.2’s account of `nli_sp` performing an internal decomposition that already resembles what GCA does externally, which should make the two conditions’ decisions converge more often under `nli_sp` than under `nli`. This pattern holds even though, as reported in Section 6.7 above, the resulting reward-model accuracy gap does not track it in the way the mode-based account of Section 7.2 predicts.

The corresponding figures for  $n = 5,000$  and  $n = 10,000$  under the intended `nli` configuration cannot be computed without rebuilding those preference files under the correct evaluator-mode flag and rescoring with `AlignScore`, which is new computation beyond what this thesis attempts (Chapter 8, Section 8.5).

#### 6.10.6. A Second Candidate Tested and Rejected: A Length/Sentence-Count Shortcut

A second candidate explanation for the reversal was tested directly. If the GCA reward model learns to rely on summary length or sentence count as a shortcut for quality, and comes to rely on it more heavily as training data grow, that shortcut would be expected to transfer poorly to the ground truth’s cross-article comparisons and could explain why GCA falls behind at large  $n$ . This requires no new training: every checkpoint already trained for RQ4 was used to score each of the 1,000 held-out ground-truth summaries, and the resulting scores were correlated against each summary’s character length and sentence count (`analysis/check_length_shortcut.py`).

Table 6.20.: Correlation between predicted reward and summary character length, by condition and training-set size, averaged over five seeds at each scale.

| $n$    | Holistic $r$ | GCA $r$ | Gap (GCA – Holistic) |
|--------|--------------|---------|----------------------|
| 2,000  | −0.037       | −0.046  | −0.009               |
| 3,000  | −0.089       | −0.132  | −0.043               |
| 10,000 | −0.198       | −0.201  | −0.003               |

The correlation is negative throughout for both conditions, meaning both reward models score shorter summaries higher, not longer ones, so this is not the length-inflation

shortcut informally expected. More importantly for the mechanism under test, the hypothesis predicts that the GCA-minus-holistic gap in this correlation should be largest at  $n = 10,000$ , the scale at which GCA starts losing. Table 6.20 shows the opposite: the gap peaks at  $n = 3,000$  ( $-0.043$ ) and nearly vanishes at  $n = 10,000$  ( $-0.003$ ), where the two conditions converge to almost identical length-reliance. This mechanism is therefore rejected on the same logic as the near-tie-training-pair hypothesis (Section 6.10.4): the pattern needed to explain a GCA-specific effect is absent from the data.

One genuine pattern does emerge from this check, independent of the rejected hypothesis: both conditions become markedly more correlated with summary length as the training set grows, from  $r \approx -0.04$  at  $n = 2,000$  to  $r \approx -0.20$  at  $n = 10,000$ , roughly a five-fold increase in magnitude for both. Because this growth affects holistic and GCA symmetrically, it cannot explain the reversal, but it is a candidate contributor to the low absolute accuracy ceiling neither condition breaks through (Section 7.4).

### 6.10.7. What the RQ4 Investigation Establishes

The sequence of experiments in this section followed a single line of enquiry, and it is worth stating what it collectively supports before Chapter 7 interprets it.

Against a held-out set that no reward model was trained on, GCA is substantially and significantly more precise than holistic at  $n = 1,000$ , provided the two summaries being compared differ in quality by enough for the comparison to be meaningful: +9.25 percentage points on the top/bottom 10% construction ( $p < 0.001$ ) and +7.38 on top/bottom 5% ( $p = 0.004$ ), with no detectable difference on same-article pairs, where the candidates are close in quality by construction (Section 6.10.1). Extending the same evaluation to larger training sets shows the advantage does not survive: it is within noise of zero at  $n = 5,000$  and significantly negative at  $n = 10,000$  on three of four constructions (Section 6.10.2). Intermediate five-seed points at  $n = 2,000$  and  $n = 3,000$  show the decline passing through a null region around  $n = 3,000$ – $5,000$ , in which no gap is distinguishable from zero in either direction (Section 6.10.3). The absolute rates locate the cause of the reversal in differing rates of improvement rather than in any degradation of GCA (Table 6.15).

Two mechanisms that would account for that difference in slope were then tested and both rejected, each because the data contradicted the hypothesis’s own prediction about where its effect should concentrate: near-tie training pairs (Section 6.10.4) and a length or sentence-count shortcut (Section 6.10.6).

Two properties of how the ground truth is constructed limit how the significance pattern above should be read, and both follow from the same fact: every scale point and every construction scores the same fixed pool of 1,000 held-out summaries (500 articles), never an independently drawn one. First, the four constructions are nested rather than independent tests: the top/bottom 5% pairs are a subset of the top/bottom 10%

pairs, which are in turn a subset of the top/bottom 25% pairs, all drawn from the same 1,000-summary ranking. No multiple-comparison correction is applied across them, and none of the “three of four constructions significant” statements in this section should be read as three independent confirmations; they are three overlapping views of the same underlying ranking at different degrees of overlap. Second, because the same 500 articles are reused at every training-set size, the five-point curve in Section 6.10.3 is not five independent samples of ground-truth precision: any idiosyncrasy of this particular held-out draw is shared by every point on it, so the curve should be read as showing how the trained models’ behaviour changes on one fixed evaluation instrument, not as an average over repeated independent draws of the ground truth itself. Chapter 8, Section 8.4 records both points.

**The reversal is characterised, not explained**

RQ4 establishes that the advantage is real at small scale, that it reverses at large scale, and roughly where the crossover falls. It does not establish *why*. Two candidate mechanisms have been eliminated, which narrows the space of remaining explanations and rules out the two most immediately available ones, but no mechanism has been confirmed. Chapter 7, Section 7.6 discusses what remains, and Chapter 8, Section 8.5 records the leading untested candidate.

## 6.11. Summary of Results

Table 6.21 summarizes how the results in this chapter bear on the research questions from Chapter 4, Section 4.1. Full interpretation is left to Chapter 7.



Table 6.21.: Summary of findings against the research questions.

| Finding    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RQ1</b> | At $n = 1,000$ , GCA leads in all 20 runs of the extended campaign (Section 6.7) with a mean difference of +3.95 percentage points, a run-level two-sided Wilcoxon $p < 0.001$ , a bootstrap 95% CI entirely above zero, and a large standardised effect ( $d = +2.22$ , $r_{rb} = +1.00$ ). An earlier six-run campaign gave a consistent but structurally underpowered estimate (+3.08pp, $p = 0.0625$ ). Twenty-run campaigns at $n = 5,000$ and $n = 10,000$ (Section 6.8), the same resolution as $n = 1,000$ , confirm the advantage is genuinely absent at those scales, not merely unreproduced: mean gaps of $-0.22\text{pp}$ ( $p = 0.3410$ ) and $-0.11\text{pp}$ ( $p = 0.5882$ ), with confidence intervals that do not overlap the $n = 1,000$ result, and equivalence tests excluding effects larger than $\pm 0.61\text{pp}$ and $\pm 0.36\text{pp}$ (Section 6.9).                                                         |
| <b>RQ2</b> | Aggregation strategy and evaluator mode are associated with substantial outcome differences. The evaluator-mode comparison is exploratory, because every mode except the subsequently selected <code>nli</code> configuration was evaluated only once.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>RQ3</b> | Stable at fixed scale, not stable across scale. All 20 runs at $n = 1,000$ favour GCA (SD 1.78pp); twenty-run campaigns at $n = 5,000$ and $n = 10,000$ are similarly consistent around a mean of approximately zero (SD 1.03pp and 0.67pp). The advantage is confirmed present at $n = 1,000$ and confirmed absent at the two larger, nested dataset sizes, with non-overlapping confidence intervals at every comparison and equivalence established at both larger scales (Section 6.9).                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>RQ4</b> | No, and the pattern is stronger than for RQ1: against an independent, held-out ground truth, GCA is significantly more precise than holistic at $n = 1,000$ on clear-cut comparisons (up to +9.25pp, $p < 0.001$ ; Section 6.10.1), indistinguishable from holistic at $n = 5,000$ , and significantly <i>less</i> precise than holistic at $n = 10,000$ on three of four constructions ( $-1.45$ to $-2.12\text{pp}$ , $p = 0.010$ – $0.036$ ; Section 6.10.2). Five-seed checks at $n = 2,000/n = 3,000$ show the decline passing through a null region around $n = 3,000$ – $5,000$ (Section 6.10.3). Two candidate explanations were tested directly and rejected: that GCA’s training pairs contain disproportionately more near-zero-margin comparisons (Section 6.10.4), and that GCA increasingly relies on a length/sentence-count shortcut relative to holistic (Section 6.10.6). The mechanism behind the reversal remains open. |

## 7. Discussion

### 7.1. What the Experiments Establish, and What They Do Not

The experiments support a narrow and conditional claim. Under one specific configuration, at one dataset size, preferences constructed by sentence-level aggregation are more consistently recoverable by a Bradley–Terry reward model than preferences constructed holistically from the same evaluator and the same candidate summaries. The difference averages +3.95 percentage points over twenty independently seeded runs and favours GCA in every one of them, with a run-level Wilcoxon  $p < 0.001$  (Chapter 6, Section 6.7).

Four claims that might appear to follow do not. The results do not establish that GCA labels are factually more accurate, since no independent annotation was collected and AlignScore supplies both sets of labels (Section 7.8). They do not establish that a summarization model trained on GCA preferences produces more faithful summaries, since the final study contains no policy-optimization stage. They do not establish that GCA is generally preferable, since Chapter 6 reports configurations and scales at which it is not. And they do not establish a strong effect in absolute terms, since both reward models remain near chance (Section 7.4). What the study contributes is evidence about *when* feedback granularity changes the structure of a preference-learning signal, and the remainder of this chapter examines the conditions under which it does.

## 7.2. Why Granularity Interacts With Evaluator Design

**This section’s account is contradicted by the confirmatory data, not merely unconfirmed**

The account originally built in this section, that GCA’s advantage specifically requires the `nli` evaluation mode because `nli` lacks the internal sentence decomposition that `nli_sp` performs, was developed from the single-run mode sweep in Section 6.5, before Chapter 5, Section 5.4 found that the twenty-run confirmatory campaign at  $n = 1,000$ , the actual headline result of this thesis, was trained under `nli_sp` throughout, not `nli`. On the account below, a robust GCA advantage under `nli_sp` should not occur. One did: +3.95 percentage points, GCA ahead in all twenty runs,  $p < 0.001$ . The account is therefore not an untested hypothesis so much as a hypothesis this thesis’s own confirmatory data speaks against. It is retained in this section, with that contradiction stated plainly throughout, because the reasoning about how AlignScore’s modes work is independently correct and documents a real empirical check, and because removing it would obscure how the interpretation of this thesis’s results changed once the discrepancy was found.

The advantage reported in Chapter 6, Section 6.6 was originally thought not to be a property of GCA in the abstract, but to appear only under a specific combination of two design choices, simple-mean aggregation ( $\alpha = 0.0$ ) and the AlignScore `nli` evaluation mode; Section 6.4 had already shown that the aggregation-formula change alone was not sufficient, since holistic still led at  $\alpha = 0.0$  under `nli_sp` in that single exploratory run. The remainder of this section reconstructs why the judge mode was thought to make the difference, before addressing, at the end, why the confirmatory data no longer support that account.

AlignScore’s implementation [41] suggests a plausible mechanism, though the present experiments do not test it directly. The `_sp` suffix in `nli_sp` enables a preprocessing and aggregation step that is internal to AlignScore itself: the context is split into roughly 350-token chunks and the claim is split into sentences, and the final score is computed as, in the library’s own `inference_per_example` function, the maximum entailment score across context chunks for each claim sentence, averaged across claim sentences (`.max(dim=0).values.mean()`). Plain `nli` mode skips this entirely and scores the full context against the full claim in one pass. The chunk-then-aggregate pattern is not unique to AlignScore: SummaC applies the same idea, segmenting document and summary and reducing a matrix of sentence-level entailment scores to a single figure [21]. What differs between evaluators is where that decomposition happens, which is precisely the distinction this section turns on.

This has a direct consequence for what the holistic and GCA conditions actually compute under each mode. Under `nli_sp`, a holistic call already decomposes the candidate summary into sentences internally, scores each one against the best-matching chunk of the article, and averages the results, which is functionally close to what GCA does *externally* at  $\alpha = 0.0$ : score each summary sentence against the article and take the mean.

When GCA is applied on top of a judge that is already doing something close to sentence-level decomposition and mean-pooling internally, its external decomposition adds little genuinely new information, which is consistent with holistic and GCA performing similarly, and even favoring holistic, under `nli_sp` (Table 6.4). Under plain `nli`, by contrast, a holistic call scores the entire summary against the (length-truncated) article as one atomic unit, with no internal sentence decomposition at all. Here, GCA’s external per-sentence scoring is the only source of sentence-level localization in the pipeline, and it is exactly under this mode that GCA shows a measurable advantage.

The stored  $n = 200$  preference data allow one component of this account to be checked directly rather than inferred. If a holistic `nli_sp` call internally splits the claim into sentences and averages, then the holistic score of a summary should closely track the mean of the separately computed per-sentence scores of that same summary. Across all 400 summaries in that set the two quantities correlate at  $r = 0.983$ , agree to within 0.001 for 234 of 400 summaries, and have a median absolute difference of 0.000 (Figure 7.1). Under `nli_sp`, holistic scoring and mean-aggregated sentence scoring are therefore close to the same computation, which is consistent with the observation that GCA at  $\alpha = 0.0$  adds little under that mode.

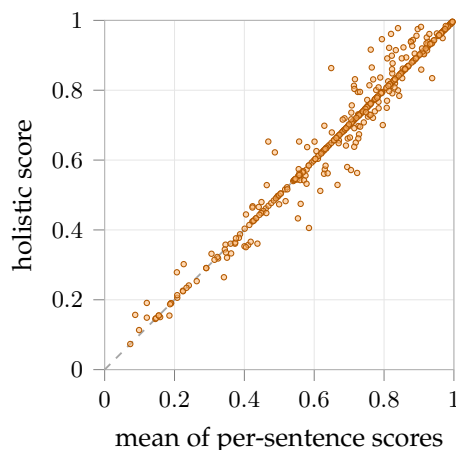


Figure 7.1.: Holistic AlignScore against the mean of the separately computed per-sentence scores, for all 400 candidate summaries in the  $n = 200$  preference set under evaluator mode `nli_sp`. The dashed line is  $y = x$ . The two quantities are near-identical ( $r = 0.983$ ), which is what the split mode’s internal sentence-averaging predicts. The corresponding files for the final `nli` configuration are not retained in the repository, so the same check could not be run for the unsplit mode.

This measurement supports the part of the account concerning `nli_sp`’s internal behaviour considered on its own, but the account as a whole was always an interpretation of an exploratory sweep, and even on its own terms three limitations constrain it. The mode comparison rests on a single run per mode; no ablation isolates the components

of the `_sp` preprocessing, such as chunk size, from the mode change as a whole; and nothing verifies empirically that `nli_sp`'s internal aggregation and GCA's external aggregation produce comparable scores rather than merely comparable rankings.

A fourth point is decisive rather than merely limiting. The claim that “only the selected `nli` configuration received repeated-run validation” is what Chapter 5, Section 5.4 found to be false: the twenty-run campaign believed to be that validation was actually run under `nli_sp`, and it reproduced a GCA advantage of similar direction and larger magnitude than the single `nli` run that motivated locking the mode in the first place (+3.95 points over twenty runs under `nli_sp`, against +6.0 points from a single `nli` run and +1.3 points from its confirmation rerun). Read alongside the six-run campaign, which genuinely was trained under `nli` (Section 6.6) and also showed a consistent GCA advantage (+3.08 points, five of six runs), the confirmatory evidence available across both modes now points the same way: GCA leads at  $n = 1,000$  under `nli` *and* under `nli_sp` once each is tested with enough runs to distinguish a real effect from noise. That is the opposite of what this section's account predicts, which requires the advantage to appear under `nli` and not under `nli_sp`. The single-run `nli_sp` result that seemed to confirm the account (−2.1 points, Table 6.4) is, on this reading, another instance of the single-run instability documented directly in Section 7.9, not a stable property of the mode.

The proposition that evaluator-internal decomposition reduces the marginal benefit of external GCA is therefore not a finding of this thesis and is not, on the evidence now available, a well-supported hypothesis either. It is retained above as an account of AlignScore's internal behaviour, which the  $r = 0.983$  correlation genuinely establishes, and as a candidate explanation that a future, correctly-configured repetition of these campaigns could still test properly; what it cannot currently explain is why the confirmatory campaign that was believed to isolate this factor shows the same effect the factor was invoked to explain away (Chapter 8, Section 8.5).

### 7.3. How Much Weight the Statistical Evidence Can Carry

The  $n = 1,000$  result was first assessed from a fold-level interval and a Wilcoxon  $p$ -value of 0.0034, computed over 30 pooled fold-level differences from six runs. It is necessary to state plainly how much that figure could support, and then to describe what changed once the campaign was extended.

Those 30 differences are not 30 independent observations. Within a single run the five cross-validation folds partition one dataset: any two folds share four fifths of their training data, and all five evaluate models trained from the same pool of preference pairs. Their errors are correlated, so a test assuming independence treats the evidence as stronger than it is. The effective sample size lies between the number of runs and

the number of folds, and closer to the former. The fold-level  $p$ -value is therefore anti-conservative and is reported as an exploratory summary, not as a hypothesis test, at either campaign size.

The appropriate unit of independent replication is the run. Applying the two-sided Wilcoxon signed-rank test to the six run-level differences from the original campaign (+0.060, +0.013, -0.010, +0.051, +0.032, +0.039) gives  $p = 0.0625$ , which does not meet the conventional 0.05 threshold. With six observations, five of them positive, 0.0625 is in fact the smallest two-sided value the test can return, so that outcome reflected the number of runs available at least as much as the size of the effect: no result with five wins and one loss out of six paired observations can reach significance, regardless of how large or small the true difference is.

This is a genuine limitation of a six-run design, not a property of the underlying effect, and Chapter 6, Section 6.7 reports what happened when it was addressed directly: the campaign was extended to twenty independently seeded runs under a corrected, fully deterministic training procedure. GCA led in all twenty, with a mean gap of +3.95 percentage points, a bootstrap 95% confidence interval of [+3.19, +4.70] points that does not cross zero, and a run-level two-sided Wilcoxon  $p < 0.001$ . This does meet the conventional significance threshold, and the effect size is close to the original six-run estimate, indicating that the earlier campaign's direction was not an artifact of having too few runs to trust, only too few to prove.

Significance alone understates how consistent the difference is. Measured against the run-to-run spread it must be distinguished from, the effect is large: Cohen's  $d = +2.22$ , and the rank-biserial correlation attains its maximum of +1.00 because no run in the campaign favours holistic (Chapter 6, Section 6.9). The  $p$ -value and the effect size are answering different questions here, and both are needed. What is large is the difference relative to the noise in the measurement; what remains small, as the next section discusses, is the difference in absolute terms.

Reaching significance changes what can be claimed, but it does not remove the need for the qualifications that surrounded the six-run result. Statistical significance at the run level establishes that GCA preferences are *consistently more learnable* than holistic preferences by this reward-model architecture, under this evaluator configuration, at this dataset size, to a degree unlikely to arise from run-to-run noise alone. It does not by itself establish a large effect (Section 7.4 addresses the absolute accuracy level), an effect that generalizes to other scales (Section 7.5), or a mechanism (Section 7.2 remains an interpretation, not confirmed by the extended campaign). What the evidence now supports is a directionally consistent, statistically significant advantage of approximately three to four percentage points under this configuration, a materially stronger claim than the six-run campaign could support, but one that is still scoped to the conditions under which it was measured.

## 7.4. Both Reward Models Are Weak in Absolute Terms

### A comparison can be sound while both conditions are weak

At  $n = 1,000$ , the holistic reward model reaches 0.5297 and the GCA reward model 0.5691 pairwise accuracy, against a chance level of 0.5. In absolute terms these models recover very little of the preference signal: the better of the two is correct on roughly 57 of every 100 held-out pairs, where guessing would be correct on 50. The comparison between conditions remains valid; what it is a comparison *of* is modest in absolute terms.

This materially qualifies what the thesis shows. The finding is that GCA preferences are *more learnable* than holistic preferences under the selected configuration, not that either produces a reward model one would want to deploy. A reward model at 0.56 accuracy would provide a very noisy training signal to any downstream policy, and the significant +3.95 percentage-point advantage established in the extended campaign (Chapter 6, Section 6.7) is an improvement to a weak signal rather than the difference between a weak and a strong one.

Several factors plausibly contribute to the low absolute numbers, and this thesis can separate them only partially. The task itself is genuinely hard: both candidate summaries in a pair are generated by the same model from the same article and differ only in sampling temperature, so many pairs are near-ties in factual quality, and a judge's preference between them may encode little learnable structure. The RoBERTa-base reward model [28] is also small, and articles are truncated to 2,000 characters before tokenization (Chapter 5, Section 5.5), so the model may simply not see enough of the source to verify many claims. The accuracies at  $n = 5,000$  and  $n = 10,000$  are higher for both conditions (0.58–0.59), which is consistent with more training data helping the reward model in general, even as the gap between conditions closes.

A preliminary ablation tested the truncation explanation directly by varying the retained article length under the RoBERTa-base backbone, three seeds per setting (Table 7.1). The result runs against the truncation hypothesis as originally stated: accuracy was *higher* with less retained context (500 and 1,000 characters) than at the thesis's own 2,000-character setting, for both conditions. A plausible reading is that 2,000 characters is not actually more informative in practice, because the concatenation of article prefix and summary already exceeds the 512-token cap at that length (Chapter 5, Section 5.5), so how much of the 2,000-character budget survives tokenization varies unpredictably with summary length across examples. A shorter, more consistently-applied budget may therefore produce a more uniform input distribution even though it discards more raw text. This ablation does not yet isolate that explanation from a simple backbone effect, since it was run only under RoBERTa-base; the two prepared but unexecuted arms of the same ablation, under `deberta-v3-base` at 512 and 1,024 tokens, would hold the backbone fixed while varying context length and are recorded as future work (Chapter 8). Taken as it stands, this result reframes rather than resolves the truncation

question: raw article length is not the limiting factor it was hypothesized to be, at least not in the straightforward way of "more characters, more evidence, higher accuracy," and the low absolute ceiling remains only partially explained.

Table 7.1.: Truncation ablation, RoBERTa-base arms (three seeds each). All three rows use the 512-token cap; `max_article_chars` is the character budget before tokenization. The `deberta-v3-base` arms (512 and 1,024 tokens) were prepared but not completed; see Chapter 8.

| Max article chars | Holistic | GCA    | Gap (GCA – Holistic, pp) |
|-------------------|----------|--------|--------------------------|
| 500               | 0.5553   | 0.5850 | +2.97                    |
| 1,000             | 0.5663   | 0.5903 | +2.40                    |
| 2,000             | 0.5290   | 0.5480 | +1.90 <sup>1</sup>       |

## 7.5. Interpreting the Larger-Scale Results

The  $n = 1,000$  advantage was initially not reproduced with comparable magnitude in single runs at  $n = 5,000$  and  $n = 10,000$  (Chapter 6, Section 6.8), and this section originally posed a question it could not answer: does that movement reflect a genuine dataset-size effect, or is it the same run-to-run variation already documented at  $n = 1,000$ , simply unmeasured at the larger sizes because only one run was performed at each? The distinguishing test was stated explicitly at the time: if the scale effect is statistical noise, repeated runs at  $n = 5,000$  should produce a confidence interval overlapping the  $n = 1,000$  estimate; if it reflects a real, scale-dependent change, they should not. Twenty-run campaigns at both  $n = 5,000$  and  $n = 10,000$ , using the same corrected, deterministic training procedure as the  $n = 1,000$  extension and matching its resolution exactly, now answer that question.

The result is unambiguous. At  $n = 5,000$ , the mean gap across twenty runs is  $-0.22$  percentage points (95% CI  $[-0.66, +0.21]$ pp, Wilcoxon  $p = 0.3410$ ); at  $n = 10,000$ ,  $-0.11$  percentage points (95% CI  $[-0.39, +0.18]$ pp,  $p = 0.5882$ ). Both intervals are centred almost exactly on zero and are indistinguishable from a null result. Neither overlaps the  $n = 1,000$  interval,  $[+3.19, +4.70]$ pp, and the margin is not close: the nearest edges are separated by more than two and a half percentage points at both scales. This

<sup>1</sup>The three seeds used here (1, 2, 3) are a subset of the twenty used in the headline campaign (Chapter 6, Section 6.7), not an independent draw. The 2,000-character row's holistic figure, 0.5290, matches the mean of seeds 1–3 in that campaign (0.5293) closely, as expected under the same configuration and seeds. Its GCA figure, 0.5480, is 1.7 percentage points below the corresponding three-seed mean there (0.5650); this size of gap is within the run-to-run spread already documented for GCA at  $n = 1,000$  (individual seeds range from 0.543 to 0.592, Table 6.8), and is expected when averaging three seeds rather than twenty. It should therefore be read as sampling noise from a small seed count, not as a discrepancy between this ablation and the headline result it is compared against.



is the pattern the genuine-scale-effect explanation predicted and the statistical-noise explanation did not.

One further step is required before the language of absence is justified, because a non-significant test result is compatible both with a negligible effect and with an effect the design was too small to see. The distinction is settled by the equivalence tests reported in Chapter 6, Section 6.9, which reverse the burden of proof and ask whether effects above a given size can be *excluded*. They can: at  $n = 5,000$  any true effect larger than  $\pm 0.61$  percentage points is ruled out at the 5% level, and at  $n = 10,000$  any effect larger than  $\pm 0.36$  points. The design was also amply powered for the effect it was testing for, detecting 0.64 and 0.42 points respectively at 80% power, against the 3.95-point effect measured at  $n = 1,000$ . The GCA advantage observed at  $n = 1,000$  is therefore not an artifact that would have appeared at any scale if only enough runs had been collected, nor an effect that persists below the resolution of these experiments; it is specific to the smaller dataset size, and it disappears, not merely shrinks, once training data becomes more plentiful.

This resolves what could previously only be raised as a hypothesis. Because this thesis holds candidate pool, evaluator, and reward-model architecture fixed while varying only preference-construction granularity and dataset size, the mechanism proposed earlier remains the most natural reading: at  $n = 1,000$  the reward model is fitting a small dataset, and how cleanly the preference relation is structured matters a great deal, since a labeling that is internally more consistent is easier to recover from few examples. As the dataset grows, the model sees more examples of the same underlying relation, and additional data can compensate for a noisier labeling, so the benefit of a cleaner preference construction shrinks toward the point of being undetectable. Consistent with this account, absolute accuracy rises for *both* conditions from  $n = 1,000$  to the larger scales (from 0.5297 and 0.5691 to roughly 0.582–0.586 at both  $n = 5,000$  and  $n = 10,000$ ), exactly as the two conditions converge. What the repeated-run evidence newly establishes is the *shape* of this pattern, a real effect at small scale that is absent at larger scale rather than merely a shrinking one, not the mechanism behind it: nothing in this design isolates article length, sentence count, or segmentation quality as an independent variable, so the account above remains the most plausible explanation rather than a demonstrated one.

One piece of direct, rather than inferred, support for it comes from how often holistic and GCA actually disagree (Chapter 6, Section 6.10.5). On any pair where the two conditions reach the same decision, both reward models train on an identical (article, chosen, rejected) triple, so the disagreeing pairs are what any gap between conditions has to come from. That fraction falls from roughly 18% of pairs at  $n = 1,000$  to 2–3% at  $n = 5,000$  and  $n = 10,000$ , under the configuration these campaigns were actually trained on. This does not by itself prove the convergence account above, since it says only how much room the two conditions have to differ, not why GCA uses that room better at small  $n$ , but it is independent, measurable evidence that the room genuinely

shrinks in step with the vanishing gap, rather than an assumption required to make the account work.

Read this way, the larger-scale results are a substantive finding rather than an unresolved caveat. GCA provides a confirmed learnability advantage when preference data are scarce, and no detectable advantage once enough data are available for holistic preferences to be recoverable on their own terms. This is also the regime in which automatic preference construction is most attractive in practice, since data scarcity is precisely where collecting human labels would be most burdensome; the result therefore bounds GCA’s practical usefulness to a specific and identifiable regime rather than leaving that boundary unmeasured, as the original single-run design had.

## 7.6. The Ground-Truth Reversal: A Stronger and Stranger Pattern

The account just given, that additional training data let holistic preferences “catch up” to a cleaner but data-hungry GCA signal, predicts *convergence*: as  $n$  grows, the gap should shrink toward zero and stay there, because both conditions are approaching the same ceiling from below. That prediction holds for learnability (Section 7.5). It does not hold for RQ4. Against the independent, held-out ground truth (Chapter 4, Section 4.8), GCA is significantly more precise than holistic at  $n = 1,000$  on comparisons with a genuine quality gap, statistically indistinguishable at  $n = 5,000$  (consistent with convergence so far), and then significantly *less* precise than holistic at  $n = 10,000$  on three of four ground-truth constructions (Chapter 6, Section 6.10.2). A convergence account has nothing to say about why the line should keep moving past the point where the two conditions meet. Something in addition to “holistic catches up” is needed to explain why GCA falls *behind*, and this section considers what that might be and what can currently be ruled out.

One clarification narrows the target before any mechanism is proposed. The reversal is not GCA deteriorating. The absolute success rates (Chapter 6, Table 6.15) show GCA improving with additional training data on every construction except the same-article one, from 86.00% to 90.00% on top/bottom 10% between  $n = 1,000$  and  $n = 10,000$ . Holistic improves over the same interval from 76.75% to 91.95%. Both conditions therefore benefit from more data, as the convergence account expects; what the account does not predict is that they benefit at such different rates, holistic gaining roughly four times as much as GCA. Any adequate explanation has to account for GCA’s flatter returns to scale rather than for a decline that does not occur, and this is the form in which the two candidate mechanisms below were posed.

One candidate was tested directly rather than left as speculation. If GCA’s uniformly margin-0 training pairs contain a growing share of near-zero-margin, effectively noise-level comparisons relative to holistic’s, a reward model trained on more of them at

larger  $n$  might be fitting that noise rather than genuine signal, which would predict exactly this pattern: fine at small  $n$ , where there is not enough data to fit the noise precisely, and increasingly harmful as  $n$  grows. Checked against the stored per-pair score gaps in the actual training data, this does not hold: the near-tie fraction is statistically indistinguishable between the two conditions at every scale, and is if anything *higher* for GCA specifically at  $n = 1,000$ , where GCA leads (Chapter 6, Section 6.10.4). Ruling this out is informative on its own terms: it means the reversal is not simply an artefact of an unfiltered margin setting that happened to interact badly with GCA’s aggregation, but it leaves the actual mechanism open.

A second candidate follows the same logic as the temperature-confound check in Section 7.10, but asks a different question of it. That section shows that summary length and sentence count do not carry the *training label*: a classifier predicting the evaluator’s decision from length or sentence count alone performs at chance (Chapter 4, Table 4.4). This does not establish that a *trained reward model* cannot come to rely on length or sentence count as a shortcut regardless of whether the label technically depends on it: a model can learn a spurious but locally predictive correlate of its training labels even when that correlate does not determine the labels outright, and it can lean on that correlate more heavily the more training examples reinforce it [12]. This is a different empirical question from the one Section 7.10 already answers, because it is about what the trained model does at inference time on unseen ground-truth summaries, not about what the label construction encodes. Unlike the mode-interaction hypothesis of Section 7.2, this one required no new training to test: every RQ4 checkpoint was already on disk, so it could be scored directly against the held-out ground-truth summaries and its predicted reward correlated with each summary’s length and sentence count (Chapter 6, Section 6.10.6). The hypothesis predicts the GCA-minus-holistic gap in this correlation should be largest at  $n = 10,000$ , where GCA loses; instead the gap peaks at  $n = 3,000$  and nearly vanishes at  $n = 10,000$ , where the two conditions converge to almost identical length-reliance. This candidate is therefore also rejected. What does emerge is a symmetric pattern that affects both conditions rather than distinguishing them: both grow markedly more correlated with summary length as  $n$  increases, a five-fold increase in magnitude from  $n = 2,000$  to  $n = 10,000$  for each. That pattern cannot explain the reversal, since it does not differentiate GCA from holistic, but it is a plausible contributor to the low absolute accuracy ceiling neither condition breaks (Section 7.4). With both mechanical candidates ruled out, the mechanism behind the reversal remains open and is recorded as future work (Chapter 8, Section 8.5).

Whatever the mechanism, the practical implication is stronger than the learnability result alone would suggest. Section 7.5’s closing statement, that GCA helps when data are scarce and does no harm once data are plentiful, has to be revised. Against the ground truth, GCA does not merely stop helping at scale; it becomes the worse choice. The regime in which GCA is advisable is narrower than the learnability result on its own implies: small training sets *and* comparisons with a genuine quality gap between the two options being ranked, not large training sets, and not the closest, most ambiguous

comparisons even at small  $n$  (Chapter 6, Table 6.14, same-article row).

## 7.7. Implications for RLAIF Preference Construction

Three implications follow for anyone constructing AI-generated preference data from a fixed factuality judge, independent of whether the specific GCA method studied here is adopted.

First, the aggregation formula used to combine sentence-level scores into a preference label is not a minor implementation detail. The original weakest-link-penalty formula ( $\alpha = 0.5$ ) actively hurt GCA relative to a plain mean (Chapter 6, Table 6.2), which runs counter to the intuition that motivated it: a formula designed to be more sensitive to localized errors was, in practice, more sensitive to noise than to signal. Aggregation-formula choices of this kind should be validated empirically rather than assumed correct from first principles. This is a qualification worth attaching to the broader case for fine-grained supervision [40]: locating errors more precisely helps only if the rule that converts those locations into a training signal is itself sound.

Second, judge configuration can matter more than the granularity decision that is nominally the point of comparison. Section 7.2’s reading of AlignScore’s mode behavior suggests that part of what looked like a question about holistic-versus-sentence-level supervision was really a question about how much sentence-level decomposition the judge was already doing internally. Anyone comparing supervision granularities on top of a judge with its own internal aggregation behavior should check what that judge already does before attributing an effect to the granularity of their own pipeline. Surveys of automatic judging emphasise sensitivity to prompt and configuration choices [13, 43]; the present result adds that a judge’s internal decomposition can also absorb the very manipulation an experiment intends to study.

Third, a result validated at one dataset scale should not be treated as general without a scale check. This is a caution this thesis can make credibly only because Section 7.5’s robustness check was actually run rather than assumed unnecessary; had the study stopped at the  $n = 1,000$  result, it would have reported a materially less qualified conclusion than the evidence supports.

## 7.8. Judge Reliability Without Human Auditing

No independent human validation of the factuality labels was collected in this study, and it is worth stating precisely what this does and does not leave open. The twenty repeated runs in Chapter 6, Section 6.7 establish that the difference between the two preference sets is repeatedly observed: it survives changes in fold assignment and appears in every run of the campaign, so it is not confined to a single favourable run.

Reproducibility and correctness are separate properties, however. That a preference relation is consistently recoverable says nothing about whether the underlying AlignScore judgments agree with what a human reader would consider faithful to the source. A judge can produce labels that are highly reproducible and systematically wrong, and nothing in this study would detect that case. The labels function here as the reference standard, not as verified ground truth, and every result is conditional on them in that sense. Establishing agreement with human factuality judgments requires annotation this study did not collect, quantified against a chance-corrected agreement measure such as Cohen’s  $\kappa$  [6], and the question is recorded as an open limitation in Chapter 8. The concern is not peculiar to this study: reviews of reinforcement learning from human feedback identify the quality of the preference data itself, and the difficulty of auditing it, as a central open problem [3], and substituting an automatic judge for human annotators relocates that problem rather than removing it.

## 7.9. Stochastic Variation Between Runs

Run-to-run variation is large enough in this setting to determine what conclusions can be drawn, and it deserves treatment as a finding rather than as an inconvenience. Two same-configuration runs differed by 4.7 percentage points (+0.060 versus +0.013), and across the six runs the gap ranged from  $-0.010$  to  $+0.060$ . A study reporting only one of these runs could honestly have concluded either that GCA helps substantially or that it does not help at all.

This is a known hazard rather than a peculiarity of the present setup. Fine-tuning pretrained language models is sensitive to random seed alone, with weight initialization and data order contributing comparably to the variance of out-of-sample performance [8], and reporting score distributions across several runs rather than a single figure has been urged for precisely this reason [34]. The repeated-run design used here follows that recommendation, and the spread it exposes is the reason this thesis reports a distribution of gaps rather than a single headline number.

This variation traced back to an implementation property documented in Chapter 5, Section 5.5: in the original six-run campaign, the seed argument fixed the cross-validation fold assignment through Python’s standard random number generator, but PyTorch’s generator was not seeded in the cross-validation path. Several sources of training stochasticity were therefore uncontrolled, including reward-head weight initialization, the order in which training batches are drawn, and dropout masks. Runs were consequently not exactly reproducible from a seed value alone. This had an interpretive consequence at the time: the two executions at seed 42 were genuine repetitions rather than duplicates, because they shared the fold assignment but not the training stochasticity, and the 4.7-point spread between them reflects that uncontrolled stochasticity rather than fold assignment alone.

This property has since been fixed: PyTorch and CUDA are now seeded per fold, with an explicit generator for the training `DataLoader` and seeded worker initialization, so a run is fully determined by its seed on fixed hardware. The extended twenty-run campaign in Chapter 6, Section 6.7 was carried out under this corrected procedure; each of its twenty seeds is consequently a genuinely independent and exactly reproducible sample, which is what made it meaningful to extend the campaign at all. The fix does not retroactively change the original six runs, which remain a documented instance of exactly the seed-sensitivity the literature describes, but it does mean the concern raised in this section no longer applies to the results this thesis reports as its headline finding.

The practical implication generalizes beyond this study regardless of the fix. At the accuracy levels and dataset sizes involved here, differences of one to two percentage points between conditions are within the range that a single under-seeded run can produce by chance. Comparisons of preference-construction methods reported from single training runs at this scale, without controlling every source of training randomness, should be treated with corresponding caution.

## 7.10. Confounds That Remain

Two confounds could not be removed within this design, and both plausibly affect the reported numbers.

The candidate pairs differ in decoding temperature, and Chapter 4, Table 4.3 shows the consequence is not small: the lower-temperature candidate is preferred in 68.5% of holistic pairs and 63.0% of GCA pairs on the data where this could be measured. The asymmetry also differs between conditions by roughly five percentage points, so the two preference sets are not equivalent with respect to this confound.

How much this threatens the reported comparison depends on whether the asymmetry is *learnable* from the summary text, and that question has now been examined rather than left open. The reward model never observes decoding temperature or candidate position; it sees only (article, summary) pairs (Chapter 5, Section 5.5). For the temperature skew to inflate its accuracy, some textual correlate of temperature would have to be recoverable from the summary. The surface-feature baselines in Chapter 4, Table 4.4 show that the two most obvious such correlates do not qualify: classifiers deciding purely from summary length or sentence count reach 0.465–0.535 accuracy against the judge’s own decisions, which is chance, and the mean length difference between chosen and rejected summaries is 2 characters under holistic scoring and –6 under GCA. Whatever the reward models are fitting, it is not summary length.

This weakens the confound but does not dispose of it. Lexical and stylistic correlates of temperature beyond length, such as word-frequency profiles, hedging, or named-entity density, were not tested, and a reward model with a pretrained encoder is well placed to pick up exactly that kind of regularity. What can be said is that the simple and most

frequently raised version of the objection, that the models are covertly learning to prefer longer or shorter summaries, is empirically excluded. Sampling both candidates at the same temperature would remove the confound at source and remains the cleanest correction available.

The reward model also sees only a truncated prefix of each source article (Chapter 5, Section 5.5), while subset selection admits articles several times longer than the retained prefix. Where the evidence bearing on a claim lies beyond that prefix, the corresponding preference is not recoverable from the model’s input even in principle. This is a plausible contributor to the low absolute accuracies discussed in Section 7.4, though its magnitude was not measured, and it should not be assumed to be the dominant cause without the ablation described there.

## 7.11. Generalizability

Every result reported here is obtained under one combination of choices: CNN/DailyMail as the corpus, AlignScore as the evaluator, Mistral-7B-Instruct-v0.3 as the candidate generator, and a RoBERTa-base Bradley–Terry model as the reward model. Each of these bounds the conclusions in a distinct way.

The evaluator is the most consequential. Section 7.2 raises the possibility that whether external decomposition helps is related to whether the evaluator already decomposes internally, which would make the finding partly *about* the evaluator’s architecture rather than simply conditional on AlignScore as a fixed instrument. If that reading is correct, a different factuality model with different internal aggregation would shift the result. The corpus matters because CNN/DailyMail summaries are short, extractive-leaning, and drawn from a single register; domains with longer or more abstractive summaries would change both the number of sentences per summary and the difficulty of the underlying factuality judgments. The reward-model backbone matters because a stronger encoder, or one with a longer context window, might recover more of the preference relation and thereby narrow or widen the difference between conditions.

Taken together, these dependencies mean the study is best read as characterizing behaviour under one specific configuration, and as motivating a hypothesis about the interaction between external and evaluator-internal decomposition, rather than as establishing a quantity that transfers directly to other setups.

## 8. Conclusion

### 8.1. Summary

This thesis asked a controlled question: does converting sentence-level factuality scores into a summary-level preference through Granular Credit Assignment produce a more learnable reward-model signal than scoring each candidate summary holistically. The two conditions were compared under an identical candidate pool, an identical factuality judge, and an identical Bradley–Terry training recipe, so that the only manipulated factor was the preference-construction procedure itself.

Averaged over an extended campaign of twenty independently seeded runs at  $n = 1,000$  (Chapter 6, Section 6.7), GCA preferences yield an advantage of approximately +3.95 percentage points in reward-model pairwise accuracy, favouring GCA in all twenty runs, with a bootstrap 95% confidence interval of  $[+3.19, +4.70]$  points and a run-level Wilcoxon  $p < 0.001$ . An earlier six-run campaign had given a consistent but structurally underpowered estimate of +3.08 points that could not, with only six paired observations, reach the conventional significance threshold; the extension confirms that direction rather than revealing a different effect. The advantage does not persist as the dataset grows: twenty-run campaigns at  $n = 5,000$  and  $n = 10,000$ , using the same corrected procedure and matching the  $n = 1,000$  campaign’s resolution exactly, give mean gaps of  $-0.22$  and  $-0.11$  percentage points respectively, both statistically indistinguishable from zero (95% CIs  $[-0.66, +0.21]$  and  $[-0.39, +0.18]$  points), with neither interval overlapping the significant  $n = 1,000$  result. Absolute accuracy improves for both conditions as the dataset grows even as their difference vanishes. Uncertainty estimates computed over cross-validation folds remain anti-conservative, because folds within a run are not independent, but the run-level estimates at all three scales are now each supported by a properly powered test.

The experiments therefore do not support a universal claim that GCA is preferable to holistic factuality supervision. They show that sentence-level aggregation produces a statistically significant, repeatedly confirmed improvement in the learnability of automatically generated pairwise preferences specifically when training data are scarce, and no detectable improvement once training data become more plentiful. In the exploratory mode comparison, the advantage appears under evaluator modes that do not themselves decompose the input, which motivates the hypothesis that external and evaluator-internal decomposition may act partly as substitutes. Taken together, the results indicate that the usefulness of granular credit assignment is bounded to a specific,



data-scarce regime, and depends within that regime on evaluator configuration and aggregation choice, rather than following from finer granularity as such.

A fourth question (RQ4, Chapter 4, Section 4.1) asked whether this learnability advantage also translates into more accurate reward-model judgments against an independent, held-out ground truth of 500 articles disjoint from every training pool used in this thesis, scored by the same locked evaluator but never used to train any reward model (Section 4.8). It does, but only in the same narrow, data-scarce regime, and the pattern is more pronounced than for learnability: at  $n = 1,000$ , GCA is significantly more precise than holistic on comparisons with a genuine quality gap (up to +9.25 percentage points,  $p < 0.001$ ), at  $n = 5,000$  the two are indistinguishable, and at  $n = 10,000$  holistic is significantly *more* precise than GCA on three of four ground-truth constructions (Chapter 6, Section 6.10). The advantage does not merely fail to persist as training data grow, as the learnability result also showed; it reverses. Five-seed checks at  $n = 2,000$  and  $n = 3,000$  show the decline passing through a null region around  $n = 3,000$ – $5,000$ , in which no gap is distinguishable from zero in either direction. Two candidate mechanisms were tested directly and both rejected. That GCA’s uniformly unfiltered training pairs contain disproportionately more near-zero-margin comparisons than holistic’s does not hold: the near-tie fraction is statistically indistinguishable between the two conditions at every scale, and is if anything *higher* for GCA specifically at  $n = 1,000$ , where GCA leads (Section 6.10.4). That GCA increasingly relies on a summary-length or sentence-count shortcut relative to holistic as training data grow also does not hold: scoring the held-out ground truth with the already-trained checkpoints shows the GCA-minus-holistic gap in this correlation peaks at  $n = 3,000$  and nearly vanishes at  $n = 10,000$ , the opposite of what the hypothesis predicts (Section 6.10.6). The mechanism behind the reversal remains open and is discussed as future work (Section 8.5).

## 8.2. Answers to the Research Questions

**RQ1** (does GCA produce a more learnable preference signal than holistic scoring?) is answered *yes at small scale, no at larger scale, both with statistical support*. An average advantage of +3.95 percentage points is observed across twenty independent runs at  $n = 1,000$ , favouring GCA in all twenty, with a run-level Wilcoxon  $p < 0.001$ , a bootstrap confidence interval that does not cross zero, and a large standardised effect ( $d = +2.22$ ). Twenty-run campaigns at  $n = 5,000$  and  $n = 10,000$ , the same resolution as  $n = 1,000$ , give mean gaps statistically indistinguishable from zero ( $p = 0.3410$  and  $p = 0.5882$ ), with confidence intervals that do not overlap the  $n = 1,000$  result; equivalence tests at those scales additionally exclude any effect larger than  $\pm 0.61$  and  $\pm 0.36$  percentage points (Section 6.9), so the absence is established positively rather than inferred from a failure to reject. The answer is therefore not merely configuration-dependent in an unresolved sense: it is a confirmed advantage at one dataset scale and a confirmed

absence of advantage at two larger ones, with both conclusions resting on repeated, adequately powered evidence rather than single runs.

**RQ2** (how do evaluator configuration and aggregation strategy affect relative learnability?) is answered *partially, and less conclusively than this thesis originally believed*. The aggregation formula clearly matters: the original weakest-link penalty measurably hurt GCA relative to the simple mean (Chapter 6, Section 6.3). Evaluator mode was believed to matter in the same way, based on an exploratory single-run sweep in which the two modes performing internal decomposition showed no GCA advantage. That belief does not survive the discovery, during final review, that the confirmatory twenty-run campaign was itself trained under one of those two modes (`nli_sp`, Chapter 5, Section 5.4) and nonetheless reproduced a GCA advantage as large as the one originally attributed to the other mode. Taken together with the six-run campaign, which genuinely used the other mode and also favoured GCA, the confirmatory evidence now available points against evaluator mode being the deciding factor, which is the opposite of what the exploratory sweep suggested (Chapter 7, Section 7.2).

**RQ3** (how stable are the differences across runs and dataset sizes?) is answered *differently, and now conclusively, across the two axes it asks about*. Across repeated runs at a fixed dataset size, the difference is highly stable: all twenty runs of the extended campaign at  $n = 1,000$  favour GCA, with a standard deviation of 1.78 percentage points around a mean of +3.95, and the twenty-run campaigns at  $n = 5,000$  and  $n = 10,000$  are similarly consistent around a mean of approximately zero (SD 1.03 and 0.67 points respectively). Across dataset sizes, stability does not hold, and this is no longer an open question left by single or few runs: twenty repeated runs at each of  $n = 5,000$  and  $n = 10,000$ , the same resolution used at  $n = 1,000$ , confirm a null result whose confidence intervals do not overlap the significant  $n = 1,000$  estimate (Section 7.5). Run-to-run stability at fixed scale is well established at all three sizes tested; stability across scale does not hold, and this absence is now itself a properly powered finding rather than an artefact of insufficient replication.

**RQ4** (does the learnability advantage also hold against an independent, held-out ground truth, and as training data grow?) is answered *yes at  $n = 1,000$ , and no at  $n = 10,000$ , with the direction reversing rather than merely vanishing*. Against a held-out set disjoint from every training pool used in this thesis (Section 4.8), GCA is significantly more precise than holistic at  $n = 1,000$  on comparisons with a genuine quality gap (+9.25 percentage points on the sharpest construction,  $p < 0.001$ ; Chapter 6, Section 6.10.1), statistically indistinguishable from holistic at  $n = 5,000$ , and significantly *less* precise than holistic at  $n = 10,000$  on three of four constructions ( $p = 0.010$ – $0.036$ ; Section 6.10.2). Five-seed checks at  $n = 2,000$  and  $n = 3,000$  show the decline passing through a null region around  $n = 3,000$ – $5,000$  (Section 6.10.3). This answer is stronger, in one sense, than RQ1's: it was obtained against data the trained reward models never saw and never influenced their selection, so it is not subject to the nesting caveat that qualifies RQ1–RQ3 (Section 4.7). It does not, however, remove the evaluator-dependence caveat that qualifies every result in this thesis: the ground-truth labels are still constructed by

AlignScore, the same evaluator used throughout (Section 8.4). Nor is the comparison as neutral as “held out” suggests: the ranking used to score both conditions is built from GCA’s own aggregation rule, so it is in-distribution for GCA and out-of-distribution for holistic in a way that disjoint articles alone do not fix, and the  $n = 1,000$  advantage in particular has not been checked against a holistic-defined criterion (Section 8.4).

### 8.3. Status of the Working Hypotheses

Chapter 4, Section 4.1 stated three working hypotheses that guided the design. Their status against the evidence collected is as follows.

**H1** (sentence-level aggregated preferences are more learnable than holistic preferences, because they localize the factuality signal within a long output) is *supported at  $n = 1,000$  and refuted at larger scale*. The predicted direction holds decisively where training data are scarce, in all twenty runs, with a large standardised effect ( $d = +2.22$ ). It does not hold at  $n = 5,000$  or  $n = 10,000$ , where equivalence tests exclude effects larger than  $\pm 0.61$  and  $\pm 0.36$  percentage points respectively (Chapter 6, Section 6.9). The hypothesis as originally stated was unconditional and is therefore not supported in that form; what survives is a scale-bounded version of it.

**H2** (the evaluator’s configuration affects whether any such difference is observable) is *not supported by the confirmatory evidence, contrary to what this thesis concluded before a final-review discovery*. The exploratory single-run sweep suggested exactly this, with GCA leading only under modes that skip AlignScore’s internal decomposition (Chapter 6, Section 6.5). But the twenty-run campaign that was meant to confirm this, and that this thesis believed had, was actually trained under the mode predicted *not* to show an advantage, and showed one anyway, as large as the single-run result it superseded (Chapter 5, Section 5.4; Chapter 7, Section 7.2). The six-run campaign, run under the other mode, also favoured GCA. H2 is retained here, marked as refuted rather than removed, because the reasoning behind it is not itself wrong and remains worth testing under a correctly configured repetition (Section 8.5); what is refuted is the specific claim that this thesis’s own confirmatory campaigns support it.

**H3** (effects observed at one dataset scale need not persist at another) is *supported, and is the hypothesis the evidence bears on most directly*. It was included as a methodological precaution rather than a substantive prediction, and it proved to be the one that determined the shape of the final result: an effect that is significant, large, and perfectly consistent at  $n = 1,000$  is confirmed absent at both larger scales. Had the study stopped at  $n = 1,000$ , as the original design would have permitted, it would have reported an unqualified positive finding that the fuller evidence does not support.

## 8.4. Limitations

1. **The confirmatory campaigns were not trained on the intended configuration.** Found during final review, after the experimental phase had concluded: the twenty-run campaigns at all three scales, the  $n = 2,000/n = 3,000$  crossover checks, and the RQ4 checkpoints were trained on preference data built under AlignScore’s `nli_sp` mode, not the `nli` mode selected in Chapter 6, Section 6.5 and described elsewhere in this thesis as locked; at  $n = 1,000$  specifically, the aggregation exponent is additionally  $\alpha = 0.5$  rather than the locked  $\alpha = 0.0$ . The comparison between holistic and GCA within each campaign remains fair, since both conditions in a given run were built under the same configuration, and every reported number is a genuine measurement of the run that produced it. What is not sound is this thesis’s account, prior to this discovery, of *why* GCA leads at  $n = 1,000$ : that account attributes the advantage to AlignScore’s `nli` mode specifically, and the confirmatory campaign that was believed to test this was not run under that mode. This is reported in full in Chapter 5, Section 5.4, and its consequences for the evaluator-mode interpretation are addressed in Chapter 7, Section 7.2. It was not corrected by re-running the affected campaigns, both for lack of time before submission and because doing so is new experimental work beyond what had been agreed with the supervisors at this stage of the project; it is recorded as the first item under future work (Section 8.5).
2. **No human factuality audit.** No independent human annotation was collected in the final study, so the evaluator’s judgments were never checked against human assessments of faithfulness (Chapter 7, Section 7.8).
3. **The labels are not independent ground truth.** AlignScore supplies both the holistic and the sentence-level scores, so it functions as the reference standard rather than as a verified one. Every result is conditional on that evaluator. RQ4 (Chapter 6, Section 6.10) evaluates trained reward models against summaries and articles they never saw during training, which removes the nesting concern discussed below, but it still scores those summaries with AlignScore-GCA, so it does not remove evaluator-dependence: a systematic bias in AlignScore itself would still be present in both the training preferences and the RQ4 ground truth alike.
4. **The RQ4 ground-truth criterion is GCA’s own rule, not a neutral one.** The ranking RQ4 scores both reward models against is built from GCA’s aggregation formula (sentence-level AlignScore, mean at  $\alpha = 0.0$ ), the same rule the GCA reward model is trained to predict; the holistic reward model is scored against a criterion built from a rule it never saw (Chapter 4, Section 4.8). The held-out *articles* are genuinely disjoint from every training pool, but the *criterion* is not neutral between the two conditions in the way this thesis originally claimed. Some part of GCA’s RQ4 advantage, including the +9.25 percentage-point result at  $n = 1,000$

(Chapter 6, Section 6.10.1), could in principle reflect this asymmetry rather than a general precision advantage, and the data collected here cannot rule that out.

5. **Preference learnability is not factual correctness.** The dependent variable measures how consistently a preference relation can be recovered by the chosen architecture, not whether the preferences are factually right (Chapter 4, Section 4.4).
6. **RQ4 models are trained on more data than RQ1–RQ3 models.** The RQ1–RQ3 accuracies are computed from fold models trained on 80% of the preference set; the RQ4 checkpoints are trained on the full 100%, since  $k$ -fold cross-validation retains no reusable single model (Chapter 5, Section 5.5; Chapter 6, Section 6.10). Comparing the two sets of results, as Chapter 7, Sections 7.5 and 7.6 do, is therefore a comparison across both a different evaluation target and a different training-data fraction at once. Nothing in this thesis separates the two, so part of the difference between the learnability and ground-truth patterns could in principle be attributable to the training-data fraction rather than to the evaluation target alone.
7. **No downstream policy evaluation.** The final study contains no DPO or other policy-optimization stage, so it does not establish whether models trained on either preference set generate more factually consistent summaries.
8. **Both reward models are weak in absolute terms.** At  $n = 1,000$  the conditions reach 0.5297 and 0.5691 pairwise accuracy against a chance level of 0.5, so the comparison is between two weak signals (Chapter 7, Section 7.4).
9. **Cross-validation folds are not independent.** Uncertainty estimates computed by pooling folds across runs treat correlated observations as independent and are therefore anti-conservative regardless of campaign size; the run-level test is the appropriate unit of replication and is what the reported significance result is based on (Section 7.3).
10. **Incomplete determinism in the original six-run campaign.** In the campaign that first produced the +3.08-point estimate, subset selection and cross-validation fold assignment were explicitly seeded, but PyTorch’s generator was not seeded in the cross-validation path, leaving weight initialization, batch ordering and dropout uncontrolled (Chapter 5, Section 5.5). This was fixed before the extended twenty-run campaign was carried out, so the headline significance result in this thesis is based on fully deterministic, exactly reproducible runs; the fix does not retroactively apply to the original six runs, which remain a documented instance of the underlying problem (Chapter 7, Section 7.9).
11. **Configuration selected on observed results.** The aggregation strategy and evaluator mode were chosen after inspecting  $n = 1,000$  outcomes (Chapter 4, Section 4.6), and the  $n = 5,000/n = 10,000$ /RQ1–RQ3 subsets are nested supersets of that same data (Section 4.7). RQ4 is the one result in this thesis that evaluates the selected configuration on data disjoint from the set it was chosen on (Section 4.8); it was

formulated specifically to close this gap, and does so for the training-data axis, though not for the evaluator-choice axis noted above.

12. **Candidate generation confounds temperature with condition.** The two candidates differ in decoding temperature, and the lower-temperature candidate is preferred substantially more often under both conditions and to differing degrees (Chapter 4, Section 4.10). Summary length and sentence count are excluded as carriers of this information, both performing at chance against the evaluator’s decisions (Table 4.4), but subtler temperature-correlated textual features were not tested and could still carry part of the label information.
13. **Source-document truncation.** The reward model receives only a truncated prefix of each article, so evidence located later in the source is unavailable to it (Chapter 5, Section 5.5).
14. **Evaluator and dataset specificity.** All findings are obtained on CNN/DailyMail with AlignScore, Mistral-7B-Instruct-v0.3 candidates, and a RoBERTa-base reward model, and need not transfer to other domains, evaluators, or architectures.
15. **Sentence segmentation was not evaluated.** Summaries are split with a regex-based splitter whose accuracy was never measured (Chapter 5, Section 5.4).
16. **Nested dataset sizes.** The  $n = 1,000$ ,  $n = 2,000$ ,  $n = 3,000$ ,  $n = 5,000$  and  $n = 10,000$  subsets overlap almost completely with one another (Chapter 4, Section 4.7), so the RQ1–RQ3 runs at each scale are not independent samples of one another. The RQ4 held-out ground truth (Section 4.8) was constructed to be disjoint from all five, so it is not affected by this limitation, but it is itself a single fixed 500-article set reused across every scale, so the five RQ4 scale points share the same evaluation sample rather than being independently sampled test sets.
17. **RQ4’s four constructions are nested, and uncorrected.** The top/bottom 5%, 10%, and 25% constructions are subsets of one another, all drawn from the same ranked pool of 1,000 held-out summaries, and no multiple-comparison correction is applied across the four constructions tested at each scale (Chapter 6, Section 6.10.7). Statements of the form “significant on three of four constructions” should accordingly be read as three overlapping views of one ranking, not as three independent replications.
18. **No error-category analysis.** No analysis links the observed effects to specific categories of factual error; this was not part of the final research questions and is listed under future work.
19. **Unequal replication at the two intermediate scales.** The  $n = 2,000$  and  $n = 3,000$  points used to locate the RQ4 crossover (Section 6.10.3) were each run at five seeds rather than the twenty used at  $n = 1,000$ ,  $n = 5,000$ , and  $n = 10,000$ , to keep the total experiment volume tractable in the time remaining before submission. They locate the crossover to a region rather than a precise value.

20. **The reversal’s mechanism is unexplained.** Two candidate explanations for why GCA’s ground-truth precision reverses at large training-set sizes were tested directly and both rejected: that its training pairs contain more near-zero-margin comparisons than holistic’s (Section 6.10.4), and that it increasingly relies on a summary-length or sentence-count shortcut relative to holistic as training data grow (Section 6.10.6). No mechanism has been confirmed; remaining candidates are discussed under future work.

## 8.5. Future Work

- **Re-run the confirmatory campaigns under the intended configuration.** Chapter 5, Section 5.4 reports that the twenty-run campaigns at all three scales, the  $n = 2,000/n = 3,000$  crossover checks, and the RQ4 checkpoints were trained under AlignScore’s `nli_sp` mode rather than the intended, selected `nli` mode, and, at  $n = 1,000$ , under  $\alpha = 0.5$  rather than  $\alpha = 0.0$ . Repeating these campaigns on the correctly-configured preference data, which already exists at `data/preferences_1000_alpha0_mode_nli/` for  $n = 1,000$  and would need to be rebuilt with the mode flag corrected for  $n = 5,000$  and  $n = 10,000$ , is the most direct way to establish whether the reported effects, and specifically the evaluator-mode interpretation in Chapter 7, Section 7.2, hold under the configuration this thesis intended to test throughout.
- **Separate optimizer steps from dataset size.** Epoch count is fixed at every training-set size, but because `total_steps = len(train_loader) * epochs` (Chapter 6, Section 6.8), the number of gradient updates scales with  $n$ : a model trained at  $n = 10,000$  receives roughly ten times the optimizer steps of one trained at  $n = 1,000$ , not only ten times the distinct data. This thesis does not separate the two. It is a plausible contributor to why absolute accuracy rises with  $n$  for both conditions alike, and, more specifically, a candidate contributor to the RQ4 reversal: if holistic and GCA converge at different rates per optimizer step, for instance because a coarser holistic signal is easier to fit with additional updates than GCA’s finer one, that would be consistent with holistic’s steeper absolute improvement documented in Section 6.10.2. Testing this means training at matched step counts rather than matched epoch counts across scales, which was not attempted here.
- **Remove the temperature confound at source.** Summary length and sentence count have been ruled out as carriers of the preference signal: classifiers using them reach chance accuracy against the evaluator’s own decisions (Chapter 4, Table 4.4). Subtler temperature correlates, such as word-frequency profiles, hedging, or named-entity density, were not tested and remain open. Generating both candidates at a single decoding temperature, and comparing the resulting preference

sets against the current ones, would settle the question for all such features at once rather than one at a time (Section 7.10).

- **Measure the effect of source truncation.** A preliminary ablation varied the retained article length under the RoBERTa-base backbone (500 / 1,000 / 2,000 characters, three seeds each) and found accuracy was, if anything, *higher* with less retained context, the opposite of what the truncation hypothesis predicts. This is plausibly because the 512-token cap already binds at 2,000 characters, so shortening the budget changes how consistently the retained prefix is truncated across examples rather than simply removing evidence; it does not yet isolate whether more context would help under a backbone that is not capped at 512 tokens. Completing the two `deberta-v3-base` arms of the same ablation (512 vs. 1,024 tokens, holding the backbone fixed) would separate that question from the backbone change and was not completed in this thesis due to an unavailable model-download mirror at the time of writing.
- **Test the decomposition-interaction mechanism directly.** Section 7.2 infers from AlignScore’s documented behavior that its split modes perform internally much of what GCA performs externally. Comparing per-sentence scores under split and unsplit modes on the same pairs would test that inference rather than rest on it.
- **Independent validation of the labels.** RQ4 (Chapter 6, Section 6.10) evaluates trained reward models against summaries and articles disjoint from every training pool, but the ground-truth labels are still constructed by AlignScore, the same evaluator used throughout. Human annotation of a stratified sample of holistic/GCA disagreement cases, or scoring the same candidate pool with a second factuality model, would provide the evaluator-independent reference this study still lacks and would indicate whether either construction procedure produces labels closer to human judgment.
- **Score the RQ4 held-out pool under a holistic ranking too.** The current ground-truth criterion is built entirely from GCA’s own aggregation rule, so it is in-distribution for the GCA reward model and out-of-distribution for holistic in a way plain data held-out-ness does not correct (Chapter 4, Section 4.8; Limitations above). Scoring the same 1,000 held-out summaries under a holistic ranking as well, and reporting each reward model’s success rate against both criteria (or their intersection), would separate a genuine precision advantage from an artefact of the criterion matching GCA’s own training objective.
- **Explain the RQ4 reversal.** Two candidate explanations for why GCA’s ground-truth precision reverses at large training-set sizes were tested and rejected: near-zero-margin training pairs (Section 6.10.4) and a length/sentence-count shortcut (Section 6.10.6). A remaining, less mechanical candidate is that GCA’s sentence-level aggregation lets the reward model fit article-specific, within-pair regularities during training (for instance, patterns tied to how a specific article’s two candidate



summaries happen to differ) that do not generalize to the ground truth’s cross-article comparisons, where the summaries being ranked come from different source articles entirely. Testing this would require comparing reward-model accuracy on held-out same-article pairs against held-out cross-article pairs drawn from the training distribution itself, which was not attempted here.

- **Downstream policy evaluation.** Training a summarization policy on each preference set and evaluating the factual consistency of its generations would test whether differences in preference learnability translate into differences in generated output, which this study deliberately does not address (Chapter 4, Section 4.9).
- **Error-category analysis.** Categorizing disagreement cases by error type would test whether any advantage concentrates in localized errors such as entity or relation mistakes, as originally hypothesized.

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## A. Data and Code Availability

All source code, configuration files, Slurm job scripts, and analysis notebooks used to produce the results reported in this thesis are publicly available at:

<https://github.com/hasnat23/master-thesis-rlaif-gca>

The repository is organized as follows. `src/` contains the pipeline implementation, split into the modules referenced throughout Chapter 5: `src/data/` (subset selection and record schemas), `src/generation/` (candidate summary generation), `src/judging/` (the AlignScore judge wrapper and both preference-construction regimes), `src/reward_model/` (Bradley–Terry training and cross-validation), and `src/eval/` and `src/analysis/` (evaluation and statistics). `configs/` holds the YAML configuration files for subset selection, generation, and judging. `slurm/` holds the batch scripts submitted on MOGON NHR, including `slurm/train_rm_1000.sh`, which produced the headline  $n = 1,000$  result. `analysis/` holds the standalone scripts used for the formula sweep, the judge-mode sweep, and the holistic-versus-GCA comparison, together with `analysis/aggregate_campaigns.py`, which computes the run-level means, bootstrap intervals and Wilcoxon tests for each seed campaign, and `analysis/effect_sizes_and_equivalence.py`, which computes the effect sizes, equivalence tests and minimum detectable effects reported in Chapter 6, Section 6.9. Both read the per-run JSON summaries under `outputs/` directly, so every reported figure can be regenerated from the stored run artifacts without retraining. `progress-updates/` contains the dated working notes kept over the course of the project.

The underlying corpus, CNN/DailyMail version 3.0.0 [16, 30], is publicly available through the Hugging Face `datasets` library and is not redistributed in the repository. The candidate summaries, preference files, and trained reward-model checkpoints generated during this project are likewise not committed, as they run to several gigabytes; they can be regenerated from the corpus using the configuration recorded in Appendix B. The factuality evaluator, `yzha/AlignScore` [41], and the candidate generator, `Mistral-7B-Instruct-v0.3`, are both publicly released model checkpoints obtained from the Hugging Face Hub.

## B. Reproduction Configuration

This appendix records the exact configuration required to reproduce the headline  $n = 1,000$  result. It restates in executable form what Chapter 4, Table 4.1 summarizes and Chapter 5 describes in prose. Under the corrected, fully-seeded training procedure described in Chapter 5, Section 5.5, a single run of these commands at a given seed value is deterministic on fixed hardware; running the full twenty-run headline campaign means repeating the reward-model training command below once per seed in  $\{1, \dots, 20\}$ , as `slurm/seed_campaign.sh` does.

### B.1. Summarization Prompt

Both candidate summaries for a given article are generated from the same fixed prompt template (`SUMMARIZATION_PROMPT` in `src/generation/candidates.py`); the two candidates differ only in the sampling parameters applied at decoding time.

```
Summarize the following news article in a concise paragraph.  
  
Article:  
{article}  
  
Summary:
```

### B.2. Pipeline Parameters

Table B.1 lists the parameters of each pipeline stage as used for the final comparison. Where a value differs from the default recorded in the corresponding YAML file, the command-line override in Section B.3 is authoritative; this is the case for the subset size, the tie margin, and the GCA exponent  $\alpha$ , each of which changed over the course of the project as described in Chapter 4, Section 4.9. The evaluation mode and  $\alpha$  below are the *intended* configuration; Chapter 5, Section 5.4 reports that the twenty-run campaign actually reproduced by the commands in Section B.3 was trained on preference data built under `nli_sp` and  $\alpha = 0.5$ , not the `nli` and  $\alpha = 0.0$  shown here, an implementation oversight found during final review. Running the commands below as written reproduces the *intended* configuration, which is `nli`/ $\alpha = 0.0$ , not the one the headline campaign was actually trained on.



Table B.1.: Pipeline parameters for the final  $n = 1,000$  comparison.

| Stage                   | Parameter           | Value                                                                           |
|-------------------------|---------------------|---------------------------------------------------------------------------------|
| Subset selection        | Corpus              | CNN/DailyMail 3.0.0, test split                                                 |
|                         | Samples             | 1,000                                                                           |
|                         | Subset seed         | 200                                                                             |
|                         | Max article length  | 8,000 characters                                                                |
|                         | Max summary length  | 2,000 characters                                                                |
| Candidate generation    | Model               | Mistral-7B-Instruct-v0.3,<br>bf16                                               |
|                         | Candidate A         | $T = 0.7$ , top- $p = 0.9$                                                      |
|                         | Candidate B         | $T = 1.0$ , top- $p = 0.95$                                                     |
|                         | Max input tokens    | 1,024                                                                           |
|                         | Max new tokens      | 256                                                                             |
|                         | Batch size          | 4                                                                               |
| Preference construction | Evaluator           | yzha/AlignScore,<br>FacebookAI/roberta-base                                     |
|                         | Evaluation mode     | nli                                                                             |
|                         | GCA exponent        | $\alpha = 0.0$ (simple mean)                                                    |
|                         | Tie margin          | 0                                                                               |
| Reward-model training   | Backbone            | FacebookAI/roberta-base                                                         |
|                         | Epochs              | 5                                                                               |
|                         | Batch size          | 8                                                                               |
|                         | Learning rate       | $2 \times 10^{-5}$ , linear warmup over 10%<br>of steps                         |
|                         | Max sequence length | 512 tokens                                                                      |
|                         | Max article chars   | 2,000                                                                           |
|                         | Cross-validation    | 5-fold; training seeds 1–20 (head-<br>line campaign, Chapter 6, Ta-<br>ble 6.8) |

## B.3. Command-Line Invocations

The three stages are run in sequence. Preference construction produces both conditions from a single invocation, and reward-model training consumes both preference files so that the holistic and GCA models are trained under an identical recipe within one process.

```
# 1. Subset selection
python scripts/01_prepare_subset.py --n-samples 1000 --seed 200

# 2. Candidate generation
python scripts/02_generate_candidates.py \
    --config configs/generation_1000.yaml

# 3. Preference construction (both conditions)
python src/judging/build_reward_preferences.py \
    --mode both --alpha 0.0 --margin 0 \
    --alignscore-evaluation-mode nli \
    --alignscore-backbone FacebookAI/roberta-base

# 4. Reward-model training (5-fold CV, both conditions)
python src/reward_model/run_training.py \
    --holistic data/preferences_1000/\
holistic_reward_preferences_1000.jsonl \
    --gca      data/preferences_1000/\
gca_reward_preferences_1000.jsonl \
    --output-dir outputs/reward_models_1000 \
    --backbone FacebookAI/roberta-base \
    --epochs 5 --batch-size 8 --lr 2e-5 \
    --max-length 512 --max-article-chars 2000 \
    --kfold 5 --seed 1
```

Step 3 above, run exactly as written, produces the intended  $nli/\alpha = 0.0$  preference data. The actual headline campaign was produced by the Slurm scripts `slurm/build_preferences_1000.sh` and `slurm/build_preferences_scale.sh` (below) instead of this exact command, and those scripts omit `-alignscore-evaluation-mode` (defaulting to `nli_sp`) and, for  $n = 1,000$ , pass `-alpha 0.5`; this is the discrepancy reported in Chapter 5, Section 5.4. A reader following the four commands above exactly as shown will reproduce the configuration this thesis intended, not the one the headline numbers in Chapter 6 were actually measured from.

This reproduces one run of the headline campaign, seed 1 of 20. Substituting each of `-seed 2` through `-seed 20` in turn reproduces the remaining nineteen.

In step 4 the two preference-file paths are shown split across lines for typesetting only; the trailing backslash inside a path is a line continuation in this listing, not part of the filename.

On the MOGON NHR cluster these are submitted as Slurm batch jobs (`slurm/generate_candidates_1000.sh`, `slurm/build_preferences_1000.sh`, `slurm/trai`

`n_rm_1000.sh`) on partition `a100dl` with one A100-SXM4-40GB GPU, 32 GB of host memory, and four CPU cores per task.

## C. Example Preference Record

Every record written by the GCA preference-construction stage retains the full sentence-level detail behind the summary-level decision, which is what makes the qualitative disagreement analysis in Chapter 6 possible. The listing below shows one such record, abridged: the article, reference summary, and the `chosen/rejected` fields are truncated, and the sentence list for summary B is cut after the first three sentences. All numeric values are reproduced verbatim.

The record shown was produced by the earlier  $n = 200$  run, which used the exploratory settings  $\alpha = 0.5$  and  $\text{margin} = 0.05$  rather than the final  $\alpha = 0.0$  and  $\text{margin} = 0$  (Chapter 4, Section 4.9). It is included to illustrate the record *schema* and the relationship between the sentence scores and the aggregate, not as an instance of the final configuration. Note in particular that `gca_score_a` = 0.382 is well below `mean_score_a` = 0.629: this is the penalty term at  $\alpha = 0.5$  pulling the aggregate down towards the summary's weakest sentence, the behaviour that Chapter 6, Section 6.3 found to perform worse than the simple mean.

```
{
  "sample_id": "cnn_00000_b0b34b2d843e",
  "article": "LONDON, England (Reuters) -- Harry Potter star Daniel ...",
  "reference_summary": "Harry Potter star Daniel Radcliffe gets ...",
  "summary_a": "Daniel Radcliffe, the 18-year-old actor who plays ...",
  "summary_b": "On Monday, July 23, 2007, Harry Potter actor Daniel ...",
  "gca_score_a": 0.382102,
  "gca_score_b": 0.169184,
  "gca_score_diff": 0.212918,
  "mean_score_a": 0.629039,
  "mean_score_b": 0.578186,
  "min_score_a": 0.232103,
  "min_score_b": 0.049505,
  "n_sentences_a": 3,
  "n_sentences_b": 6,
  "n_low_faithfulness_a": 1,
  "n_low_faithfulness_b": 3,
  "decision": "A",
  "judge_method": "gca",
  "judge_model": "yzha/AlignScore [FacebookAI/roberta-base]",
  "margin": 0.05,
  "alpha": 0.5,
  "sentence_details_a": [
    { "index": 0,
      "text": "Daniel Radcliffe, the 18-year-old actor who plays Harry
```

```

        Potter, has gained access to a 20 million pound fortune
        on his 18th birthday.",
        "score": 0.748840, "label": "faithful" },
    { "index": 1,
      "text": "Despite the newfound wealth, he has stated that he does
              not plan to be extravagant, and will instead use his
              money to buy books, CDs, and DVDs.",
        "score": 0.232103, "label": "hallucinated" },
    { "index": 2,
      "text": "Radcliffe has filmed a TV movie and an Australian film,
              and has made his stage debut in \"Equus.\" He is expected
              to face closer media scrutiny now that he is legally an
              adult.",
        "score": 0.906173, "label": "faithful" }
  ],
  "sentence_details_b": [
    { "index": 0, "text": "On Monday, July 23, 2007, ...",
      "score": 0.475803, "label": "hallucinated" },
    { "index": 1, "text": "Radcliffe has stated he has no ...",
      "score": 0.997014, "label": "faithful" },
    { "index": 2, "text": "Instead, he plans to use the money ...",
      "score": 0.049505, "label": "hallucinated" }
  ]
}

```

Two features of this record are worth drawing out, because they illustrate mechanisms discussed in the main text. First, the `label` field is a threshold applied at a score of 0.5 for reporting convenience only; it plays no part in the aggregation, which operates on the continuous scores. Second, summary B contains both the highest-scoring sentence in the pair (0.997) and the lowest (0.050). Under the holistic condition this internal variation is not observable at all: the summary receives a single score. This is precisely the information that sentence-level decomposition exposes and holistic scoring discards, and whether exposing it yields a more learnable preference signal is the question the thesis sets out to answer.